

SYSTEM V
APPLICATION
BINARY INTERFACE

Intel386™ Architecture
Processor Supplement

Fourth Edition

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The Intel386 Architecture and the System V ABI

The *System V Application Binary Interface*, or ABI, defines a system interface for compiled application programs. Its purpose is to establish a standard binary interface for application programs on systems that implement the interfaces defined in the *System V Interface Definition, Edition 4*. This includes systems that have implemented UnixWare® 2.0. M

This document is a supplement to the generic *System V ABI*, and it contains information specific to System V implementations built on the Intel386 processor architecture. Together, these two specifications, the generic *System V ABI* and the *Intel386 Architecture System V ABI Supplement* (hereafter referred to as the *Intel386 ABI*), constitute a complete *System V Application Binary Interface* specification for systems that implement the processor architecture of the Intel386 microprocessors.

Note that, because the Intel486 and Pentium processor are compatible members of the Intel386 architecture, this *Intel386 ABI* also applies to any system built with the Intel486 or the Pentium processor chips. M M

How to Use the Intel386 Architecture ABI Supplement

This document is a supplement to the generic *System V ABI* and contains information referenced in the generic specification that may differ when System V is implemented on different processors. Therefore, the generic ABI is the prime reference document, and this supplement is provided to fill gaps in that specification.

As with the *System V ABI*, this specification references other publicly-available reference documents, especially the *Intel 80386 Programmer's Reference Manual*. All the information referenced by this supplement should be considered part of this specification, and just as binding as the requirements and data explicitly included here.

Evolution of the ABI Specification

The *System V Application Binary Interface* will evolve over time to address new technology and market requirements, and will be reissued at intervals of approximately three years. Each new edition of the specification is likely to contain extensions and additions that will increase the potential capabilities of applications that are written to conform to the ABI.

As with the *System V Interface Definition*, the ABI will implement **Level 1** and **Level 2** support for its constituent parts. **Level 1** support indicates that a portion of the specification will continue to be supported indefinitely, while **Level 2** support means that a portion of the specification may be withdrawn or altered after the next edition of the ABI is made available. That is, a portion of the specification moved to **Level 2** support in an edition of the ABI specification will remain in effect at least until the following edition of the specification is published.

These **Level 1** and **Level 2** classifications and qualifications apply to this Supplement, as well as to the generic specification. All components of the ABI and of this supplement have **Level 1** support unless they are explicitly labelled as **Level 2**.

The following documents may be of interest to the reader of this specification:

- *i486 MICROPROCESSOR Programmer's Reference Manual* (Intel Literature order number 240486)

- *80386 Programmer's Reference Manual* (Intel Literature order number 230985)
- *80387 Programmer's Reference Manual* (Intel Literature order number 231917)
- *UnixWare® 2.0 Command Reference (a-l)*
- *UnixWare® 2.0 Command Reference (m-z)*
- *UnixWare® 2.0 Operating System API Reference: System Calls*
- *UnixWare® 2.0 Operating System API Reference: Library Functions*
- *UnixWare® 2.0 System Administration: Volumes I and II*
- *System V Interface Definition, Edition 4*

NOTE

Diffmarkings have been retained in the text of this book to indicate in which revisions of System V certain modifications were made to the *ABI*.

A "G" character in the right hand margin indicates a change in the *ABI* made in UNIX System V Release 4.2.

A "M" character in the right hand margin indicates a change in the *ABI* made in UnixWare® 2.0. M

2 SOFTWARE INSTALLATION

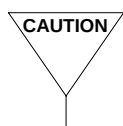
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Software Distribution Formats

Physical Distribution Media

Approved media for physical distribution of ABI-conforming software are listed below. Inclusion of a particular medium on this list does not require an ABI-conforming system to accept that medium. For example, a conforming system may install all software through its network connection and accept none of the listed media.

- 1.44MB 3 1/2" floppy disk: quad-density, double-sided, 80 tracks/side, 18 sectors/track, 512 bytes/sector.
- 1.2MB 5 1/4" floppy disk: quad-density, double-sided, 80 tracks/side, 15 sectors/track, 512 bytes/sector.
- 360KB 5 1/4" floppy disk: double-density, double-sided, 40 tracks/side, 9 sectors/track, 512 bytes/sector.
- 60 MB quarter-inch cartridge tape in QIC-24 format. G
- CD-ROM optical disks. G
- 150 MB quarter-inch tape.



The use of 360KB 5 1/4" floppy disk, and 60 MB quarter inch cartridge tape as media for application distribution is moved to Level 2 as of January 1, 1993. G

File System Formats

Every file system storage volume must conform to a supported format. Two formats are supported: s5 and ufs.

s5 File System

The first physical block on the medium should be empty, and the second contains the device's *superblock*. The third contains an inode list, and remaining blocks on the device contain data. The *superblock* has the following format:

```

#define NICFREE          50
#define NICINOD          100

struct filsys {
    u_short    s_isize;
    daddr_t    s_fsize;
    short      s_nfree;
    daddr_t    s_free[NICFREE];
    short      s_ninode;
    ushort_t   s_inode[NICINOD];
    char        s_flock;
    char        s_iloc;
    char        s_fmod;
    char        s_ronly;
    time_t      s_time;
    short      s_dinfo[4];
    daddr_t     s_tfree;
    ushort_t    s_tinode;
    char        s_fname[6];
    char        s_fpack[6];
    long        s_fill[12];
    long        s_state;
    long        s_magic;
    long        s_type;
};

#define FsMAGIC          0xfd187e20

#define Fs1b              1
#define Fs2b              2
#define Fs4b              3

#define FsOKAY            0x7c269d38
#define FsACTIVE          0x5e72d81a
#define FsBAD             0xcb096f43
#define FsBADBLK          0xbadbc14b

```

s_type indicates the file system type. Currently, three types of file systems are supported: the original 512-byte logical block, the 1024-byte logical block, and the 2048-byte logical block. *s_magic* is used to distinguish the original 512-byte oriented file systems from the newer file systems. If this field is not equal to the magic number, *FsMAGIC*, the type is assumed to be *Fs1b*, otherwise the *s_type* field is used.

s_state indicates the state of the file system. A cleanly unmounted, undamaged file system is indicated by the *FsOKAY* state. After a file system has been mounted for update, the state changes to *FsACTIVE*.

s_ysize is the address of the first data block after the i-list; the i-list starts just after the super-block, namely in block 2; thus the i-list is *s_ysize*-2 blocks long.

s_fsize is the first block not potentially available for allocation to a file.

The free list for each volume is maintained as follows. The *s_free* array contains up to 49 numbers of free blocks. *s_free*[0] is the block number of the head of a chain of blocks constituting the free list. The first long in each free-chain block is the number (up to 50) of free-block numbers listed in the next 50 longs of this chain member. The first of these 50 blocks is the link to the next member of the chain.

s_tfree is the total free blocks available in the file system.

s_ninode is the number of free i-numbers in the *s_inode* array.

s_tinode is the total free i-nodes available in the file system.

s_flock and *s_iloc* are flags maintained in the core copy of the file system. *s_fmod* is a flag that indicates that the super-block has changed and should be copied to the disk during the next periodic update of file system information.

s_roonly is a read-only flag to indicate write-protection.

s_time is the last time the super-block of the file system was changed, and is the number of seconds that have elapsed since 00:00 Jan. 1, 1970 (GMT).

s_fname is the name of the file system and *s_fpack* is the name of the pack.

I-numbers begin at 1, and the storage for i-nodes begins in block 2. I-node 1 is reserved for future use. I-node 2 is reserved for the root directory of the file system, but no other i-number has a built-in meaning. Each i-node represents one file.

UFS File System

In the UFS file system, the first physical block on the device should be empty, and the second contains the *superblock* for the file system. Remaining blocks contain data.

The ufs *superblock* contains an *fs* data structure. This structure, and other relevant data objects are defined below.

```

struct csum {
    long    cs_ndir;
    long    cs_nbfree;
    long    cs_nifree;
    long    cs_nffree;
};

struct fs {
    struct fs    *fs_link;
    struct fs    *fs_rlink;
    daddr_t      fs_sblkno;
    daddr_t      fs_cblkno;
    daddr_t      fs_iblkno;
    daddr_t      fs_dblkno;
    long         fs_cgoffset;
    long         fs_cgmask;
    time_t       fs_time;
    long         fs_size;
    long         fs_dsize;
    long         fs_ncg;
    long         fs_bsize;
    long         fs_fsize;
    long         fs_frag;
    long         fs_minfree;
    long         fs_rotdelay;
    long         fs_rps;
    long         fs_bmask;
    long         fs_fmask;
    long         fs_bshift;
    long         fs_fshift;
    long         fs_maxcontig;
    long         fs_maxbpg;
    long         fs_fragshift;
    long         fs_fsbtodb;
    long         fs_sbsize;
    long         fs_csmask;
    long         fs_csshift;
    long         fs_nindir;
    long         fs_inopb;
    long         fs_nspf;
    long         fs_optim;
    long         fs_state;
    long         fs_sparecon[2];
    long         fs_id[2];
    daddr_t      fs_csaddr;
    long         fs_cssize;
    long         fs_cgsize;
    long         fs_ntrak;
    long         fs_nsect;
    long         fs_spc;
    long         fs_ncyl;

```

(continued on next page)

```

    long      fs_cpg;
    long      fs_ipg;
    long      fs_fpg;
    struct csum fs_cstotal;
    char      fs_fmod;
    char      fs_clean;
    char      fs_ronly;
    char      fs_flags;
    char      fs_fsmnt[MAXMNTLEN];
    long      fs_cgrotor;
    struct csum *fs_csp[MAXCSBUFS];
    long      fs_cpc;
    short     fs_postbl[MAXCPG][NRPOS];
    long      fs_magic;
    u_char    fs_rotbl[1];
};

struct cg {
    struct cg *cg_link;
    struct cg *cg_rlink;
    time_t    cg_time;
    long      cg_cgx;
    short     cg_ncyl;
    short     cg_niblk;
    long      cg_ndblk;
    struct csum cg_cs;
    long      cg_rotor;
    long      cg_frotor;
    long      cg_irotor;
    long      cg_frsum[MAXFRAG];
    long      cg_btot[MAXCPG];
    short     cg_b[MAXCPG][NRPOS];
    char      cg_iused[MAXIPG/NBBY];
    long      cg_magic;
    u_char    cg_free[1];
};

#define FS_MAGIC      0x011954
#define BBSIZE        8192
#define SBSIZE        8192
#define BBLOCK        ((daddr_t)(0))
#define SBLOCK        ((daddr_t)(BBLOCK + BBSIZE / DEV_BSIZE))
#define UFSROOTINO    ((ino_t)2)
#define LOSTFOUNDINO  (UFSROOTINO + 1)
#define NRPOS         8
#define MAXIPG        2048
#define MINBSIZE      4096
#define MAXCPG        32
#define MAXMNTLEN     512
#define MAXCSBUFS     32
#define FS_OPTTIME    0

```

(continued on next page)

```
#define FS_OPTSPACE    1
#define MAXBPC         (SBSIZE - sizeof (struct fs))
#define CG_MAGIC        0x090255
```

CAUTION

The distribution of software in filesystem format is Level 2 as of January 1, 1993. G

3 LOW-LEVEL SYSTEM INFORMATION

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Machine Interface

Processor Architecture

The Intel *80386 Programmer's Reference Manual* (Intel Literature order number 230985) and the Intel *80387 Programmer's Reference Manual* (Intel Literature order number 231917) together define the processor architecture. The architecture of the combined Intel386/Intel 387 processors is hereafter referred to as the Intel386 architecture. Programs intended to execute directly on the processor use the instruction set, instruction encodings, and instruction semantics of the architecture. Three points deserve explicit mention.

- A program may assume all documented instructions exist.
- A program may assume all documented instructions work.
- A program may use only the instructions defined by the architecture.

In other words, *from a program's perspective*, the execution environment provides a complete and working implementation of the Intel386 architecture.

This does not imply that the underlying implementation provides all instructions in hardware, only that the instructions perform the specified operations and produce the specified results. The ABI neither places performance constraints on systems nor specifies what instructions must be implemented in hardware. A software emulation of the architecture could conform to the ABI.

Some processors might support the Intel386 architecture as a subset, providing additional instructions or capabilities. Programs that use those capabilities explicitly do not conform to the Intel386 ABI. Executing those programs on machines without the additional capabilities gives undefined behavior.

Data Representation

Within this specification, the term *halfword* refers to a 16-bit object, the term *word* refers to a 32-bit object, and the term *doubleword* refers to a 64-bit object.

Fundamental Types

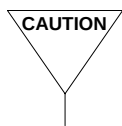
Figure 3-1 shows the correspondence between ANSI C's scalar types and the processor's.

Figure 3-1: Scalar Types

Type	C	sizeof	Alignment (bytes)	Intel386 Architecture
Integral	char	1	1	signed byte
	signed char	1	1	signed byte
	unsigned char	1	1	unsigned byte
	short	2	2	signed halfword
	signed short	2	2	signed halfword
	unsigned short	2	2	unsigned halfword
Integral	int	4	4	signed word
	signed int			
Integral	long			
	signed long			
Integral	enum			
	unsigned int	4	4	unsigned word
Integral	unsigned long	4	4	unsigned word
Pointer	<i>any-type</i> * <i>any-type</i> (*)()	4	4	unsigned word
Floating-point	float	4	4	single-precision (IEEE)
	double	8	4	double-precision (IEEE)
	long double	12	4	extended-precision (IEEE)

NOTE

The Intel386 architecture does not require doubleword alignment for double-precision values. Nevertheless, for data structure compatibility with other Intel architectures, compilers may provide a method to align double-precision values on doubleword boundaries.



A compiler that provides the doubleword alignment mentioned above can generate code (data structures and function calling sequences) that do not conform to the Intel386 ABI. Programs built with the doubleword alignment facility can thus violate conformance to the Intel386 ABI. See “Aggregates and Unions” below and “Function Calling Sequence” later in this chapter for more information.

A null pointer (for all types) has the value zero.

The Intel386 architecture does not require all data access to be properly aligned. For example, double-precision values occupy 1 doubleword (8-bytes), and their natural alignment is a word boundary, meaning their addresses are multiples of 4. Compilers should allocate independent data objects with the proper alignment; examples include global arrays of double-precision variables, FORTRAN COMMON blocks, and unconstrained stack objects. However, some language facilities (such as FORTRAN EQUIVALENCE statements) may create objects with only byte alignment. Consequently, arbitrary data accesses, such as pointers dereference or reference arguments, might or might not be properly aligned. Accessing misaligned data will be slower than accessing properly aligned data, but otherwise there is no difference.

Aggregates and Unions

Aggregates (structures and arrays) and unions assume the alignment of their most strictly aligned component. The size of any object, including aggregates and unions, is always a multiple of the object’s alignment. An array uses the same alignment as its elements. Structure and union objects can require padding to meet size and alignment constraints. The contents of any padding is undefined.

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- An entire structure or union object is aligned on the same boundary as its most strictly aligned member.
- Each member is assigned to the lowest available offset with the appropriate alignment. This may require *internal padding*, depending on the previous member.
- A structure’s size is increased, if necessary, to make it a multiple of the alignment. This may require *tail padding*, depending on the last member.

NOTE

ABI conformant code may not read or modify anything marked reserved or padding.

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In the following examples, members' byte offsets appear in the upper right corners.

Figure 3-2: Structure Smaller Than a Word



Figure 3-3: No Padding

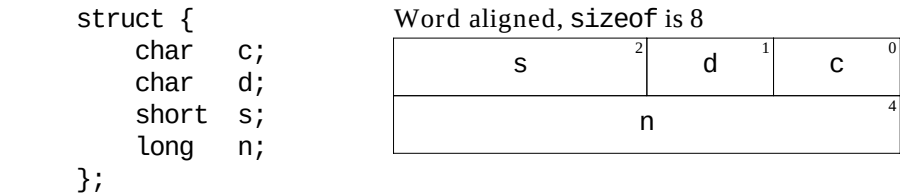


Figure 3-4: Internal Padding

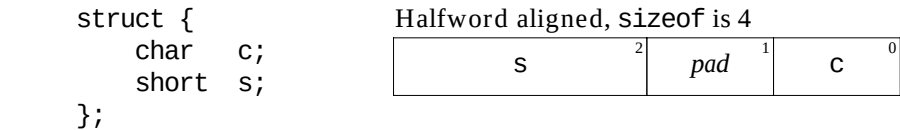
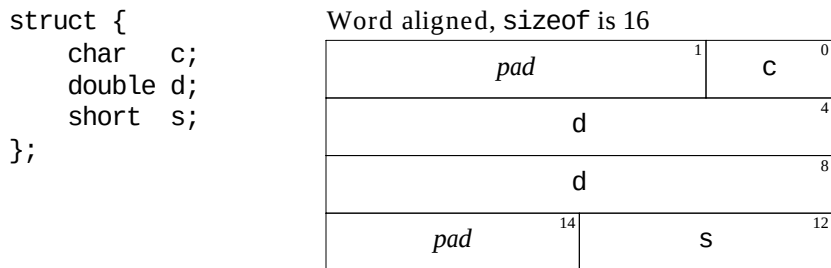


Figure 3-5: Internal and Tail Padding



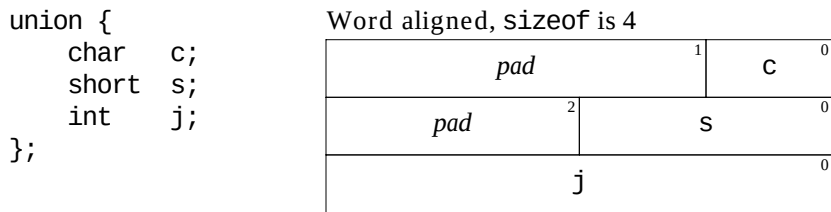
NOTE

The Intel386 architecture does not require doubleword alignment for double-precision values. Nevertheless, for data structure compatibility with other Intel architectures, compilers may provide a method to align double-precision values on doubleword boundaries.

CAUTION

A compiler that provides the doubleword alignment mentioned above would arrange the preceding structure differently. Programs built with the doubleword alignment facility would not conform to the Intel386 ABI, and they would not be data-compatible with conforming Intel386 programs.

Figure 3-6: union Allocation



Bit-Fields

C struct and union definitions may have *bit-fields*, which define integral objects with a specified number of bits.

Figure 3-7: Bit-Field Ranges

Bit-field Type	Width w	Range
signed char	1 to 8	-2^{w-1} to $2^{w-1} - 1$
char		0 to $2^w - 1$
unsigned char		0 to $2^w - 1$
signed short	1 to 16	-2^{w-1} to $2^{w-1} - 1$
short		0 to $2^w - 1$
unsigned short		0 to $2^w - 1$
signed int	1 to 32	-2^{w-1} to $2^{w-1} - 1$
int		0 to $2^w - 1$
enum		0 to $2^w - 1$
unsigned int		0 to $2^w - 1$
signed long	1 to 32	-2^{w-1} to $2^{w-1} - 1$
long		0 to $2^w - 1$
unsigned long		0 to $2^w - 1$

“Plain” bit-fields (that is, those neither signed nor unsigned) always have non-negative values. Although they may have type `char`, `short`, `int`, or `long` (which can have negative values), these bit-fields have the same range as a bit-field of the same size with the corresponding unsigned type. Bit-fields obey the same size and alignment rules as other structure and union members, with the following additions:

- Bit-fields are allocated from right to left (least to most significant).
- A bit-field must entirely reside in a storage unit appropriate for its declared type. Thus a bit-field never crosses its unit boundary.
- Bit-fields may share a storage unit with other struct/union members, including members that are not bit-fields. Of course, struct members occupy different parts of the storage unit.
- Unnamed bit-fields’ types do not affect the alignment of a structure or union, although individual bit-fields’ member offsets obey the alignment constraints.

The following examples show `struct` and `union` members' byte offsets in the upper right corners; bit numbers appear in the lower corners.

Figure 3-8: Bit Numbering

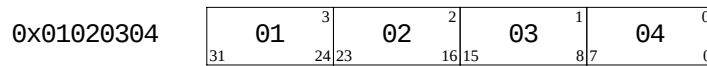


Figure 3-9: Right-to-Left Allocation

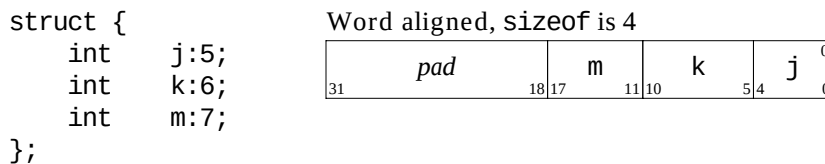


Figure 3-10: Boundary Alignment

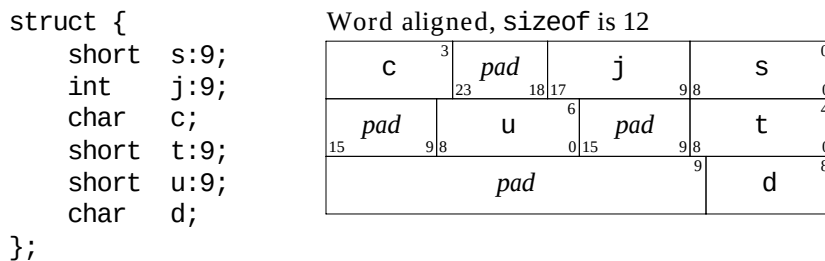


Figure 3-11: Storage Unit Sharing

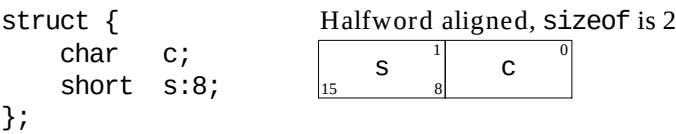


Figure 3-12: union Allocation

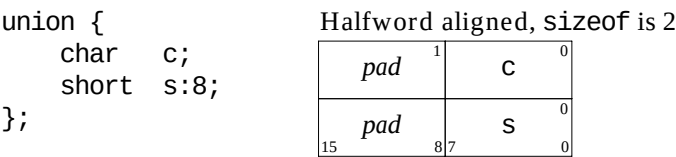
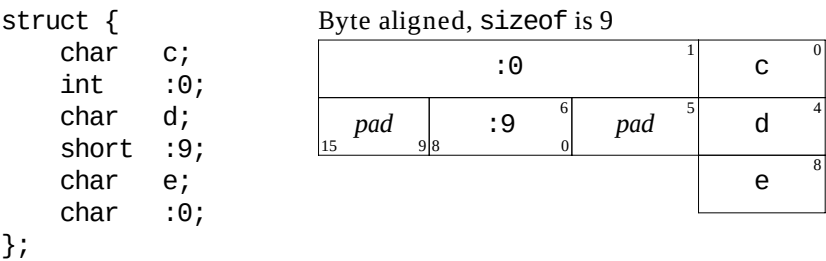


Figure 3-13: Unnamed Bit-Fields



As the examples show, `int` bit-fields (including signed and unsigned) pack more densely than smaller base types. One can use `char` and `short` bit-fields to force particular alignments, but `int` is generally more efficient.

Function Calling Sequence

This section discusses the standard function calling sequence, including stack frame layout, register usage, parameter passing, and so on. The system libraries described in Chapter 6 require this calling sequence.

NOTE

The standard calling sequence *requirements* apply only to global functions. Local functions that are not reachable from other compilation units may use different conventions. Nonetheless, it is recommended that all functions use the standard calling sequence when possible.

NOTE

C programs follow the conventions given here. For specific information on the implementation of C, see “Coding Examples” in this chapter.

Registers and the Stack Frame

The Intel386 architecture provides a number of registers. All the integer registers and all the floating-point registers are global to all procedures in a running program.

Brief register descriptions appear in Figure 3-14 more complete information appears later.

Figure 3-14: Processor Registers

Type	Name	Usage
General	%eax	Return value
	%edx	Dividend register (divide operations)
	%ecx	Count register (shift and string operations)
	%ebx	Local register variable
	%ebp	Stack frame pointer (optional)
	%esi	Local register variable
	%edi	Local register variable
	%esp	Stack pointer
Floating-point	%st(0)	floating-point stack top, return value
	%st(1)	floating-point next to stack top
	...	
	%st(7)	floating-point stack bottom

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In addition to registers, each function has a frame on the run-time stack. This stack grows downward from high addresses. Figure 3-15 shows the stack frame organization.

Figure 3-15: Standard Stack Frame

Position	Contents	Frame	
$4n+8(\%ebp)$	argument word n	Previous	<i>High addresses</i>
...	...		
$8(\%ebp)$	argument word 0	Current	<i>Low addresses</i>
$4(\%ebp)$	return address		
$0(\%ebp)$	previous %ebp (optional)		
$-4(\%ebp)$	unspecified		
...	...		
$0(\%esp)$	variable size		

Several key points about the stack frame deserve mention.

- The stack is word aligned. Although the architecture does not require any alignment of the stack, software convention and the operating system requires that the stack be aligned on a word boundary.

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- Argument words are pushed onto the stack in reverse order (that is, the rightmost argument in C call syntax has the highest address), preserving the stack's word alignment. All incoming arguments appear on the stack, residing in the stack frame of the caller.
- An argument's size is increased, if necessary, to make it a multiple of words. This may require tail padding, depending on the size of the argument.
- Other areas depend on the compiler and the code being compiled. The standard calling sequence does not define a maximum stack frame size, nor does it restrict how a language system uses the "unspecified" area of the standard stack frame.

All registers on the Intel386 are global and thus visible to both a calling and a called function. Registers `%ebp`, `%ebx`, `%edi`, `%esi`, and `%esp` "belong" to the calling function. In other words, a called function must preserve these registers' values for its caller. Remaining registers "belong" to the called function. If a calling function wants to preserve such a register value across a function call, it must save the value in its local stack frame.

Some registers have assigned roles in the standard calling sequence:

<code>%esp</code>	The <i>stack pointer</i> holds the limit of the current stack frame, which is the address of the stack's bottom-most, valid word. At all times, the stack pointer should point to a word-aligned area.	
<code>%ebp</code>	The <i>frame pointer</i> optionally holds a base address for the current stack frame. Consequently, a function has registers pointing to both ends of its frame. Incoming arguments reside in the previous frame, referenced as positive offsets from <code>%ebp</code> , while local variables reside in the current frame, referenced as negative offsets from <code>%ebp</code> . A function must preserve this register's value for its caller.	G
<code>%eax</code>	<i>Integral and pointer return values</i> appear in <code>%eax</code> . A function that returns a <code>struct</code> or <code>union</code> value places the address of the result in <code>%eax</code> . Otherwise this is a scratch register.	
<code>%ebx</code>	As described below, this register serves as the <i>global offset table base register</i> for position-independent code. For absolute code, <code>%ebx</code> serves as a local register and has no specified role in the function calling sequence. In either case, a function must preserve the register value for the caller.	
<code>%esi</code> and <code>%edi</code>	These <i>local registers</i> have no specified role in the function calling sequence. A function must preserve their values for the caller.	

Function Calling Sequence

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%ecx and %edx	<i>Scratch registers</i> have no specified role in the standard calling sequence. Functions do not have to preserve their values for the caller.	
%st(0)	<i>Floating-point return values</i> appear on the top of the floating-point register stack; there is no difference in the representation of single- or double-precision values in floating-point registers. If the function does not return a floating-point value, then this register must be empty. This register must be empty before entry to a function.	G
%st(1) through %st(7)	<i>Floating-point scratch registers</i> have no specified role in the standard calling sequence. These registers must be empty before entry and upon exit from a function.	
EFLAGS	The <i>flags register</i> contains the system flags, such as the direction flag and the carry flag. The direction flag must be set to the “forward” (that is, zero) direction before entry and upon exit from a function. Other user flags have no specified role in the standard calling sequence and are not preserved.	
Floating-Point Control Word	The Intel387 <i>control word</i> contains the floating-point flags, such as the rounding mode and exception masking.	*

Signals can interrupt processes [see `signal(BA_OS)`]. Functions called during signal handling have no unusual restrictions on their use of registers. Moreover, if a signal handling function returns, the process resumes its original execution path with registers restored to their original values. Thus, programs and compilers may freely use all registers without the danger of signal handlers changing their values.

Functions Returning Scalars or No Value

A function that returns an integral or pointer value places its result in register `%eax`.

A floating-point return value appears on the top of the Intel387 register stack. The caller then must remove the value from the Intel387 stack, even if it doesn't use the value. Failure of either side to meet its obligations leads to undefined program behavior. The standard calling sequence does not include any method to detect such failures nor to detect return value type mismatches. Therefore the user must declare all functions properly. There is no difference in the representation of

single-, double- or extended-precision values in floating-point registers.

Functions that return no value (also called procedures or `void` functions) put no particular value in any register.

A `call` instruction pushes the address of the next instruction (the return address) onto the stack. The `ret` instruction pops the address off the stack and effectively continues execution at the next instruction after the `call` instruction. A function that returns a scalar or no value must preserve the caller's registers as described earlier. Additionally, the called function must remove the return address from the stack, leaving the stack pointer (`%esp`) with the value it had before the `call` instruction was executed.

To illustrate, the following function prologue allocates 80 bytes of local stack space and saves the local registers `%ebx`, `%esi`, and `%edi`.

Figure 3-16: Function Prologue

```
prologue:
    pushl %ebp          / save frame pointer
    movl  %esp, %ebp    / set new frame pointer
    subl  $80, %esp     / allocate stack space
    pushl %edi          / save local register
    pushl %esi          / save local register
    pushl %ebx          / save local register
```

An epilogue for the example that restores the state for the caller. This example returns the value in `%edi` by moving it to `%eax`.

Figure 3-17: Function Epilogue

```
        movl    %edi, %eax    / set up return value
epilogue:
        popl    %ebx          / restore local register
        popl    %esi          / restore local register
        popl    %edi          / restore local register
        leave   / restore frame pointer
        ret      / pop return address
```

NOTE

Although some functions can be optimized to eliminate the save and restore of the frame pointer, the general case uses the standard prologue and epilogue.

Sections below describe where arguments appear on the stack. The examples are written as if the function prologue described above had been used.

Position-independent code uses the `%ebx` register to hold the address of the global offset table. If a function needs the global offset table's address, either directly or indirectly, it is responsible for computing the value. See "Coding Examples" later in this chapter and "Dynamic Linking" in Chapter 5 for more information.

Functions Returning Structures or Unions

If a function returns a structure or union, then the caller provides space for the return value and places its address on the stack as argument word zero. In effect, this address becomes a "hidden" first argument. Having the caller supply the return object's space allows re-entrancy.

NOTE

Structures and unions in this context have fixed sizes. The ABI does not specify how to handle variable sized objects.

A function that returns a structure or union also sets `%eax` to the value of the original address of the caller's area before it returns. Thus when the caller receives control again, the address of the returned object resides in register `%eax` and can be used to access the object. Both the calling and the called functions must cooperate to pass the return value successfully:

- The calling function must supply space for the return value and pass its address in the stack frame;
- The called function must use the address from the frame and copy the return value to the object so supplied;
- The called function must remove this address from the stack before returning.

Failure of either side to meet its obligations leads to undefined program behavior. The standard function calling sequence does not include any method to detect such failures nor to detect structure and union type mismatches. Therefore the user must declare all functions properly.

Figure 3-18 illustrates the stack contents when the function receives control (after the `call` instruction) and when the calling function again receives control (after the `ret` instruction).

Figure 3-18: Stack Contents for Functions Returning struct/union

Position	After <code>call</code>	After <code>ret</code>	Position
$4n+4(\%esp)$	argument word n	argument word n	$4n-4(\%esp)$
	...	...	
$8(\%esp)$	argument word 1	argument word 1	$0(\%esp)$
$4(\%esp)$	value address	<i>undefined</i>	
$0(\%esp)$	return address		

To illustrate, the following function prologue allocates 80 bytes of local stack space and saves the local registers `%ebx`, `%esi`, and `%edi`. Additionally, it removes the “hidden” argument from the stack and saves it in the highest word of the local stack frame.

Figure 3-19: Function Prologue (Returning struct/union)

```
prologue:
    popl    %eax           / pop return address
    xchgl   %eax, 0(%esp)  / swap return address
                        / and return value address
    pushl   %ebp           / save frame pointer
    movl    %esp, %ebp     / set new frame pointer
    subl    $80, %esp      / allocate local space
    pushl   %edi           / save local register
    pushl   %esi           / save local register
    pushl   %ebx           / save local register
    movl    %eax, -4(%ebp) / save return value address
```

An epilogue for the example that restores the state for the caller.

Figure 3-20: Function Epilogue

```
epilogue:
    movl    -4(%ebp), %eax  / set up return value
    popl    %ebx           / restore local register
    popl    %esi           / restore local register
    popl    %edi           / restore local register
    leave   / restore frame pointer
    ret      / pop return address
```

NOTE

Although some functions can be optimized to eliminate the save and restore of the frame pointer, the general case uses the standard prologue and epilogue.

Sections below describe where arguments appear on the stack. The examples are written as if the function prologue described above had been used.

Position-independent code uses the `%ebx` register to hold the address of the global offset table. If a function needs the global offset table's address, either directly or indirectly, it is responsible for computing the value. See "Coding Examples" later in this chapter and "Dynamic Linking" in Chapter 5 for more information.

Integral and Pointer Arguments

As mentioned, a function receives all its arguments through the stack; the last argument is pushed first. In the standard calling sequence, the first argument is at offset 8(`%ebp`), the second argument is at offset 12(`%ebp`), and so on. Functions pass all integer-valued arguments as words, expanding or padding signed or unsigned bytes and halfwords as needed.

Figure 3-21: Integral and Pointer Arguments

Call	Argument	Stack address
g(1, 2, 3, (void *)0);	1	8(%ebp)
	2	12(%ebp)
	3	16(%ebp)
	(void *)0	20(%ebp)

Floating-Point Arguments

The stack also holds floating-point arguments: single-precision values use one word, double-precision use two, and extended-precision use three. See "Coding Examples" for information about floating-point arguments and variable argument lists. The example below uses only double-precision arguments. Single- and extended-precision arguments behave as specified above.

Figure 3-22: Floating-Point Arguments

Call	Argument	Stack address
h(1.414, 1, 2.998e10);	word 0, 1.414	8(%ebp)
	word 1, 1.414	12(%ebp)
	1	16(%ebp)
	word 0, 2.998e10	20(%ebp)
	word 1, 2.998e10	24(%ebp)

NOTE The Intel386 architecture does not require doubleword alignment for double-precision values. Nevertheless, for data structure compatibility with other Intel architectures, compilers may provide a method to align double-precision values on doubleword boundaries.

CAUTION A compiler that provides the doubleword alignment mentioned above would have to maintain doubleword alignment for the stack. Moreover, the arguments in the preceding example would appear in different positions. Programs built with the doubleword alignment facility would not conform to the Intel386 ABI, and their function calling sequence would not be compatible with conforming Intel386 programs.

Structure and Union Arguments

As described in the data representation section, structures and unions can have byte, halfword, or word alignment, depending on the constituents. An argument's size is increased, if necessary, to make it a multiple of words. This may require tail padding, depending on the size of the argument. To ensure that data in the stack is properly aligned, the stack pointer should always point to a word boundary. Structure and union arguments are pushed onto the stack in the same manner as integral arguments, described above. This provides call-by-value semantics, letting the called function modify its arguments without affecting the calling function's object.

Figure 3-23: Structure and Union Arguments

Call	Argument	Callee
i(1, s);	1	8(%ebp)
	word 0, s	12(%ebp)
	word 1, s	16(%ebp)
	...	...

Operating System Interface

Virtual Address Space

Processes execute in a 32-bit virtual address space. Memory management translates virtual addresses to physical addresses, hiding physical addressing and letting a process run anywhere in the system's real memory. Processes typically begin with three logical segments, commonly called text, data, and stack. As Chapter 5 describes, dynamic linking creates more segments during execution, and a process can create additional segments for itself with system services.

Page Size

Memory is organized by pages, which are the system's smallest units of memory allocation. Page size can vary from one system to another, depending on the processor, memory management unit and system configuration. Processes may call `sysconf(BA_OS)` to determine the system's current page size.

*

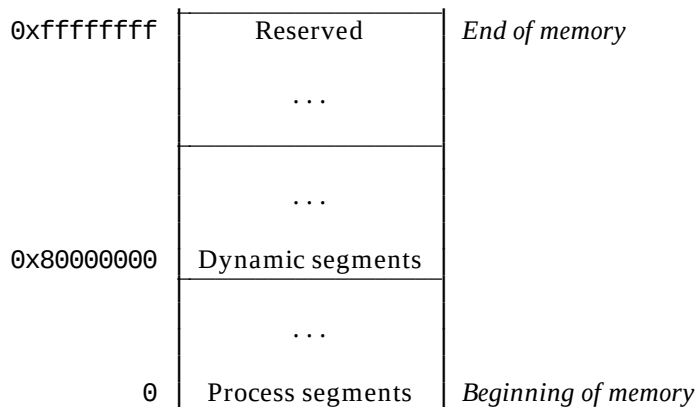
Virtual Address Assignments

Conceptually, processes have the full 32-bit address space available. In practice, however, several factors limit the size of a process.

- The system reserves a configuration-dependent amount of virtual space.
- The system reserves a configuration dependent amount of space per process.
- A process whose size exceeds the system's available, combined physical memory and secondary storage cannot run. Although some physical memory must be present to run any process, the system can execute processes that are bigger than physical memory, paging them to and from secondary storage. Nonetheless, both physical memory and secondary storage are shared resources. System load, which can vary from one program execution to the next, affects the available amounts.

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Figure 3-24: Virtual Address Configuration



CAUTION Programs that dereference null pointers are erroneous. although an implementation is not obliged to detect such erroneous behavior. Such programs may or may not fail on a particular system. To enhance portability, programmers are strongly cautioned not to rely on this behavior.

Process segments

Processes' loadable segments and stack may begin at 0. The exact addresses depend on the executable file format [see further information below and in Chapters 4 and 5]. Processes can control the amount of virtual memory allotted for stack space, as described below.

Dynamic segments

A process's dynamic segments reside below the reserved area.

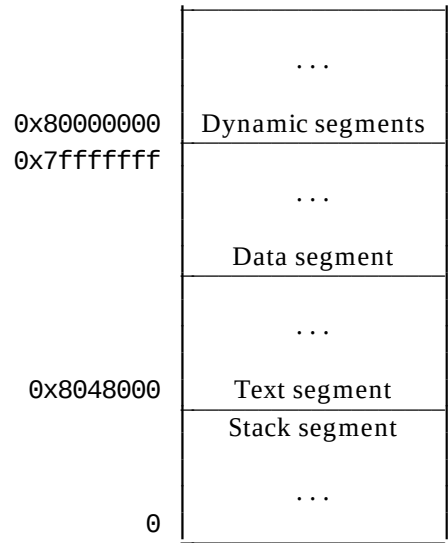
Reserved

A reserved area resides at the top of virtual space.

As the figure shows, the system reserves the high end of virtual address space, with a process's dynamic segments below that. Although the exact boundary between the reserved area and a process depends on the system's configuration, the reserved area shall not consume more than 1 GB of the address space. Thus the user virtual address range has a minimum upper bound of 0xc0000000. Individual systems may reserve less space, increasing processes' virtual memory range.

Although applications may control their memory assignments, the typical arrangement appears below.

Figure 3-25: Conventional Segment Arrangements



The process’s text segment resides at 0x8048000. The data segment follows immediately, and dynamic segments occupy the higher range. When applications let the system choose addresses for dynamic segments (including shared object segments), it chooses high addresses. This leaves the “middle” of the address spectrum available for dynamic memory allocation with facilities such as `malloc(BA_OS)`. Processes should *not* depend on finding their dynamic segments at particular virtual addresses. Facilities exist to let the system choose dynamic segment virtual addresses. The stack resides immediately below the text segment, growing toward lower addresses. This arrangement provides a little over 128 MB for the stack and about 2 GB for text and data.

Managing the Process Stack

Section “Process Initialization” in this chapter describes the initial stack contents. Stack addresses can change from one system to the next—even from one process execution to the next on the same system. Processes, therefore, should *not* depend on finding their stack at a particular virtual address.

A tunable configuration parameter controls the system maximum stack size. A process also can use `setrlimit(BA_OS)`, to set its own maximum stack size, up to the system limit. On the Intel386, the stack segment has read and write permissions.

Coding Guidelines

Operating system facilities, such as `mmap(KE_OS)`, allow a process to establish address mappings in two ways. First, the program can let the system choose an address. Second, the program can force the system to use an address the program supplies. This second alternative can cause application portability problems, because the requested address might not always be available. Differences in virtual address space can be particularly troublesome between different architectures, but the same problems can arise within a single architecture.

Processes' address spaces typically have three segment areas that can change size from one execution to the next: the stack [through `setrlimit(BA_OS)`], the data segment [through `malloc(BA_OS)`], and the dynamic segment area [through `mmap(KE_OS)`]. Changes in one area may affect the virtual addresses available for another. Consequently, an address that is available in one process execution might not be available in the next. A program that used `mmap(KE_OS)` to request a mapping at a specific address thus could appear to work in some environments and fail in others. For this reason, programs that wish to establish a mapping in their address space should let the system choose the address.

Despite these warnings about requesting specific addresses, the facility is both useful and can be used in a controlled manner. For example, a multiprocess application might map several files into the address space of each process and build relative pointers among the files' data. This could be done by having each process ask for a certain amount of memory at an address chosen by the system. After each process receives its own, private address from the system, it would map the desired files into memory, at specific addresses within the original area. This collection of mappings could be at different addresses in each process but their *relative* positions would be fixed. Without the ability to ask for specific addresses, the application could not build shared data structures, because the relative positions for files in each process would be unpredictable.

Processor Execution Modes

Four execution modes exist in the Intel386 architecture: ring 3 (or user mode) and three privileged rings. User processes run in user mode ring (the least privileged). The operating system kernel runs in a privileged mode ring, although the ABI does not specify which one. A program executes the `lcall` instruction through a system call gate to change execution modes, and thus the `lcall` instruction provides the low-level interface to system calls. For the Intel386, one low-level interface is defined: `_exit(BA_OS)`.

To ensure a process has a way to terminate itself, the system treats `_exit` as a special case. The ABI does not specify the implementation of other system services. Instead, programs should use the system libraries that Chapter 6 describes. Programs with other embedded `lcall` instructions do not conform to the ABI.

Figure 3-26: `_exit` System Trap

```
.globl _exit
_exit:
    movl    $1, %eax
    lcall   $7, $0
```

Exception Interface

As the Intel386 architecture manuals describe, the processor changes mode to handle *exceptions*, which may be synchronous, floating-point/coprocessor, or asynchronous. Synchronous and floating-point/coprocessor exceptions, being caused by instruction execution, can be explicitly generated by a process. This section, therefore, specifies those exception types with defined behavior. The Intel386 architecture classifies exceptions as *faults*, *traps*, and *aborts*. See the Intel 80386 *Programmer's Reference Manual* for more information about their differences.

Hardware Exception Types

The operating system defines the following correspondence between hardware exceptions and the signals specified by `signal(BA_OS)`.

Figure 3-27: Hardware Exceptions and Signals

Number	Exception Name	Signal
0	divide error fault	SIGFPE
1	single step trap / fault	SIGTRAP
2	nonmaskable interrupt	none
3	breakpoint trap	SIGTRAP
4	overflow trap	SIGSEGV
5	bounds check fault	SIGSEGV
6	invalid opcode fault	SIGILL
7	no coprocessor fault	SIGFPE
8	double fault abort	none
9	coprocessor overrun abort	SIGSEGV
10	invalid TSS fault	none
11	segment not present fault	none
12	stack exception fault	SIGSEGV
13	general protection fault / abort	SIGSEGV
14	page fault	SIGSEGV
15	(reserved)	
16	coprocessor error fault	SIGFPE
other	(unspecified)	SIGILL

Floating-point instructions exist in the architecture, but they may be implemented either in hardware (via the Intel387 chip) or in software (via the Intel387 emulator). In the case of “no coprocessor” exception, if the Intel387 emulator is configured into the kernel, the process receives no signal. Instead, the system intercepts the exception, emulates the instruction, and returns control to the process. A process receives SIGFPE for the “no coprocessor” exception only when the indicated floating-point instruction is illegal (invalid operands, and so on).

Software Trap Types

Because the `int` instruction generates traps, some hardware exceptions can be generated by software. However, the `int` instruction generates only traps and not faults; so it is not possible to match the exact hardware generated faults in software.

Process Initialization

This section describes the machine state that `exec(BA_OS)` creates for “infant” processes, including argument passing, register usage, stack frame layout, and so on. Programming language systems use this initial program state to establish a standard environment for their application programs. As an example, a C program begins executing at a function named `main`, conventionally declared in the following way.

Figure 3-28: Declaration for `main`

```
extern int main(int argc, char *argv[], char *envp[]);
```

Briefly, `argc` is a non-negative argument count; `argv` is an array of argument strings, with `argv[argc]==0`; and `envp` is an array of environment strings, also terminated by a null pointer.

Although this section does not describe C program initialization, it gives the information necessary to implement the call to `main` or to the entry point for a program in any other language.

Special Registers

As the Intel386 architecture defines, several state registers control and monitor the processor: the Machine Status Word register (MSW, also known as register `%cr0`), EFLAGS register, the floating-point status register, and the floating-point control register. Application programs cannot access the full EFLAGS register directly; because they run in the processor’s *user mode*, and the instructions to write some of the bits of the EFLAGS register are privileged. Nonetheless, a program has access to many of the flags in the EFLAGS register. Flags identified with an “*” below are not modifiable by a user mode process, they either have unspecified values or do not affect user program behavior. At process initialization, the EFLAGS register contains the following values.

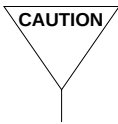
Figure 3-29: EFLAGS Register Fields

Flag	Value	Note
CF	unspecified	Carry flag
PF	unspecified	Parity flag
AF	unspecified	Auxiliary carry flag
ZF	unspecified	Zero flag
SF	unspecified	Sign flag
TF	unspecified	Trap flag
IF*	unspecified	Interrupt enable
DF	0	Direction flag low to high
OF	unspecified	Overflow flag
IOPL*	unspecified	I/O privilege level
NT*	unspecified	Nested task
RF*	unspecified	Resume flag
VM*	unspecified	Virtual 8086 mode

The Intel386 architecture defines floating-point instructions, and those instructions work whether the processor has a hardware floating-point unit or not. (A system may provide hardware or software floating-point facilities.) Consequently, the contents of the MSW register is not specified, letting the system set it according to the hardware configuration. In any case, however, the processor presents a working floating-point implementation, including the Intel387 status and control word registers with the following values at process initialization.

Figure 3-30: Floating-Point Control Word

Field	Value	Note	
IC	1	Affine infinity (for compatibility)	
RC	00	Round to nearest or even	
PC	11	53-bit (double precision)	G
PM	1	Precision masked	
UM	1	Underflow masked	
OM	1	Overflow	G
ZM	1	Zero divide	G
DM	1	Denormalized operand masked	
IM	1	Invalid operation	G



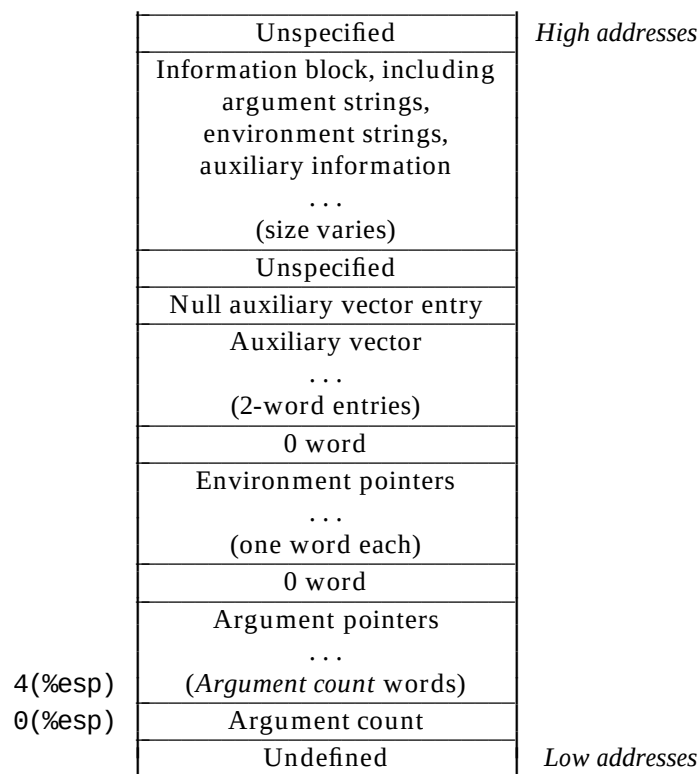
The initial floating-point state should be changed with care. In particular, many floating-point routines may produce undefined behavior if the precision control is set to less than 53 bits. The `_fpstart` routine (see Chapter 6) changes the precision control to 64 bits and sets all exceptions to be asked. This is the default state required for conformance to the ANSI C standard and to the IEEE 754 Floating-point standard.

G
G
G
G

Process Stack and Registers

When a process receives control, its stack holds the arguments and environment from `exec(BA_OS)`.

Figure 3-31: Initial Process Stack



Argument strings, environment strings, and the auxiliary information appear in no specific order within the information block; the system makes no guarantees about their arrangement. The system also may leave an unspecified amount of memory between the null auxiliary vector entry and the beginning of the information block.

General and floating-point register values are unspecified at process entry, with the exceptions appearing below. Consequently, a program that requires registers to have specific values must set them explicitly during process initialization. It should *not* rely on the operating system to set all registers to 0.

<code>%ebp</code>	The content of this register is unspecified at process initialization time, but the user code should mark the deepest stack frame by setting the frame pointer to zero. No other frame's <code>%ebp</code> should have a zero value.
<code>%esp</code>	Performing its usual job, the stack pointer holds the address of the bottom of the stack, which is guaranteed to be word aligned.
<code>%edx</code>	In a conforming program, this register contains a function pointer that the application should register with <code>atexit(BA_OS)</code> . This function is used for shared object termination code [see “Dynamic Linking” in Chapter 5 of the <i>System V ABI</i>].
<code>%cs, %ds, %es, %ss</code>	The segment registers are initialized so that the user process can address the code, data, and stack segments using a 32-bit virtual address. A program that alters their values does not conform to the ABI and has undefined behavior.

Every process has a stack, but the system defines *no* fixed stack address. Furthermore, a program's stack address can change from one system to another—even from one process invocation to another. Thus the process initialization code must use the stack address in `%esp`. Data in the stack segment at addresses below the stack pointer contain undefined values.

Whereas the argument and environment vectors transmit information from one application program to another, the auxiliary vector conveys information from the operating system to the program. This vector is an array of the following structures, interpreted according to the `a_type` member.

Figure 3-32: Auxiliary Vector

```
typedef struct
{
    int    a_type;
    union {
        long    a_val;
        void    *a_ptr;
        void    (*a_fcn)();
    } a_un;
} auxv_t;
```

Figure 3-33: Auxiliary Vector Types, a_type

Name	Value	a_un	
AT_NULL	0	ignored	
AT_IGNORE	1	ignored	
AT_EXECD	2	a_val	
AT_PHDR	3	a_ptr	
AT_PHENT	4	a_val	
AT_PHNUM	5	a_val	
AT_PAGESZ	6	a_val	
AT_BASE	7	a_ptr	
AT_FLAGS	8	a_val	
AT_ENTRY	9	a_ptr	
AT_LIBPATH	10	a_val	M
AT_FPHW	11	a_val	M
AT_INTP_DEVICE	12	a_val	M
AT_INTP_INODE	13	a_val	M

AT_NULL The auxiliary vector has no fixed length; instead its last entry's a_type member has this value.

AT_IGNORE	This type indicates the entry has no meaning. The corresponding value of <code>a_un</code> is undefined.
AT_EXECD	As Chapter 5 describes, <code>exec(BA_OS)</code> may pass control to an interpreter program. When this happens, the system places either an entry of type <code>AT_EXECD</code> or one of type <code>AT_PHDR</code> in the auxiliary vector. The entry for type <code>AT_EXECD</code> uses the <code>a_val</code> member to contain a file descriptor open to read the application program's object file.
AT_PHDR	Under some conditions, the system creates the memory image of the application program before passing control to the interpreter program. When this happens, the <code>a_ptr</code> member of the <code>AT_PHDR</code> entry tells the interpreter where to find the program header table in the memory image. If the <code>AT_PHDR</code> entry is present, entries of types <code>AT_PHENT</code> , <code>AT_PHNUM</code> , and <code>AT_ENTRY</code> must also be present. See Chapter 5 in both the System V ABI and the processor supplement for more information about the program header table.
AT_PHENT	The <code>a_val</code> member of this entry holds the size, in bytes, of one entry in the program header table to which the <code>AT_PHDR</code> entry points.
AT_PHNUM	The <code>a_val</code> member of this entry holds the number of entries in the program header table to which the <code>AT_PHDR</code> entry points.
AT_PAGESZ	If present, this entry's <code>a_val</code> member gives the system page size, in bytes. The same information also is available through <code>sysconf(BA_OS)</code> .
AT_BASE	The <code>a_ptr</code> member of this entry holds the base address at which the interpreter program was loaded into memory. See "Program Header" in the System V ABI for more information about the base address.
AT_FLAGS	If present, the <code>a_val</code> member of this entry holds one-bit flags. Bits with undefined semantics are set to zero.
AT_ENTRY	The <code>a_ptr</code> member of this entry holds the entry point of the application program to which the interpreter program should transfer control.
AT_LIBPATH	The <code>a_val</code> member of this entry is non-zero if the dynamic linker should examine <code>LD_LIBRARY_PATH</code> when searching for shared objects of the process based on the security considerations in the Shared Object Dependency section in Chapter 5 of the gABI.

AT_FPHW The `a_val` member of this entry will be set to

Figure 3-34: AT_FPHW values

Value	Meaning
0	if no floating point support exists
1	if floating point software emulation exists
2	if it has a 80287 chip
3	if it has a 80387 chip or a 80487 chip

AT_INTP_DEVICE

The `a_val` member of this entry holds the device number of the file from which the dynamic linker is loaded.

AT_INTP_INODE

The `a_val` member of this entry holds the inode of the file from which the dynamic linker is loaded.

Other auxiliary vector types are reserved. No flags are currently defined for AT_FLAGS, on the Intel386 architecture.

To illustrate, suppose an example process receives two arguments.

- `echo`
- `abi`

It also inherits two environment strings (this example is not intended to show a fully configured execution environment).

- `HOME=/home/dir`
- `PATH=/usr/bin:`

Its one non-null auxiliary vector entry holds a file descriptor.

- `{AT_EXECFD, 13}`

The resulting stack resides below `0x8048000`, growing toward lower addresses.

Figure 3-35: Example Process Stack

	n	:	\0	<i>pad</i>	<i>High addresses</i>
	r	/	b	i	
	=	/	u	s	
0x8047ff0	P	A	T	H	
	d	i	r	\0	
	o	m	e	/	
	E	=	/	h	
0x8047fe0	\0	H	O	M	
	\0	a	b	i	
	e	c	h	o	
	0				
0x8047fd0	0				
	13				
	2				Auxiliary vector
	0				
0x8047fc0	0x8047ff0				
	0x8047fe1				Environment vector
	0				
	0x8047fdd				
0x8047fb0	0x8047fd8				Argument vector
0(%esp), 0x8047fac	2				Argument count
	<i>Undefined</i>				<i>Low addresses</i>

Coding Examples

This section discusses example code sequences for fundamental operations such as calling functions, accessing static objects, and transferring control from one part of a program to another. Previous sections discuss how a program may use the machine or the operating system, and they specify what a program may and may not assume about the execution environment. Unlike previous material, the information here illustrates how operations *may* be done, not how they *must* be done.

As before, examples use the ANSI C language. Other programming languages may use the same conventions displayed below, but failure to do so does *not* prevent a program from conforming to the ABI. Two main object code models are available.

- *Absolute code*. Instructions can hold absolute addresses under this model. To execute properly, the program must be loaded at a specific virtual address, making the program's absolute addresses coincide with the process's virtual addresses.
- *Position-independent code*. Instructions under this model hold relative addresses, *not* absolute addresses. Consequently, the code is not tied to a specific load address, allowing it to execute properly at various positions in virtual memory.

Following sections describe the differences between these models. Code sequences for the models (when different) appear together, allowing easier comparison.

NOTE

Examples below show code fragments with various simplifications. They are intended to explain addressing modes, not to show optimal code sequences nor to reproduce compiler output.

NOTE

When other sections of this document show assembly language code sequences, they typically show only the absolute versions. Information in this section explains how position-independent code would alter the examples.

Code Model Overview

When the system creates a process image, the executable file portion of the process has fixed addresses, and the system chooses shared object library virtual addresses to avoid conflicts with other segments in the process. To maximize text sharing, shared objects conventionally use position-independent code, in which instructions contain no absolute addresses. Shared object text segments can be loaded at various virtual addresses without having to change the segment images. Thus multiple processes can share a single shared object text segment, even though the segment resides at a different virtual address in each process.

Position-independent code relies on two techniques.

- Control transfer instructions hold offsets relative to the extended instruction pointer (EIP). An EIP-relative branch or function call computes its destination address in terms of the current instruction pointer, *not* relative to any absolute address.
- When the program requires an absolute address, it computes the desired value. Instead of embedding absolute addresses in the instructions, the compiler generates code to calculate an absolute address during execution.

Because the Intel386 architecture provides EIP-relative call and branch instructions, compilers can satisfy the first condition easily.

A *global offset table* provides information for address calculation. Position-independent object files (executable and shared object files) have this table in their data segment. When the system creates the memory image for an object file, the table entries are relocated to reflect the absolute virtual addresses as assigned for an individual process. Because data segments are private for each process, the table entries can change—unlike text segments, which multiple processes share.

Assembly language examples below show the explicit notation needed for position-independent code.

`name@GOT(%ebx)`

This expression denotes an `%ebx`-relative reference to the global offset table entry for the symbol `name`. The `%ebx` register contains the absolute address of the global offset table, as explained below.

`name@GOTOFF(%ebx)`

This expression denotes an `%ebx`-relative reference to the symbol `name`. Again, `%ebx` holds the global offset table address. Note this expression references `name`, not the global offset table entry for `name`.

`name@PLT` This expression denotes an EIP-relative reference to the procedure linkage table entry for the symbol *name*.

`_GLOBAL_OFFSET_TABLE_`

The symbol `_GLOBAL_OFFSET_TABLE_` is used to access the global offset table. When an instruction uses the symbol, it sees the offset between the current instruction and the global offset table as the symbol value.

Position-Independent Function Prologue

This section describes the function prologue for position-independent code. A function's prologue allocates the local stack space, saves any registers it must preserve, and sets register `%ebx` to the global offset table's address. Because `%ebx` is private for each function and preserved across function calls, a function calculates its value once at the entry.

Figure 3-36: Calculating Global Offset Table Address

Line	Code
1	<code>call .L1</code>
2	<code>.L1: popl %ebx</code>
3	<code>addl \$_GLOBAL_OFFSET_TABLE_+[.-.L1], %ebx</code>

These three lines accomplish the following.

1. The `call` instruction pushes the *absolute* address of the next instruction onto the stack.
2. Consequently, the `popl` instruction pops the absolute address of `.L1` into register `%ebx`.
3. The last instruction computes the desired absolute value into `%ebx`. This works because `_GLOBAL_OFFSET_TABLE_` in the expression gives the distance from the `addl` instruction to the global offset table; `[.-.L1]` gives the distance from `.L1` to the `addl` instruction. Adding their sum to the absolute address of `.L1`, already in `%ebx`, gives the absolute address of the global offset table.

This computation can be added to the standard function prologue, giving the standard prologue for position-independent code. To illustrate, the following function prologue allocates 80 bytes of local stack space and saves the local registers `%ebx`, `%esi`, and `%edi`.

Figure 3-37: Position-Independent Function Prologue

```
prologue:
    pushl %ebp
    movl  %esp, %ebp
    subl  $80, %esp
    pushl %edi
    pushl %esi
    pushl %ebx
    call  .L1
.L1:    popl  %ebx
    addl  $_GLOBAL_OFFSET_TABLE_+[.-.L1], %ebx
```

Position-independent and absolute code use the same function epilogue.

Data Objects

This discussion excludes stack-resident objects, because programs always compute their virtual addresses relative to the stack and frame pointers. Instead, this section describes objects with static storage duration.

In the Intel386 architecture, all memory reference instructions can address any location within the 32-bit address space. Symbolic references in absolute code put the symbols' values—or absolute virtual addresses—into instructions.

Figure 3-38: Absolute Data Access

C	Assembly
<pre>extern int src; extern int dst; extern int *ptr; ptr = &dst; *ptr = src;</pre>	<pre>.globl src, dst, ptr movl \$dst, ptr movl ptr, %eax movl src, %edx movl %edx, (%eax)</pre>

Position-independent instructions cannot contain absolute addresses. Instead, instructions that reference symbols hold the symbols' offsets into the global offset table. Combining the offset with the global offset table address in `%ebx` gives the absolute address of the table entry holding the desired address.

Figure 3-39: Position-Independent Data Access

C	Assembly
<pre>extern int src; extern int dst; extern int *ptr; ptr = &dst; *ptr = src;</pre>	<pre>.globl src, dst, ptr movl ptr@GOT(%ebx), %eax movl dst@GOT(%ebx), %edx movl %edx, (%eax) movl ptr@GOT(%ebx), %eax movl (%eax), %eax movl src@GOT(%ebx), %edx movl (%edx), %edx movl %edx, (%eax)</pre>

Finally, position-independent references to static data may be optimized. Because `%ebx` holds a known address, the global offset table, a program may use it as a base register. External references should use the global offset table entry, because dynamic linking may bind the entry to a definition outside the current object file's scope.

Figure 3-40: Position-Independent Static Data Access

C	Assembly
<pre>static int src; static int dst; static int *ptr; ptr = &dst; *ptr = src;</pre>	<pre>leal ptr@GOTOFF(%ebx), %eax leal dst@GOTOFF(%ebx), %edx movl %edx, (%eax) movl ptr@GOTOFF(%ebx), %eax movl src@GOTOFF(%ebx), %edx movl %edx, (%eax)</pre>

Function Calls

Programs use the `call` instruction to make direct function calls. A `call` instruction's destination is an EIP-relative value that can reach any address in the 32-bit virtual space. Even when the code for a function resides in a shared object, the caller uses the same assembly language instruction sequence, although in that case control passes from the original call, through an indirection sequence, to the desired destination. See "Procedure Linkage Table" in Chapter 5 for more information on the indirection sequence.

Figure 3-41: Absolute Direct Function Call

C	Assembly
<pre>extern void function(); function();</pre>	<pre>.globl function call function</pre>

Dynamic linking may redirect a function call outside the current object file's scope; so position-independent calls should use the procedure linkage table explicitly.

Figure 3-42: Position-Independent Direct Function Call

C	Assembly
<pre>extern void function(); function();</pre>	<pre>.globl function call function@PLT</pre>

Indirect function calls use the indirect `call` instruction.

Figure 3-43: Absolute Indirect Function Call

C	Assembly
<pre>extern void (*ptr)(); extern void name(); ptr = name; (*ptr)();</pre>	<pre>.globl ptr, name; movl \$name, ptr call *ptr</pre>

For position-independent code, the global offset table supplies absolute addresses for all required symbols, whether the symbols name objects or functions.

Figure 3-44: Position-Independent Indirect Function Call

C	Assembly
<pre>extern void (*ptr)(); extern void name(); ptr = name; (*ptr)();</pre>	<pre>.globl ptr, name movl ptr@GOT(%ebx), %eax movl name@GOT(%ebx), %edx movl %edx, (%eax) movl ptr@GOT(%ebx), %eax call *(%eax)</pre>

Branching

Programs use branch instructions to control their execution flow. As defined by the Intel386 architecture, branch instructions hold an EIP-relative value with a signed 32-bit range, allowing a jump to any location within the virtual address space.

Figure 3-45: Branch Instruction, All Models

C	Assembly
<pre>label: . . . goto label;</pre>	<pre>.L01: . . . jmp .L01</pre>

C switch statements provide multiway selection. When the case labels of a switch statement satisfy grouping constraints, the compiler implements the selection with an address table. The following examples use several simplifying conventions to hide irrelevant details:

- The selection expression resides in register `%eax`;
- case label constants begin at zero;
- case labels, `default`, and the address table use assembly names `.Lcasei`, `.Ldef`, and `.Ltab`, respectively.

Address table entries for absolute code contain virtual addresses; the selection code extracts an entry's value and jumps to that address. Position-independent table entries hold offsets; the selection code computes a destination's absolute address.

Figure 3-46: Absolute switch Code

C	Assembly
switch (j)	cmpl \$3, %eax
{	ja .Ldef
case 0:	jmp *.Ltab(,%eax,4)
. . .	.Ltab: .long .Lcase0
case 2:	.long .Ldef
. . .	.long .Lcase2
case 3:	.long .Lcase3
. . .	
default:	
. . .	
}	

Figure 3-47: Position-Independent switch Code

C	Assembly
switch (j)	cmpl \$3, %eax
{	ja .Ldef
case 0:	leal .Ltab@GOTOFF(%ebx), %edx
. . .	movl (%edx,%eax,4), %eax
case 2:	movl .Ltab@GOTOFF(%ebx,%eax,4), %eax
. . .	call .Ljmp
case 3:	.Ljmp:
. . .	popl %ecx
default:	addl %ecx, %eax
. . .	jmp %eax
}	.Ltab:
	.long .Lcase0 - .Ljmp
	.long .Ldef - .Ljmp
	.long .Lcase2 - .Ljmp
	.long .Lcase3 - .Ljmp

C Stack Frame

Figure 3-48 shows the C stack frame organization. It conforms to the standard stack frame with designated roles for unspecified areas in the standard frame. This represents one possible organization of the C stack frame. Usage of `%ebp` as a frame pointer, the exact positions of the callee saved registers, and space for local storage is implementation specific.

Figure 3-48: C Stack Frame

Base	Offset	Contents	
<code>%ebp</code>	$4n+8$	argument word n	<i>High addresses</i>
		...	
	8	argument word 0	
	4	return address	
<code>%ebp</code>	0	caller's <code>%ebp</code>	
<code>%ebp</code>	-4	x words local space: automatic variables, temporaries, etc.	
<code>%ebp</code>	$-4x$		
<code>%esp</code>	12		
<code>%esp</code>	8	caller's <code>%edi</code>	
	4	caller's <code>%esi</code>	
<code>%esp</code>	0	caller's <code>%ebx</code>	<i>Low addresses</i>

A C stack frame doesn't normally change size during execution. The exception is dynamically allocated stack memory, discussed below. By convention, a function allocates automatic (local) variables in the middle of its frame and references them as negative offsets from `%ebp`. Its incoming arguments reside in the previous frame, referenced as positive offsets from `%ebp`. If necessary, a function saves the values of `%edi`, `%esi`, and `%ebx` in the positions shown and restores their values before returning to the caller. The positions may be different from the diagram above, depending on which of these three registers the function saves and restores.

Variable Argument List

Previous sections describe the rules for passing arguments. Unfortunately, some otherwise portable C programs depend on the argument passing scheme, implicitly assuming that 1) all arguments reside on the stack, and 2) arguments appear in increasing order on the stack. Programs that make these assumptions never have been portable, but they have worked on many machines, including the Intel386. Nonetheless, portable C programs should use the facilities defined in the header files `<stdarg.h>` or `<varargs.h>` to deal with variable argument lists.

Allocating Stack Space Dynamically

Unlike some other languages, C does not need dynamic stack allocation *within* a stack frame. Frames are allocated dynamically on the program stack, depending on program execution, but individual stack frames can have static sizes. Nonetheless, the architecture supports dynamic allocation for those languages that require it, and the standard calling sequence and stack frame support it as well. Thus languages that need dynamic stack frame sizes can call C functions, and vice versa.

Figure 3-48 shows the layout of the C stack frame. The double line divides the area referenced from `%ebp` from the area referenced from `%esp`. Dynamic space is allocated below the line, as a downward growing heap whose size changes as required. Typical C functions have no space in the heap. All areas above the heap in the current frame have a known size to the compiler. Dynamic stack allocation thus takes the following steps.

1. Stack frames are word aligned; dynamic allocation should preserve this property. Thus the program rounds (up) the desired byte count to a multiple of 4.
2. The program decreases the stack pointer by the rounded byte count, increasing its frame size. At this point, the “new” space resides just below the register save area at the bottom of the stack.
3. The program copies the register save area (three or fewer words) to the bottom of the stack, effectively moving the new space up into the frame.

NOTE

The register save area is reserved and should not be used for purposes outside of this document.

G

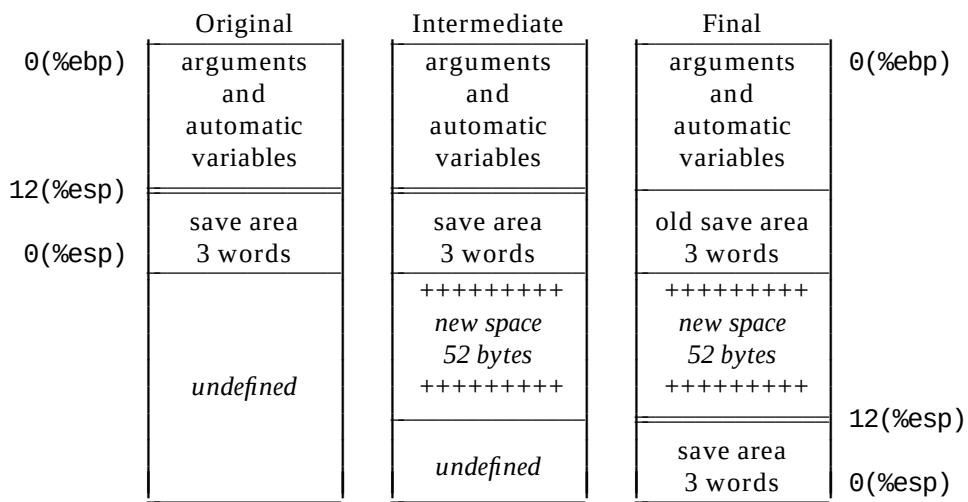
Even in the presence of signals, dynamic allocation is “safe.” If a signal interrupts allocation, one of three things can happen.

- The signal handler can return. The process then resumes the dynamic allocation from the point of interruption.
- The signal handler can execute a non-local goto, or `longjmp` [see `setjmp(BA_LIB)`]. This resets the process to a new context in a previous stack frame, automatically discarding the dynamic allocation.
- The process can terminate.

Regardless of when the signal arrives during dynamic allocation, the result is a consistent (though possibly dead) process.

To illustrate, assume a program wants to allocate 50 bytes, and it has saved three registers in the bottom of the frame. The first step is rounding 50 to 52, making it a multiple of 4. Figure 3-49 shows how the stack frame changes.

Figure 3-49: Dynamic Stack Allocation



New space starts at 12(%esp). As described, every dynamic allocation in *this* function will return a new area starting at 12(%esp), leaving previous heap objects untouched (other functions could have different heap addresses).

Consequently, the compiler should compute the absolute address for each area, avoiding relative references. Otherwise, future allocations in the same frame would destroy the heap's integrity.

Existing stack objects reside at fixed offsets from the frame pointer (`%ebp`). Dynamic allocation preserves those offsets, because the frame pointer does not change and the objects relative to it do not move. Objects relative to the stack pointer (`%esp`) move, but their `%esp`-relative positions do not change. Accordingly, compilers arrange not to publicize the absolute address of any object in the bottom half of the stack frame (in a way that violates the scope rules). `%esp`-relative references stay valid after dynamic allocation, but absolute addresses do not.

No special code is needed to free dynamically allocated stack memory. The function return resets the stack pointer and removes the entire stack frame, including the heap, from the stack. Naturally, a program should not reference heap objects after they have gone out of scope.

4 OBJECT FILES

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ELF Header

Machine Information

For file identification in `e_ident`, the Intel386 architecture requires the following values.

Figure 4-1: Intel386 Identification, `e_ident`

Position	Value
<code>e_ident[EI_CLASS]</code>	ELFCLASS32
<code>e_ident[EI_DATA]</code>	ELFDATA2LSB

Processor identification resides in the ELF header's `e_machine` member and must have the value `EM_386`.

The ELF header's `e_flags` member holds bit flags associated with the file. The Intel386 architecture defines no flags; so this member contains zero.

Sections

Special Sections

Various sections hold program and control information. Sections in the list below are used by the system and have the indicated types and attributes.

Figure 4-2: Special Sections

Name	Type	Attributes
.got	SHT_PROGBITS	SHF_ALLOC + SHF_WRITE
.plt	SHT_PROGBITS	SHF_ALLOC + SHF_EXECINSTR

.got	This section holds the global offset table. See “Coding Examples” in Chapter 3 and “Global Offset Table” in Chapter 5 for more information.
.plt	This section holds the procedure linkage table. See “Procedure Linkage Table” in Chapter 5 for more information.

Symbol Table

Symbol Values

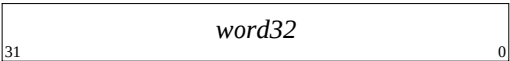
If an executable file contains a reference to a function defined in one of its associated shared objects, the symbol table section for that file will contain an entry for that symbol. The `st_shndx` member of that symbol table entry contains `SHN_UNDEF`. This signals to the dynamic linker that the symbol definition for that function is not contained in the executable file itself. If that symbol has been allocated a procedure linkage table entry in the executable file, and the `st_value` member for that symbol table entry is non-zero, the value will contain the virtual address of the first instruction of that procedure linkage table entry. Otherwise, the `st_value` member contains zero. This procedure linkage table entry address is used by the dynamic linker in resolving references to the address of the function. See “Function Addresses” in Chapter 5 for details.

Relocation

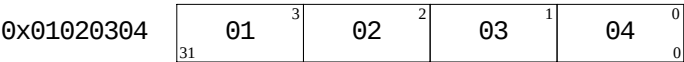
Relocation Types

Relocation entries describe how to alter the following instruction and data fields (bit numbers appear in the lower box corners).

Figure 4-3: Relocatable Fields



word32 This specifies a 32-bit field occupying 4 bytes with arbitrary byte alignment. These values use the same byte order as other word values in the Intel386 architecture.



Calculations below assume the actions are transforming a relocatable file into either an executable or a shared object file. Conceptually, the link editor merges one or more relocatable files to form the output. It first decides how to combine and locate the input files, then updates the symbol values, and finally performs the relocation. Relocations applied to executable or shared object files are similar and accomplish the same result. Descriptions below use the following notation.

- A This means the addend used to compute the value of the relocatable field.
- B This means the base address at which a shared object has been loaded into memory during execution. Generally, a shared object file is built with a 0 base virtual address, but the execution address will be different. See “Program Header” in the System V ABI for more information about the base address.
- G This means the offset into the global offset table at which the address of the relocation entry’s symbol will reside during execution. See “Coding Examples” in Chapter 3 and “Global Offset Table” in Chapter 5 for more information.

- GOT** This means the address of the global offset table. See “Coding Examples” in Chapter 3 and “Global Offset Table” in Chapter 5 for more information.
- L** This means the place (section offset or address) of the procedure linkage table entry for a symbol. A procedure linkage table entry redirects a function call to the proper destination. The link editor builds the initial procedure linkage table, and the dynamic linker modifies the entries during execution. See “Procedure Linkage Table” in Chapter 5 for more information.
- P** This means the place (section offset or address) of the storage unit being relocated (computed using `r_offset`).
- S** This means the value of the symbol whose index resides in the relocation entry.

A relocation entry’s `r_offset` value designates the offset or virtual address of the first byte of the affected storage unit. The relocation type specifies which bits to change and how to calculate their values. The Intel386 architecture uses only `Elf32_Rel` relocation entries, the field to be relocated holds the addend. In all cases, the addend and the computed result use the same byte order.

Figure 4-4: Relocation Types

Name	Value	Field	Calculation
R_386_NONE	0	none	none
R_386_32	1	word32	$S + A$
R_386_PC32	2	word32	$S + A - P$
R_386_GOT32	3	word32	$G + A - P$
R_386_PLT32	4	word32	$L + A - P$
R_386_COPY	5	none	none
R_386_GLOB_DAT	6	word32	S
R_386_JMP_SLOT	7	word32	S
R_386_RELATIVE	8	word32	$B + A$
R_386_GOTOFF	9	word32	$S + A - GOT$
R_386_GOTPC	10	word32	$GOT + A - P$

Some relocation types have semantics beyond simple calculation.

- R_386_GOT32** This relocation type computes the distance from the base of the global offset table to the symbol’s global offset table entry. It additionally instructs the link editor to build a global offset table.

R_386_PLT32	This relocation type computes the address of the symbol's procedure linkage table entry and additionally instructs the link editor to build a procedure linkage table.
R_386_COPY	The link editor creates this relocation type for dynamic linking. Its offset member refers to a location in a writable segment. The symbol table index specifies a symbol that should exist both in the current object file and in a shared object. During execution, the dynamic linker copies data associated with the shared object's symbol to the location specified by the offset.
R_386_GLOB_DAT	This relocation type is used to set a global offset table entry to the address of the specified symbol. The special relocation type allows one to determine the correspondence between symbols and global offset table entries.
R_386_JMP_SLOT	The link editor creates this relocation type for dynamic linking. Its offset member gives the location of a procedure linkage table entry. The dynamic linker modifies the procedure linkage table entry to transfer control to the designated symbol's address [see "Procedure Linkage Table" in Chapter 5].
R_386_RELATIVE	The link editor creates this relocation type for dynamic linking. Its offset member gives a location within a shared object that contains a value representing a relative address. The dynamic linker computes the corresponding virtual address by adding the virtual address at which the shared object was loaded to the relative address. Relocation entries for this type must specify 0 for the symbol table index.
R_386_GOTOFF	This relocation type computes the difference between a symbol's value and the address of the global offset table. It additionally instructs the link editor to build the global offset table.
R_386_GOTPC	This relocation type resembles R_386_PC32, except it uses the address of the global offset table in its calculation. The symbol referenced in this relocation normally is <code>_GLOBAL_OFFSET_TABLE_</code> , which additionally instructs the link editor to build the global offset table.

5 PROGRAM LOADING AND DYNAMIC LINKING

Program Loading	5-1
------------------------	-----

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Program Loading

As the system creates or augments a process image, it logically copies a file's segment to a virtual memory segment. When—and if—the system physically reads the file depends on the program's execution behavior, system load, and so on. A process does not require a physical page unless it references the logical page during execution, and processes commonly leave many pages unreferenced. Therefore delaying physical reads frequently obviates them, improving system performance. To obtain this efficiency in practice, executable and shared object files must have segment images whose file offsets and virtual addresses are congruent, modulo the page size.

Virtual addresses and file offsets for the Intel386 architecture segments are congruent modulo 4 KB (0x1000) or larger powers of 2. Because 4 KB is the maximum page size, the files will be suitable for paging regardless of physical page size.

Figure 5-1: Executable File

File Offset	File	Virtual Address
0	ELF header	
	Program header table	
	Other information	
0x100	Text segment	0x8048100
	...	
	0x2be00 bytes	0x8073eff
0x2bf00	Data segment	0x8074f00
	...	
	0x4e00 bytes	0x8079cff
0x30d00	Other information	
	...	

Figure 5-2: Program Header Segments

Member	Text	Data
p_type	PT_LOAD	PT_LOAD
p_offset	0x100	0x2bf00
p_vaddr	0x8048100	0x8074f00
p_paddr	unspecified	unspecified
p_filesz	0x2be00	0x4e00
p_memsz	0x2be00	0x5e24
p_flags	PF_R+PF_X	PF_R+PF_W+PF_X
p_align	0x1000	0x1000

Although the example's file offsets and virtual addresses are congruent modulo 4 KB for both text and data, up to four file pages hold impure text or data (depending on page size and file system block size).

- The first text page contains the ELF header, the program header table, and other information.
- The last text page holds a copy of the beginning of data.
- The first data page has a copy of the end of text.
- The last data page may contain file information not relevant to the running process.

Logically, the system enforces the memory permissions as if each segment were complete and separate; segments' addresses are adjusted to ensure each logical page in the address space has a single set of permissions. In the example above, the region of the file holding the end of text and the beginning of data will be mapped twice: at one virtual address for text and at a different virtual address for data.

The end of the data segment requires special handling for uninitialized data, which the system defines to begin with zero values. Thus if a file's last data page includes information not in the logical memory page, the extraneous data must be set to zero, not the unknown contents of the executable file. "Impurities" in the other three pages are not logically part of the process image; whether the system expunges them is unspecified. The memory image for this program follows, assuming 4 KB (0x1000) pages.

Figure 5-3: Process Image Segments

Virtual Address	Contents	Segment
0x8048000	<i>Header padding</i> 0x100 bytes	Text
0x8048100	Text segment ...	
	0x2be00 bytes	
0x8073f00	<i>Data padding</i> 0x100 bytes	
0x8074000	<i>Text padding</i> 0xf00 bytes	Data
0x8074f00	Data segment ...	
	0x4e00 bytes	
0x8079d00	Uninitialized data 0x1024 zero bytes	
0x807ad24	<i>Page padding</i> 0x2dc zero bytes	

One aspect of segment loading differs between executable files and shared objects. Executable file segments typically contain absolute code (see “Coding Examples” in Chapter 3). To let the process execute correctly, the segments must reside at the virtual addresses used to build the executable file. Thus the system uses the `p_vaddr` values unchanged as virtual addresses.

On the other hand, shared object segments typically contain position-independent code. This lets a segment’s virtual address change from one process to another, without invalidating execution behavior. Though the system chooses virtual addresses for individual processes, it maintains the segments’ *relative positions*. Because position-independent code uses relative addressing between segments, the difference between virtual addresses in memory must match the difference between virtual addresses in the file. The following table shows possible shared object virtual address assignments for several processes, illustrating constant relative positioning. The table also illustrates the base address computations.

Figure 5-4: Example Shared Object Segment Addresses

Source	Text	Data	Base Address
File	0x200	0x2a400	0x0
Process 1	0x80000200	0x8002a400	0x80000000
Process 2	0x80081200	0x800ab400	0x80081000
Process 3	0x900c0200	0x900ea400	0x900c0000
Process 4	0x900c6200	0x900f0400	0x900c6000

Dynamic Linking

Dynamic Section

Dynamic section entries give information to the dynamic linker. Some of this information is processor-specific, including the interpretation of some entries in the dynamic structure.

DT_PLTGOT On the Intel386 architecture, this entry's `d_ptr` member gives the address of the first entry in the global offset table. As mentioned below, the first three global offset table entries are reserved, and two are used to hold procedure linkage table information.

Global Offset Table

Position-independent code cannot, in general, contain absolute virtual addresses. Global offset tables hold absolute addresses in private data, thus making the addresses available without compromising the position-independence and sharability of a program's text. A program references its global offset table using position-independent addressing and extracts absolute values, thus redirecting position-independent references to absolute locations.

Initially, the global offset table holds information as required by its relocation entries [see "Relocation" in Chapter 4]. After the system creates memory segments for a loadable object file, the dynamic linker processes the relocation entries, some of which will be type `R_386_GLOB_DAT` referring to the global offset table. The dynamic linker determines the associated symbol values, calculates their absolute addresses, and sets the appropriate memory table entries to the proper values. Although the absolute addresses are unknown when the link editor builds an object file, the dynamic linker knows the addresses of all memory segments and can thus calculate the absolute addresses of the symbols contained therein.

If a program requires direct access to the absolute address of a symbol, that symbol will have a global offset table entry. Because the executable file and shared objects have separate global offset tables, a symbol's address may appear in several tables. The dynamic linker processes all the global offset table relocations before giving control to any code in the process image, thus ensuring the absolute addresses are available during execution.

The table's entry zero is reserved to hold the address of the dynamic structure, referenced with the symbol `_DYNAMIC`. This allows a program, such as the dynamic linker, to find its own dynamic structure without having yet processed its relocation entries. This is especially important for the dynamic linker, because it must initialize itself without relying on other programs to relocate its memory image. On the Intel386 architecture, entries one and two in the global offset table also are reserved. "Procedure Linkage Table" below describes them.

The system may choose different memory segment addresses for the same shared object in different programs; it may even choose different library addresses for different executions of the same program. Nonetheless, memory segments do not change addresses once the process image is established. As long as a process exists, its memory segments reside at fixed virtual addresses.

A global offset table's format and interpretation are processor-specific. For the Intel386 architecture, the symbol `_GLOBAL_OFFSET_TABLE_` may be used to access the table.

Figure 5-5: Global Offset Table

```
extern Elf32_Addr  _GLOBAL_OFFSET_TABLE_[];
```

The symbol `_GLOBAL_OFFSET_TABLE_` may reside in the middle of the `.got` section, allowing both negative and non-negative "subscripts" into the array of addresses.

Function Addresses

References to the address of a function from an executable file and the shared objects associated with it might not resolve to the same value. References from within shared objects will normally be resolved by the dynamic linker to the virtual address of the function itself. References from within the executable file to a function defined in a shared object will normally be resolved by the link editor to the address of the procedure linkage table entry for that function within the executable file.

To allow comparisons of function addresses to work as expected, if an executable file references a function defined in a shared object, the link editor will place the address of the procedure linkage table entry for that function in its associated symbol table entry. [See "Symbol Values" in Chapter 4]. The dynamic linker

treats such symbol table entries specially. If the dynamic linker is searching for a symbol, and encounters a symbol table entry for that symbol in the executable file, it normally follows the rules below.

1. If the `st_shndx` member of the symbol table entry is not `SHN_UNDEF`, the dynamic linker has found a definition for the symbol and uses its `st_value` member as the symbol's address.
2. If the `st_shndx` member is `SHN_UNDEF` and the symbol is of type `STT_FUNC` and the `st_value` member is not zero, the dynamic linker recognizes this entry as special and uses the `st_value` member as the symbol's address.
3. Otherwise, the dynamic linker considers the symbol to be undefined within the executable file and continues processing.

Some relocations are associated with procedure linkage table entries. These entries are used for direct function calls rather than for references to function addresses. These relocations are not treated in the special way described above because the dynamic linker must not redirect procedure linkage table entries to point to themselves.

Procedure Linkage Table

Much as the global offset table redirects position-independent address calculations to absolute locations, the procedure linkage table redirects position-independent function calls to absolute locations. The link editor cannot resolve execution transfers (such as function calls) from one executable or shared object to another. Consequently, the link editor arranges to have the program transfer control to entries in the procedure linkage table. On the Intel386 architecture, procedure linkage tables reside in shared text, but they use addresses in the private global offset table. The dynamic linker determines the destinations' absolute addresses and modifies the global offset table's memory image accordingly. The dynamic linker thus can redirect the entries without compromising the position-independence and sharability of the program's text. Executable files and shared object files have separate procedure linkage tables.

Figure 5-6: Absolute Procedure Linkage Table

```
.PLT0: pushl  got_plus_4
        jmp    *got_plus_8
        nop; nop
        nop; nop
.PLT1: jmp    *name1_in_GOT
        pushl  $offset
        jmp    .PLT0@PC
.PLT2: jmp    *name2_in_GOT
        pushl  $offset
        jmp    .PLT0@PC
...
```

Figure 5-7: Position-Independent Procedure Linkage Table

```
.PLT0: pushl  4(%ebx)
        jmp    *8(%ebx)
        nop; nop
        nop; nop
.PLT1: jmp    *name1@GOT(%ebx)
        pushl  $offset
        jmp    .PLT0@PC
.PLT2: jmp    *name2@GOT(%ebx)
        pushl  $offset
        jmp    .PLT0@PC
...
```


NOTE

As the figures show, the procedure linkage table instructions use different operand addressing modes for absolute code and for position-independent code. Nonetheless, their interfaces to the dynamic linker are the same.

Following the steps below, the dynamic linker and the program “cooperate” to resolve symbolic references through the procedure linkage table and the global offset table.

1. When first creating the memory image of the program, the dynamic linker sets the second and the third entries in the global offset table to special values. Steps below explain more about these values.
2. If the procedure linkage table is position-independent, the address of the global offset table must reside in `%ebx`. Each shared object file in the process image has its own procedure linkage table, and control transfers to a procedure linkage table entry only from within the same object file. Consequently, the calling function is responsible for setting the global offset table base register before calling the procedure linkage table entry.
3. For illustration, assume the program calls `name1`, which transfers control to the label `.PLT1`.
4. The first instruction jumps to the address in the global offset table entry for `name1`. Initially, the global offset table holds the address of the following `pushl` instruction, not the real address of `name1`.
5. Consequently, the program pushes a relocation offset (*offset*) on the stack. The relocation offset is a 32-bit, non-negative byte offset into the relocation table. The designated relocation entry will have type `R_386_JMP_SLOT`, and its offset will specify the global offset table entry used in the previous `jmp` instruction. The relocation entry also contains a symbol table index, thus telling the dynamic linker what symbol is being referenced, `name1` in this case.
6. After pushing the relocation offset, the program then jumps to `.PLT0`, the first entry in the procedure linkage table. The `pushl` instruction places the value of the second global offset table entry (*got_plus_4* or `4(%ebx)`) on the stack, thus giving the dynamic linker one word of identifying information. The program then jumps to the address in the third global offset table entry (*got_plus_8* or `8(%ebx)`), which transfers control to the dynamic linker.
7. When the dynamic linker receives control, it unwinds the stack, looks at the designated relocation entry, finds the symbol’s value, stores the “real” address for `name1` in its global offset table entry, and transfers control to the desired destination.

8. Subsequent executions of the procedure linkage table entry will transfer directly to `name1`, without calling the dynamic linker a second time. That is, the `jmp` instruction at `.PLT1` will transfer to `name1`, instead of “falling through” to the `pushl` instruction.

The `LD_BIND_NOW` environment variable can change dynamic linking behavior. If its value is non-null, the dynamic linker evaluates procedure linkage table entries before transferring control to the program. That is, the dynamic linker processes relocation entries of type `R_386_JMP_SLOT` during process initialization. Otherwise, the dynamic linker evaluates procedure linkage table entries lazily, delaying symbol resolution and relocation until the first execution of a table entry.

NOTE

Lazy binding generally improves overall application performance, because unused symbols do not incur the dynamic linking overhead. Nevertheless, two situations make lazy binding undesirable for some applications. First, the initial reference to a shared object function takes longer than subsequent calls, because the dynamic linker intercepts the call to resolve the symbol. Some applications cannot tolerate this unpredictability. Second, if an error occurs and the dynamic linker cannot resolve the symbol, the dynamic linker will terminate the program. Under lazy binding, this might occur at arbitrary times. Once again, some applications cannot tolerate this unpredictability. By turning off lazy binding, the dynamic linker forces the failure to occur during process initialization, before the application receives control.

Program Interpreter

G

There is one valid program interpreter for programs conforming to the Intel386 ABI:

G

`/usr/lib/libc.so.1`

G

6 LIBRARIES

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-----------------------------	-----

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Shared Library Names

M

The version number of the libraries named in the *System V Generic ABI* is specified below.

M

M

Figure 6-1: Shared Library Names

M

<u>Library Reference Name</u>	M
libc.so.1	M
libthread.so.1	M
libdl.so.1	M
libnsl.so.1	M
libX11.so.5.0	M
libXt.so.5.0	M
libXext.so.5.0	M
libXm.so.1.2	M
libMrm.so.1.2	M

Additional Entry Points

The following routines are included in the **libc** library to provide entry points for the required source-level interface listed in the *System V ABI*. A description and syntax summary for each function follows the table. M

Figure 6-2: libc Additional Required Entry Points

```
_fxstat  _lxstat  _xmknod  _xstat  nuname
_nuname
```

```
int _fxstat(int, int, struct stat *);
```

The semantics of this function are identical to those of the `fstat(BA_OS)` function described in the *System V Interface Definition, Edition 4*. Its only difference is that it requires an extra first argument whose value must be 2.

```
int _lxstat(int, char *, struct stat *);
```

The semantics of this function are identical to those of the `lstat(BA_OS)` function described in the *System V Interface Definition, Edition 4*. Its only difference is that it requires an extra first argument whose value must be 2.

```
int nuname(struct utsname *);
```

The semantics and syntax of this function are identical to those of the `uname(BA_OS)` function described in the *System V Interface Definition, Edition 4*. The symbol `_nuname` is also available with the same semantics.

```
int _xmknod(int, char *, mode_t, dev_t);
```

The semantics of this function are identical to those of the `mknod(BA_OS)` function described in the *System V Interface Definition, Edition 4*. Its only difference is that it requires an extra first argument whose value must be 2.

```
int _xstat(int, char *, struct stat *);
```

The semantics of this function are identical to those of the `stat(BA_OS)` function described in the *System V Interface*

Definition, Edition 4. Its only difference is that it requires an extra first argument whose value must be 2.

Support Routines

Besides operating system services, **libc** contains the following processor-specific support routines. M

Figure 6-3: libc, Support Routines

`_fpstart` `__fpstart` `sbrk` `_sbrk`

`char *sbrk(int incr);`

This function adds *incr* bytes to the *break value* and changes the allocated space accordingly. *Incr* can be negative, in which case the amount of allocated space is decreased. The break value is the address of the first allocation beyond the end of the data segment. The amount of allocated space increases as the break value increases. Newly allocated space is set to zero. If, however, the same memory space is reallocated to the same process, its contents are undefined. Upon successful completion, `sbrk` returns the old break value. Otherwise, it returns -1 and sets `errno` to indicate the error. The symbol `_sbrk` is also available with the same semantics.

`void __fpstart(void);`

This function calls `_fpstart()`, to initialize the floating-point environment.

`void _fpstart(void);`

This function initializes the floating-point execution environment. It sets `_fp_hw` to the appropriate value. It sets the rounding mode to “nearest.” It also resets the Intel387 control word to the default state.

Global Data Symbols

The `libc` library requires that some global external data objects be defined for the routines to work properly. In addition to the corresponding data symbols listed in the *System V ABI*, the following symbols must be provided in the system library on all ABI-conforming systems implemented with the Intel386 architecture. Declarations for the data objects listed below can be found in the Data Definitions section of this chapter or immediately following the table.

Figure 6-4: `libc`, Global External Data Symbols

`__flt_rounds` `_fp_hw` `__huge_val`

`extern int _fp_hw;`
This variable describes the floating-point hardware available. If the value is zero, no floating-point support is present. If the value is 1, the floating-point support is provided by an Intel387 software emulator. If the value is 2, an 80287 chip is available. If the value is 3, an Intel387 chip is available. System software sets the value appropriately, before transferring control to `main`.

Application Constraints

As described above, `libc` provides symbols for applications. In a few cases, however, an application is obliged to provide symbols for the library. In addition to the application-provided symbols listed in this section of the *System V ABI*, conforming applications on the Intel386 architecture are also required to provide the following symbols.

`extern _end;`
This symbol refers neither to a routine nor to a location with interesting contents. Instead, its address must correspond to the beginning of a program's dynamic allocation area, called the heap. Typically, the heap begins immediately after the data segment of the program's executable file.

`extern const int _lib_version;`
This variable's value specifies the compilation and execution mode for the program. If the value is zero, the program wants to preserve the semantics of older (pre-ANSI) C, where conflicts exist with ANSI. Otherwise, the value is non-zero, and the program wants ANSI C semantics.

System Data Interfaces

Data Definitions

This section contains standard data definitions that describe system data. These files are referred to by their names in angle brackets: `<name.h>` and `<sys/name.h>`. Included in these data definitions are macro definitions and data definitions.

The data objects described in this section are part of the interface between an ABI-conforming application and the underlying ABI-conforming system where it will run. While an ABI-conforming system must provide these interfaces, it is not required to contain the actual data definitions referenced here. Programmers should observe that the sources of the structures defined in these data definitions are defined in SVID.

ANSI C serves as the ABI reference programming language, and data definitions are specified in ANSI C format. The C language is used here as a convenient notation. Using a C language description of these data objects does *not* preclude their use by other programming languages. M

Reentrancy Considerations

M

New conventions have been added to accomodate the new requirements of reentrancy. Some historic binary code sequences are inherently non-reentrant. Unless great care is taken, multi-threaded applications cannot safely use such sequences. The most portable (i.e. those guaranteed to work in all cases) are those that are marked as reentrant in this chapter. For the ABI, this sometimes requires that two definitions exist for these interfaces, one that is reentrant and one that is not. These are indicated by comments that define which of the alternate definitions is reentrant. These alternatives are not selected at run-time, but are intended to be bound at application build time. M
M
M
M
M
M

NOTE

All information presented in the figures marked with * are new to the Fourth Edition of the psABI. M

Figure 6-5: <aio.h>*

```
struct aiocb {
    int          aio_fildes;
    volatile void* aio_buf;
    size_t       aio_nbytes;
    off_t        aio_offset;
    int          aio_reqprio;
    struct sigevent aio_sigevent;
    int          aio_lio_opcode;
    ssize_t      ;
    int          ;
    int          ;
    void         ;
    int          ;
} ;

#define AIO_CANCELED    (0)
#define AIO_ALLDONE    (1)
#define AIO_NOTCANCELED (2)

#define LIO_NOWAIT      (0)
#define LIO_WAIT       (1)
#define LIO_NOP         (0)
#define LIO_READ        (1)
#define LIO_WRITE       (2)
```

Figure 6-6: <assert.h>

```
extern void __assert(const char *, const char *, int);
#define assert(EX) (void)((EX)||(__assert(#EX, __FILE__, __LINE__), 0))
```

Figure 6-7: <ctype.h>

```
#define _U    01
#define _L    02
#define _N    04
#define _S    010
#define _P    020
#define _C    040
#define _B    0100
#define _X    0200

extern unsigned char      __ctype[];

#define isalpha(c) ((__ctype+1)[c]&(_U|_L))
#define isupper(c) ((__ctype+1)[c]&_U)
#define islower(c) ((__ctype+1)[c]&_L)
#define isdigit(c) ((__ctype+1)[c]&_N)
#define isxdigit(c) ((__ctype+1)[c]&_X)
#define isalnum(c) ((__ctype+1)[c]&(_U|_L|_N))
#define isspace(c) ((__ctype+1)[c]&_S)
#define ispunct(c) ((__ctype+1)[c]&_P)
#define isprint(c) ((__ctype+1)[c]&(_P|_U|_L|_N|_B))
#define isgraph(c) ((__ctype+1)[c]&(_P|_U|_L|_N))
#define iscntrl(c) ((__ctype+1)[c]&_C)
#define isascii(c) (!(c)&~0177)
#define _toupper(c) ((__ctype+258)[c])
#define _tolower(c) ((__ctype+258)[c])
#define toascii(c) ((c)&0177)
```

CAUTION

The data definitions in ctype.h are moved to Level 2 as of January 1, 1993. In order to correctly function in an internationalized environment, applications are encouraged to use the functions in libc instead.

M

Figure 6-8: <dirent.h>

```
typedef struct {
    int      dd_fd;
    int      dd_loc;
    int      dd_size;
    char     *dd_buf;
} DIR;

struct dirent {
    ino_t     d_ino;
    off_t     d_off;
    unsigned short d_reclen;
    char      d_name[1];
};
```

Figure 6-9: <dlfcn.h>*

```
#define RTLD_LAZY      1
#define RTLD_NOW      2
#define RTLD_GLOBAL    4
```

Figure 6-10: <elf.h>*, Part 1 of 6

```
#define ELF32_FSZ_ADDR      4
#define ELF32_FSZ_HALF      2
#define ELF32_FSZ_OFF       4
#define ELF32_FSZ_SWORD     4
#define ELF32_FSZ_WORD      4

#define EI_NIDENT    16

typedef struct {
    unsigned char e_ident[EI_NIDENT];
    Elf32_Half    e_type;
    Elf32_Half    e_machine;
    Elf32_Word    e_version;
    Elf32_Addr    e_entry;
    Elf32_Off     e_phoff;
    Elf32_Off     e_shoff;
    Elf32_Word    e_flags;
    Elf32_Half    e_ehsize;
    Elf32_Half    e_phentsize;
    Elf32_Half    e_phnum;
    Elf32_Half    e_shentsize;
    Elf32_Half    e_shnum;
    Elf32_Half    e_shstrndx;
} Elf32_Ehdr;

#define ELFMAG0      0x7f
#define ELFMAG1      'E'
#define ELFMAG2      'L'
#define ELFMAG3      'F'
#define ELFMAG       "\177ELF"
#define SELFMAG      4
```

Figure 6-11: <elf.h>*, Part 2 of 6

```
#define EI_MAG0      0
#define EI_MAG1      1
#define EI_MAG2      2
#define EI_MAG3      3
#define EI_CLASS      4
#define EI_DATA      5
#define EI_VERSION    6
#define EI_PAD        7

#define ELFCLASSNONE  0
#define ELFCLASS32    1
#define ELFCLASS64    2
#define ELFCLASSNUM   3
#define ELFDATANONE   0
#define ELFDATA2LSB   1
#define ELFDATA2MSB   2
#define ELFDATANUM    3

#define ET_NONE       0
#define ET_REL        1
#define ET_EXEC       2
#define ET_DYN        3
#define ET_CORE       4
#define ET_NUM        5

#define ET_LOPROC     0xff00
#define ET_HIPROC     0xffff

#define EM_NONE       0
#define EM_M32        1
#define EM_SPARC      2
#define EM_386        3
#define EM_68K        4
#define EM_88K        5
#define EM_486        6
#define EM_860        7
#define EM_NUM        8
```

Figure 6-12: <elf.h>*, Part 3 of 6

```
#define EV_NONE          0
#define EV_CURRENT      1
#define EV_NUM          2

typedef struct {
    Elf32_Word    p_type;
    Elf32_Off     p_offset;
    Elf32_Addr    p_vaddr;
    Elf32_Addr    p_paddr;
    Elf32_Word    p_filesz;
    Elf32_Word    p_memsz;
    Elf32_Word    p_flags;
    Elf32_Word    p_align;
} Elf32_Phdr;

#define PT_NULL          0
#define PT_LOAD          1
#define PT_DYNAMIC       2
#define PT_INTERP        3
#define PT_NOTE          4
#define PT_SHLIB         5
#define PT_PHDR          6
#define PT_NUM           7

#define PT_LOPROC        0x70000000
#define PT_HIPROC        0x7fffffff

#define PF_R             0x4
#define PF_W             0x2
#define PF_X             0x1

#define PF_MASKPROC      0xf0000000
```

Figure 6-13: <elf.h>*, Part 4 of 6

```
typedef struct {
    Elf32_Word    sh_name;
    Elf32_Word    sh_type;
    Elf32_Word    sh_flags;
    Elf32_Addr    sh_addr;
    Elf32_Off     sh_offset;
    Elf32_Word    sh_size;
    Elf32_Word    sh_link;
    Elf32_Word    sh_info;
    Elf32_Word    sh_addralign;
    Elf32_Word    sh_entsize;
} Elf32_Shdr;

#define SHT_NULL          0
#define SHT_PROGBITS      1
#define SHT_SYMTAB        2
#define SHT_STRTAB        3
#define SHT_RELA          4
#define SHT_HASH          5
#define SHT_DYNAMIC        6
#define SHT_NOTE          7
#define SHT_NOBITS        8
#define SHT_REL           9
#define SHT_SHLIB         10
#define SHT_DYNSYM        11
#define SHT_NUM           12

#define SHT_LOUSER        0x80000000
#define SHT_HIUSER        0xffffffff
#define SHT_LOPROC        0x70000000
#define SHT_HIPROC        0x7fffffff
#define SHF_MASKPROC      0xf0000000

#define SHF_WRITE          0x1
#define SHF_ALLOC          0x2
#define SHF_EXECINSTR      0x4
```

Figure 6-14: <elf.h>*, Part 5 of 6

```
#define SHN_UNDEF          0
#define SHN_LORESERVE     0xfff00
#define SHN_ABS           0xffff1
#define SHN_COMMON        0xffff2
#define SHN_HIRESERVE     0xfffff
#define SHN_LOPROC        0xff00
#define SHN_HIPROC        0xff1f

typedef struct {
    Elf32_Word      st_name;
    Elf32_Addr      st_value;
    Elf32_Word      st_size;
    unsigned char    st_info;
    unsigned char    st_other;
    Elf32_Half      st_shndx;
} Elf32_Sym;

#define STN_UNDEF  0

#define ELF32_ST_BIND(info)      ((info) >> 4)
#define ELF32_ST_TYPE(info)     ((info) & 0xf)
#define ELF32_ST_INFO(bind, type) (((bind)<<4)+((type)&0xf))

#define STB_LOCAL  0
#define STB_GLOBAL 1
#define STB_WEAK   2
#define STB_NUM     3
#define STB_LOPROC 13
#define STB_HIPROC 15
```


Figure 6-15: <elf.h>*, Part 6 of 6

```
#define STT_NOTYPE 0
#define STT_OBJECT 1
#define STT_FUNC 2
#define STT_SECTION 3
#define STT_FILE 4
#define STT_NUM 5
#define STT_LOPROC 13
#define STT_HIPROC 15

typedef struct {
    Elf32_Addr r_offset;
    Elf32_Word r_info;
} Elf32_Rel;

typedef struct {
    Elf32_Addr r_offset;
    Elf32_Word r_info;
    Elf32_Sword r_addend;
} Elf32_Rela;

#define ELF32_R_SYM(info) ((info)>>8)
#define ELF32_R_TYPE(info) ((unsigned char)(info))
#define ELF32_R_INFO(sym, type) (((sym)<<8)+(unsigned char)(type))
```

Figure 6-16: <errno.h>, Part 1 of 3

#define EPERM	1
#define ENOENT	2
#define ESRCH	3
#define EINTR	4
#define EIO	5
#define ENXIO	6
#define E2BIG	7
#define ENOEXEC	8
#define EBADF	9
#define ECHILD	10
#define EAGAIN	11
#define ENOMEM	12
#define EACCES	13
#define EFAULT	14
#define ENOTBLK	15
#define EBUSY	16
#define EEXIST	17
#define EXDEV	18
#define ENODEV	19
#define ENOTDIR	20
#define EISDIR	21
#define EINVAL	22
#define ENFILE	23
#define EMFILE	24
#define ENOTTY	25
#define ETXTBSY	26
#define EFBIG	27

Figure 6-17: <errno.h>, Part 2 of 3

#define ENOSPC	28
#define ESPIPE	29
#define EROFS	30
#define EMLINK	31
#define EPIPE	32
#define EDOM	33
#define ERANGE	34
#define ENOMSG	35
#define EIDRM	36
#define ECHRNG	37
#define EL2NSYNC	38
#define EL3HLT	39
#define EL3RST	40
#define ELNRNG	41
#define EUNATCH	42
#define ENOCSI	43
#define EL2HLT	44
#define EDEADLK	45
#define ENOLCK	46
#define ENOSTR	60
#define ENODATA	61
#define ETIME	62
#define ENOSR	63
#define ENONET	64
#define ENOPKG	65
#define EREMOTE	66
#define ENOLINK	67

Figure 6-18: <errno.h>, Part 3 of 3

#define EADV	68	
#define ESRMNT	69	
#define ECOMM	70	
#define EPROTO	71	
#define EMULTIHOP	74	
#define EBADMSG	77	
#define ENAMETOOLONG	78	
#define EOVERFLOW	79	
#define ENOTUNIQ	80	
#define EBADFD	81	
#define EREMCHG	82	
#define ENOSYS	89	
#define ELOOP	90	
#define ERESTART	91	
#define ESTRPIPE	92	
#define ENOTEMPTY	93	
#define EUSERS	94	
#define ECONNABORTED	130	G
#define CONNRESET	131	G
#define ECONNREFUSED	146	G
#define EINPROGRESS	150	M
#define ESTALE	151	
#define ECANCELED	158	M
/* Non-reentrant */		M
extern int errno;		M
/* Reentrant */		M
#define errno (*__thr_errno())		M

Figure 6-19: <fcntl.h>, Part 1 of 2

```
#define O_RDONLY      0
#define O_WRONLY      1
#define O_RDWR        2
#define O_NDELAY      0x04
#define O_APPEND      0x08
#define O_SYNC         0x10
#define O_NONBLOCK     0x80
#define O_CREAT        0x100
#define O_TRUNC        0x200
#define O_EXCL         0x400
#define O_NOCTTY       0x800

#define F_DUPFD        0
#define F_GETFD        1
#define F_SETFD        2
#define F_GETFL        3
#define F_SETFL        4
#define F_GETLK        14
#define F_SETLK        6
#define F_SETLKW       7
```

Figure 6-20: <fcntl.h>, Part 2 of 2

```
typedef struct flock {
    short  l_type;
    short  l_whence;
    off_t  l_start;
    off_t  l_len;
    long   l_sysid;
    pid_t  l_pid;
    long   pad[4];
} flock_t;

#define F_RDLCK      01
#define F_WRLCK      02
#define F_UNLCK      03

#define O_ACCMODE     3
#define FD_CLOEXEC    1
```

Figure 6-21: <float.h>, Single-Precision

```
extern int      __flt_rounds;
#define FLT_ROUNDS      __flt_rounds

#define FLT_RADIX      2
#define FLT_MANT_DIG    24
#define FLT_EPSILON     1.19209290E-07F
#define FLT_DIG         6
#define FLT_MIN_EXP     (-125)
#define FLT_MIN         1.17549435E-38F
#define FLT_MIN_10_EXP  (-37)
#define FLT_MAX_EXP     (+128)
#define FLT_MAX         3.40282347E+38F
#define FLT_MAX_10_EXP  (+38)
```

Figure 6-22: <float.h>, Double-Precision

```
#define DBL_MANT_DIG    53
#define DBL_EPSILON     2.2204460492503131E-16
#define DBL_DIG         15
#define DBL_MIN_EXP     (-1021)
#define DBL_MIN         2.2250738585072014E-308
#define DBL_MIN_10_EXP  (-307)
#define DBL_MAX_EXP     (+1024)
#define DBL_MAX         1.7976931348623157E+308
#define DBL_MAX_10_EXP  (+308)
```

Figure 6-23: <float.h>, Extended-Precision

```
#define LDBL_MANT_DIG      64
#define LDBL_EPSILON      1.084202172485504434e-19
#define LDBL_DIG          18
#define LDBL_MIN_EXP      -16381
#define LDBL_MIN          3.362103143112093506e-4932
#define LDBL_MIN_10_EXP   -4931
#define LDBL_MAX_EXP      16384
#define LDBL_MAX          1.189731495347231765e+4932
#define LDBL_MAX_10_EXP   4932
```

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Figure 6-24: <fmtmsg.h>, Part 1 of 2

```
#define MM_NULL           0L

#define MM_HARD           0x00000001L
#define MM_SOFT          0x00000002L
#define MM_FIRM          0x00000004L
#define MM_RECOVER       0x00000100L
#define MM_NRECOV        0x00000200L
#define MM_APPL          0x00000008L
#define MM_UTIL          0x00000010L
#define MM_OPSYS         0x00000020L
#define MM_PRINT         0x00000040L
#define MM_CONSOLE       0x00000080L
```


Figure 6-25: <fmtmsg.h>, Part 2 of 2

```
#define MM_NOSEV          0
#define MM_HALT           1
#define MM_ERROR          2
#define MM_WARNING        3
#define MM_INFO           4

#define MM_NULLLBL        ((char *) NULL)
#define MM_NULLSEV        MM_NOSEV
#define MM_NULLMC         MM_NULL
#define MM_NULLTXT        ((char *) NULL)
#define MM_NULLACT        ((char *) NULL)
#define MM_NULLTAG        ((char *) NULL)

#define MM_NOTOK          -1
#define MM_OK             0x00
#define MM_NOMSG          0x01
#define MM_NOCON          0x04
```

Figure 6-26: <fnmatch.h>*

```
#define FNM_PATHNAME      0x001
#define FNM_PERIOD        0x002
#define FNM_NOESCAPE      0x004
#define FNM_BADRANGE      0x008
#define FNM_EXTENDED      0x020

#define FNM_NOSYS         (-1)
#define FNM_NOMATCH       (-2)
```

Figure 6-27: <ftw.h>

```
#define FTW_F          0
#define FTW_D          1
#define FTW_DNR        2
#define FTW_NS         3
#define FTW_SL         4
#define FTW_DP         6
#define FTW_SLN        7

#define FTW_PHYS        01
#define FTW_MOUNT      02
#define FTW_CHDIR      04
#define FTW_DEPTH      010

struct FTW {
    int    quit;
    int    base;
    int    level;
};

#define FTW_SKD         1
#define FTW_FOLLOW      2
#define FTW_PRUNE       4
```

Figure 6-28: <glob.h> *

```
#define GLOB_APPEND      0x0001
#define GLOB_DOOFFS      0x0002
#define GLOB_ERR          0x0004
#define GLOB_MARK         0x0008
#define GLOB_NOCHECK      0x0010
#define GLOB_NOSORT       0x0020
#define GLOB_NOESCAPE     0x0040
#define GLOB_OKAYDOT      0x0200
#define GLOB_BADRANGE     0x0400
#define GLOB_EXTENDED     0x1000

#define GLOB_NOSYS        (-1)
#define GLOB_ABORTED      (-2)
#define GLOB_NOSPACE      (-3)
#define GLOB_NOMATCH      (-4)

typedef struct
{
    void          *;
    char          **gl_pathv;
    size_t        gl_pathc;
    size_t        gl_offs;
} glob_t;
```

Figure 6-29: <grp.h>

```
struct group {  
    char    *gr_name;  
    char    *gr_passwd;  
    gid_t   gr_gid;  
    char    **gr_mem;  
};
```

Figure 6-30: <iconv.h>*

```
typedef void *iconv_t;
```

Figure 6-31: <sys/ipc.h>

```
struct ipc_perm {
    uid_t    uid;
    gid_t    gid;
    uid_t    cuid;
    gid_t    cgid;
    mode_t   mode;
    ulong    seq;
    key_t    key;
    long     pad[4];
};

#define IPC_CREAT  0001000
#define IPC_EXCL   0002000
#define IPC_NOWAIT 0004000

#define IPC_PRIVATE (key_t)0

#define IPC_RMID    10
#define IPC_SET     11
#define IPC_STAT    12
```

Figure 6-32: <langinfo.h>, Part 1 of 2

```
#define DAY_1          1
#define DAY_2          2
#define DAY_3          3
#define DAY_4          4
#define DAY_5          5
#define DAY_6          6
#define DAY_7          7

#define ABDAY_1        8
#define ABDAY_2        9
#define ABDAY_3       10
#define ABDAY_4       11
#define ABDAY_5       12
#define ABDAY_6       13
#define ABDAY_7       14

#define MON_1         15
#define MON_2         16
#define MON_3         17
#define MON_4         18
#define MON_5         19
#define MON_6         20
#define MON_7         21
#define MON_8         22
#define MON_9         23
#define MON_10        24
#define MON_11        25
#define MON_12        26
```

Figure 6-33: <langinfo.h>, Part 2 of 2

#define ABMON_1	27	
#define ABMON_2	28	
#define ABMON_3	29	
#define ABMON_4	30	
#define ABMON_5	31	
#define ABMON_6	32	
#define ABMON_7	33	
#define ABMON_8	34	
#define ABMON_9	35	
#define ABMON_10	36	
#define ABMON_11	37	
#define ABMON_12	38	
#define RADIXCHAR	39	
#define THOUSEP	40	
#define YESSTR	41	
#define NOSTR	42	
#define CRNCYSTR	43	
#define D_T_FMT	44	
#define D_FMT	45	
#define T_FMT	46	
#define AM_STR	47	
#define PM_STR	48	
#define CODESET	49	M
#define T_FMT_AMPM	50	M
#define ERA	51	M
#define ERA_D_FMT	52	M
#define ERA_D_T_FMT	53	M
#define ERA_T_FMT	54	M
#define ALT_DIGITS	55	M
#define YESEXPR	56	M
#define NOEXPR	57	M

Figure 6-34: <limits.h>, Part 1 of 2

```
#define CHAR_BIT            8
#define SCHAR_MIN          (-128)
#define SCHAR_MAX          127
#define UCHAR_MAX          255
#define MB_LEN_MAX         5

#define CHAR_MIN            SCHAR_MIN
#define CHAR_MAX            SCHAR_MAX

#define SHRT_MIN            (-32768)
#define SHRT_MAX            32767
#define USHRT_MAX           65535
#define INT_MIN             (-2147483647-1)
#define INT_MAX             2147483647
#define UINT_MAX            4294967295
#define LONG_MIN            (-2147483647-1)
#define LONG_MAX            2147483647
#define ULONG_MAX           4294967295

#define ARG_MAX              *
#define LINK_MAX              *
#define MAX_CANON             *
#define MAX_INPUT             *
#define NGROUPS_MAX           *
#define PATH_MAX              *
#define PIPE_BUF              *
#define TMP_MAX               *
#define PASS_MAX              *
#define CHILD_MAX             *

/* starred values vary and should be
   retrieved using sysconf() or pathconf() */
```


Figure 6-35: <limits.h>, Part 2 of 2

```
#define NL_ARGMAX      9
#define NL_LANGMAX     14
#define NL_MSGMAX     32767
#define NL_NMAX        1
#define NL_SETMAX      255
#define NL_TEXTMAX     255
#define NZERO          20

#define WORD_BIT       32
#define LONG_BIT       32

#define DBL_DIG        15
#define DBL_MAX        1.7976931348623157E+308
#define DBL_MIN        2.2250738585072014E-308
#define FLT_DIG        6
#define FLT_MAX        3.40282347E+38F
#define FLT_MIN        1.17549435E-38F

#define FCHR_MAX       1048576
```

Figure 6-36: <locale.h>

```
struct lconv {
    char    *decimal_point;
    char    *thousands_sep;
    char    *grouping;
    char    *int_curr_symbol;
    char    *currency_symbol;
    char    *mon_decimal_point;
    char    *mon_thousands_sep;
    char    *mon_grouping;
    char    *positive_sign;
    char    *negative_sign;
    char    int_frac_digits;
    char    frac_digits;
    char    p_cs_precedes;
    char    p_sep_by_space;
    char    n_cs_precedes;
    char    n_sep_by_space;
    char    p_sign_posn;
    char    n_sign_posn;
};

#define LC_CTYPE      0
#define LC_NUMERIC    1
#define LC_TIME       2
#define LC_COLLATE    3
#define LC_MONETARY   4
#define LC_MESSAGES   5
#define LC_ALL        6
```

Figure 6-37: <lwpsynch.h>*

```
typedef volatile struct {
    char        wanted;
    _simplelock_t lock;
} lwp_mutex_t;

typedef volatile struct {
    char        wanted;
} lwp_cond_t;
```

Figure 6-38: <machlock.h>*

```
typedef volatile unsigned char _simplelock_t;
```

Figure 6-39: <math.h>

```
extern const double __huge_val;
#define HUGE_VAL    __huge_val
```

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Figure 6-40: <sys/mman.h>

```
#define PROT_READ      0x1
#define PROT_WRITE     0x2
#define PROT_EXEC      0x4
#define PROT_NONE      0x0

#define MAP_SHARED      1
#define MAP_PRIVATE     2
#define MAP_FIXED      0x10

#define MS_SYNC         0x0
#define MS_ASYNC        0x1
#define MS_INVALIDATE   0x2

#define PROC_TEXT       (PROT_EXEC | PROT_READ)
#define PROC_DATA       (PROT_READ | PROT_WRITE | PROT_EXEC)

#define SHARED          0x10
#define PRIVATE         0x20

#define MC_SYNC         1
#define MC_LOCK         2
#define MC_UNLOCK       3
#define MC_LOCKAS       5
#define MC_UNLOCKAS     6

#define MCL_CURRENT     0x1
#define MCL_FUTURE      0x2
```

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Figure 6-41: <sys/mod.h>*

```
#define VOID          void
#define MAXPATHLEN    1024
#define MODMAXLINKINFOLEN 32

struct modspecific_stat {
    char    mss_linkinfo[MODMAXLINKINFOLEN];
    int     mss_type;
    int     mss_p0[2];
    int     mss_p1[2];
};

#define MODMAXLINK      4

struct modstatus {
    int             ms_id;
    VOID            *ms_base;
    unsigned int    ms_size;
    int             ms_rev;
    char            ms_path[MAXPATHLEN];
    time_t          ms_unload_delay;
    int             ms_refcnt;
    int             ms_depcnt;
    struct modspecific_stat ms_msinfo[MODMAXLINK];
};
```

Figure 6-42: <sys/mount.h>

#define MS_RDONLY	0x01
#define MS_FSS	0x02
#define MS_DATA	0x04
#define MS_HADBAD	0x08
#define MS_NOSUID	0x10
#define MS_REMOUNT	0x20
#define MS_NOTRUNC	0x40

Figure 6-43: <sys/msg.h>

```
#define MSG_NOERROR      010000

struct msqid_ds {
    struct ipc_perm    msg_perm;
    struct msg         *msg_first;
    struct msg         *msg_last;
    ulong              msg_cbytes;
    ulong              msg_qnum;
    ulong              msg_qbytes;
    pid_t              msg_lspid;
    pid_t              msg_lrpid;
    time_t             msg_stime;
    long               msg_pad1;
    time_t             msg_rtime;
    long               msg_pad2;
    time_t             msg_ctime;
    long               msg_pad3;
    long               msg_pad4[4];
};

struct msg {
    struct msg         *msg_next;
    long               msg_type;
    ushort             msg_ts;
    short              msg_spot;
};
```

Figure 6-44: <netconfig.h>, Part 1 of 2

```
struct netconfig {
    char          *nc_netid;
    unsigned long nc_semantics;
    unsigned long nc_flag;
    char          *nc_protobuf;
    char          *nc_proto;
    char          *nc_device;
    unsigned long nc_nlookups;
    char          **nc_lookups;
    unsigned long nc_unused[8];
};

#define NC_TPI_CLTS      1
#define NC_TPI_COTS     2
#define NC_TPI_COTS_ORD 3
#define NC_TPI_RAW      4

#define NC_NOFLAG      00
#define NC_VISIBLE     01
```

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Figure 6-45: <netconfig.h>, Part 2 of 2

```
#define NC_NOPROTOFMLY    "-"
#define NC_LOOPBACK      "loopback"
#define NC_INET           "inet"
#define NC_IMPLINK       "implink"
#define NC_PUP            "pup"
#define NC_CHAOS          "chaos"
#define NC_NS             "ns"
#define NC_NBS            "nbs"
#define NC_ECMA           "ecma"
#define NC_DATAKIT       "datakit"
#define NC_CCITT          "ccitt"
#define NC_SNA            "sna"
#define NC_DECNET        "decnet"
#define NC_DLI            "dli"
#define NC_LAT            "lat"
#define NC_HYLINK        "hylink"
#define NC_APPLETALK      "appletalk"
#define NC_NIT            "nit"
#define NC_IEEE802        "ieee802"
#define NC_OSI            "osi"
#define NC_X25            "x25"
#define NC_OSINET        "osinet"
#define NC_GOSIP          "gossip"
#define NC_NETWARE        "netware"

#define NC_NOPROTO        "-"
#define NC_TCP            "tcp"
#define NC_UDP            "udp"
#define NC_ICMP           "icmp"
#define NC_IPX            "ipx"
#define NC_SPX            "spx"
```

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Figure 6-46: <netdir.h>, Part 1 of 2

```
struct nd_addrlist {
    int          n_cnt;
    struct netbuf *n_addrs;
};

struct nd_hostservlist {
    int          h_cnt;
    struct nd_hostserv *h_hostservs;
};

struct nd_hostserv {
    char          *h_host;
    char          *h_serv;
};

#define ND_HOSTSERV          0
#define ND_HOSTSERVLIST     1
#define ND_ADDR              2
#define ND_ADDRLIST         3

#define ND_BADARG            -2
#define ND_NOMEM             -1
#define ND_OK                0
#define ND_NOHOST            1
#define ND_NOSERV            2
#define ND_NOSYM             3
#define ND_OPEN              4
#define ND_ACCESS            5
#define ND_UKNWN             6
#define ND_NOCTRL            7
#define ND_FAILCTRL          8
#define ND_SYSTEM            9
#define ND_NOERRMEM          10
#define ND_NOLIB             11
#define ND_XTIERROR          12
#define ND_BADSTATE          13
```

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Figure 6-47: <netdir.h>, Part 2 of 2

#define ND_SET_BROADCAST	1	
#define ND_SET_RESERVEDPORT	2	
#define ND_CHECK_RESERVEDPORT	3	
#define ND_MERGEADDR	4	
#define ND_CLEAR_BROADCAST	5	M
#define ND_SET_REUSEADDR	6	M
#define ND_CLEAR_REUSEADDR	7	M
#define HOST_SELF	"\\1"	
#define HOST_ANY	"\\2"	
#define HOST_BROADCAST	"\\3"	

Figure 6-48: <nl_types.h>

#define NL_SETD	1
typedef int nl_item;	
typedef void *nl_catd;	

Figure 6-49: <sys/param.h>

#define CANBSIZ	256
#define HZ	100
#define TICK	10000000
#define NGROUPS_UMIN	0
#define NBPSCTR	512
#define MAXPATHLEN	1024
#define MAXSYMLINKS	20
#define MAXNAMELEN	256
#define NADDR	13
#define PIPE_MAX	5120
#define NBBY	8
#define MAXFRAG	8

Figure 6-50: <poll.h>

```
struct pollfd {
    int    fd;
    short  events;
    short  revents;
};

#define POLLIN          0x0001
#define POLLPRI         0x0002
#define POLLOUT         0x0004
#define POLLRDNORM      0x0040
#define POLLWRNORM      POLLOUT
#define POLLRDBAND      0x0080
#define POLLWRBAND      0x0100

#define POLLNORM         POLLRDNORM

#define POLLERR          0x0008
#define POLLHUP          0x0010
#define POLLNVAL         0x0020
```

Figure 6-51: <sys/priocntl.h>*

```
#define PC_GETCID          0
#define PC_GETCLINFO      1
#define PC_SETPARMS       2
#define PC_GETPARMS       3

#define PC_CLNULL         -1

#define PC_CLNMSZ    16
#define PC_CLINFOSZ (32 / sizeof(long))
#define PC_CLPARMSZ (32 / sizeof(long))

typedef struct pcinfo {
    id_t    pc_cid;
    char    pc_clname[PC_CLNMSZ];
    long    pc_clinfo[PC_CLINFOSZ];
} pcinfo_t;

typedef struct pcparms {
    id_t    pc_cid;
    long    pc_clparms[PC_CLPARMSZ];
} pcparms_t;
```

Figure 6-52: <sys/procset.h>

```
#define P_INITPID      1
#define P_INITUID      0
#define P_INITPGID     0

typedef enum idtype {
    P_PID,
    P_PPID,
    P_PGID,
    P_SID,
    P_CID,
    P_UID,
    P_GID,
    P_ALL
} idtype_t;

typedef enum idop {
    POP_DIFF,
    POP_AND,
    POP_OR,
    POP_XOR
} idop_t;

typedef struct procset {
    idop_t      p_op;
    idtype_t    p_lidtype;
    id_t        p_lid;
    idtype_t    p_ridtype;
    id_t        p_rid;
} procset_t;
```

Figure 6-53: <pwd.h>

```
struct passwd {  
    char      *pw_name;  
    char      *pw_passwd;  
    uid_t     pw_uid;  
    gid_t     pw_gid;  
    char      *pw_age;  
    char      *pw_comment;  
    char      *pw_gecos;  
    char      *pw_dir;  
    char      *pw_shell;  
};
```


Figure 6-54: <regex.h>*, Part 1 of 2

```
#define REG_NOTBOL      0x000001
#define REG_NOTEOL      0x000002
#define REG_NONEMPTY    0x000004

#define REG_OR          0x000001
#define REG_PLUS        0x000002
#define REG_REQUEST     0x000004
#define REG_BRACES      0x000008
#define REG_PARENS      0x000010
#define REG_ANCHORS     0x000020
#define REG_NOBACKREF    0x000040
#define REG_NOAUTOQUOTE 0x000080

#define REG_EXTENDED    (REG_OR | REG_PLUS | REG_REQUEST |
                        REG_BRACES | REG_PARENS | REG_ANCHORS |
                        REG_NOBACKREF | REG_NOAUTOQUOTE)
#define REG_ICASE       0x000100
#define REG_NOSUB       0x000200
#define REG_NEWLINE     0x000400
#define REG_ONESUB      0x000800
#define REG_BADRANGE    0x004000
#define REG_ANGLES      0x040000
#define REG_ESCNL       0x080000
#define REG_OLDBRE      (REG_BADRANGE | REG_ANGLES | REG_ESCNL)
```

Figure 6-55: <regex.h>*, Part 2 of 2

```
#define REG_ENOSYS      (-1)
#define REG_NOMATCH     1
#define REG_BADPAT      2
#define REG_ECOLLATE    3
#define REG_ETYPE       4
#define REG_EESCAPE     7
#define REG_ESUBREG     8
#define REG_EBRACK      9
#define REG_NOPAT      12
#define REG_EPAREN     13
#define REG_EBRACE     14
#define REG_BADBR      15
#define REG_ERANGE     16
#define REG_ESPACE     17
#define REG_BADRPT     18

typedef struct
{
    size_t      re_nsub;
    unsigned long re_flags;
    void        *[4];
} regex_t;

typedef ssize_t regoff_t;

typedef struct
{
    regoff_t     rm_so;
    regoff_t     rm_eo;
} regmatch_t;
```

Figure 6-56: <sys/resource.h>

```
#define RLIMIT_CPU          0
#define RLIMIT_FSIZE       1
#define RLIMIT_DATA        2
#define RLIMIT_STACK       3
#define RLIMIT_CORE        4
#define RLIMIT_NOFILE      5
#define RLIMIT_VMEM        6
#define RLIM_NLIMITS      7
#define RLIMIT_AS          RLIMIT_VMEM
#define RLIM_INFINITY      0x7fffffff

typedef unsigned long      rlim_t;

struct rlimit {
    rlim_t      rlim_cur;
    rlim_t      rlim_max;
};
```

Figure 6-57: <rpc.h>, Part 1 of 16

```
#define bool_t      int
#define enum_t      int

enum xdr_op {
    XDR_ENCODE=0,
    XDR_DECODE=1,
    XDR_FREE=2
};

typedef bool_t      (*xdrproc_t)();

typedef struct {
    enum xdr_op  x_op;
    struct xdr_ops {
        bool_t (*x_getlong)();
        bool_t (*x_putlong)();
        bool_t (*x_getbytes)();
        bool_t (*x_putbytes)();
        u_int  (*x_getpostn)();
        bool_t (*x_setpostn)();
        long * (*x_inline)();
        void   (*x_destroy)();
    } *x_ops;
    caddr_t      x_public;
    caddr_t      x_private;
    caddr_t      x_base;
    int          x_handy;
} XDR;
```

Figure 6-58: <rpc.h>, Part 2 of 16

```
#define xdr_getpos(xdrs)          \
    (*(xdrs)->x_ops->x_getpostn)(xdrs) \
#define xdr_setpos(xdrs, pos)    \
    (*(xdrs)->x_ops->x_setpostn)(xdrs, pos) \
#define xdr_inline(xdrs, len)    \
    (*(xdrs)->x_ops->x_inline)(xdrs, len) \
#define xdr_destroy(xdrs)        \
    (*(xdrs)->x_ops->x_destroy)(xdrs) \
                                     M
                                     M

#define NULL_xdrproc_t    ((xdrproc_t)0)
struct xdr_discrim {
    int      value;
    xdrproc_t proc;
};
```

Figure 6-59: <rpc.h>, Part 3 of 16

```
#define MAX_AUTH_BYTES      400
#define MAXNETNAMELEN      255
#define HEXKEYBYTES        48

enum auth_stat {
    AUTH_OK=0,
    AUTH_BADCRED=1,
    AUTH_REJECTEDDCRED=2,
    AUTH_BADVERF=3,
    AUTH_REJECTEDVERF=4,
    AUTH_TOOWEAK=5,
    AUTH_INVALIDRESP=6,
    AUTH_FAILED=7
};

typedef u_long      u_int32      M

union des_block {
    struct {
        u_int32      high;      M
        u_int32      low;       M
    } key;
    char    c[8];
};
typedef union des_block      des_block;
```

Figure 6-60: <rpc.h>, Part 4 of 16

```
struct opaque_auth {
    enum_t      oa_flavor;
    caddr_t     oa_base;
    u_int       oa_length;
};

typedef struct {
    struct opaque_auth ah_cred;
    struct opaque_auth ah_verf;
    union des_block    ah_key;
    struct auth_ops {
        void (*ah_nextverf)();
        int  (*ah_marshall)();
        int  (*ah_validate)();
        int  (*ah_refresh)();
        void (*ah_destroy)();
    } *ah_ops;
    caddr_t ah_private;
} AUTH;

#define auth_destroy(auth) \
    ((*((auth)->ah_ops->ah_destroy))(auth))

#define AUTH_NONE      0
#define AUTH_NULL      0
#define AUTH_SYS        1
#define AUTH_UNIX      AUTH_SYS
#define AUTH_SHORT      2
#define AUTH_DES        3
#define AUTH_ESV        200004
```

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Figure 6-61: <rpc.h>, Part 5 of 16

```
enum clnt_stat {
    RPC_SUCCESS=0,
    RPC_CANTENCODEARGS=1,
    RPC_CANTDECODERES=2,
    RPC_CANTSEND=3,
    RPC_CANTRECV=4,
    RPC_TIMEDOUT=5,
    RPC_INTR=18,
    RPC_UDERROR=23,
    RPC_VERSMISMATCH=6,
    RPC_AUTHERROR=7,
    RPC_PROGUNAVAIL=8,
    RPC_PROGVERSMISMATCH=9,
    RPC_PROCUNAVAIL=10,
    RPC_CANTDECODEARGS=11,
    RPC_SYSTEMERROR=12,
    RPC_UNKNOWNHOST=13,
    RPC_UNKNOWNPROTO=17,
    RPC_UNKNOWNADDR=19,
    RPC_NOBROADCAST=21,
    RPC_RPCBFAILURE=14,
    RPC_PROGNOTREGISTERED=15,
    RPC_N2AXLATEFAILURE=22,
    RPC_TLIERROR=20,
    RPC_FAILED=16
};
#define RPC_PMAPFAILURE    RPC_RPCBFAILURE
```


Figure 6-62: <rpc.h>, Part 6 of 16

```
struct rpc_err {
    enum clnt_stat    re_status;
    union {
        struct {
            int    errno;
            int    t_errno;
        } RE_err;
        enum auth_stat    RE_why;
        struct {
            u_long    low;
            u_long    high;
        } RE_vers;
        struct {
            long    s1;
            long    s2;
        } RE_lb;
    } ru;
};
typedef struct {
    AUTH    *cl_auth;
    struct clnt_ops {
        enum clnt_stat    (*cl_call)();
        void    (*cl_abort)();
        void    (*cl_geterr)();
        bool_t    (*cl_freeres)();
        void    (*cl_destroy)();
        bool_t    (*cl_control)();
    } *cl_ops;
    caddr_t    cl_private;
    char    *cl_netid;
    char    *cl_tp;
} CLIENT;
```

Figure 6-63: <rpc.h>, Part 7 of 16

```
#define FEEDBACK_REXMIT1 1
#define FEEDBACK_OK      2

#define clnt_call(rh, proc, xargs, argsp, xres, resp, secs)\
    ((*rh)->cl_ops->cl_call) \
    (rh, proc, xargs, argsp, xres, resp, secs)
#define clnt_abort(rh) \
    ((*rh)->cl_ops->cl_abort)(rh)
#define clnt_geterr(rh, errp) \
    ((*rh)->cl_ops->cl_geterr)(rh, errp)
#define clnt_freeres(rh,xres,resp) \
    ((*rh)->cl_ops->cl_freeres)(rh,xres,resp)
#define clnt_control(cl, rq, in) \
    ((*cl)->cl_ops->cl_control)(cl, rq, in)
#define clnt_destroy(rh) \
    ((*rh)->cl_ops->cl_destroy)(rh)

#define CLSET_TIMEOUT      1
#define CLGET_TIMEOUT      2
#define CLGET_SERVER_ADDR 3
#define CLGET_FD           6
#define CLGET_SVC_ADDR     7
#define CLSET_FD_CLOSE     8
#define CLSET_FD_NCLOSE    9
#define CLSET_RETRY_TIMEOUT 4
#define CLGET_RETRY_TIMEOUT 5
```

Figure 6-64: <rpc.h>, Part 8 of 16

typedef struct {		M
enum clnt_stat	cf_stat;	M
struct rpc_err	cf_error;	M
} rpc_createerr_t;		M
extern rpc_createerr_t	rpc_createerr;	M

Figure 6-65: <rpc.h>, Part 9 of 16

```
enum msg_type {
    CALL=0,
    REPLY=1
};

enum reply_stat {
    MSG_ACCEPTED=0,
    MSG_DENIED=1
};

enum accept_stat {
    SUCCESS=0,
    PROG_UNAVAIL=1,
    PROG_MISMATCH=2,
    PROC_UNAVAIL=3,
    GARBAGE_ARGS=4,
    SYSTEM_ERR=5
};

enum reject_stat {
    RPC_MISMATCH=0,
    AUTH_ERROR=1
};
```

Figure 6-66: <rpc.h>, Part 10 of 16

```
struct accepted_reply {
    struct opaque_auth      ar_verf;
    enum accept_stat        ar_stat;
    union {
        struct {
            u_long          low;
            u_long          high;
        } AR_versions;
        struct {
            caddr_t         where;
            xdrproc_t       proc;
        } AR_results;
    } ru;
};

struct rejected_reply {
    enum reject_stat        rj_stat;
    union {
        struct {
            u_long          low;
            u_long          high;
        } RJ_versions;
        enum auth_stat      RJ_why;
    } ru;
};
```

Figure 6-67: <rpc.h>, Part 11 of 16

```
struct reply_body {
    enum reply_stat    rp_stat;
    union {
        struct accepted_reply RP_ar;
        struct rejected_reply RP_dr;
    } ru;
};

struct call_body {
    u_long             cb_rpcvers;
    u_long             cb_prog;
    u_long             cb_vers;
    u_long             cb_proc;
    struct opaque_auth cb_cred;
    struct opaque_auth cb_verf;
};

struct rpc_msg {
    u_long             rm_xid;
    enum msg_type      rm_direction;
    union {
        struct call_body  RM_cmb;
        struct reply_body RM_rmb;
    } ru;
};
```

Figure 6-68: <rpc.h>, Part 12 of 16

```
struct authsys_parms {
    u_long      aup_time;
    char        *aup_machname;
    uid_t       aup_uid;
    gid_t       aup_gid;
    u_int       aup_len;
    gid_t       *aup_gids;
};
```

Figure 6-69: <rpc.h>, Part 13 of 16

```
enum authdes_namekind {
    ADN_FULLNAME,
    ADN_NICKNAME
};

struct authdes_fullname {
    char        *name;
    des_block   key;
    u_long      window;
};

struct authdes_cred {
    enum authdes_namekind  adc_namekind;
    struct authdes_fullname adc_fullname;
    u_long                 adc_nickname;
};
```

Figure 6-70: <rpc.h>, Part 14 of 16

```
enum xpirt_stat {
    XPRT_DIED,
    XPRT_MOREREQS,
    XPRT_IDLE
};

typedef struct {
    int          xp_fd;
    u_short      xp_port;
    struct xp_ops {
        bool_t    (*xp_rcv)();
        enum xpirt_stat (*xp_stat)();
        bool_t    (*xp_getargs)();
        bool_t    (*xp_reply)();
        bool_t    (*xp_freeargs)();
        void      (*xp_destroy)();
    } *xp_ops;
    int          xp_addrlen;
    char         *xp_tp;
    char         *xp_netid;
    struct netbuf xp_ltaddr;
    struct netbuf xp_rtaddr;
    char         xp_raddr[16];
    struct opaque_auth xp_verf;
    caddr_t      xp_p1;
    caddr_t      xp_p2;
    caddr_t      xp_p3;
    int          xp_type;
} SVCXPRT;
```

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Figure 6-71: <rpc.h>, Part 15 of 16

```
#define svc_getrpccaller(x) (&(x)->xp_rtaddr)
#define svc_getargs(xprt, xargs, argsp) \
    (*(xprt)->xp_ops->xp_getargs)((xprt), (xargs), (argsp))
#define svc_freeargs(xprt, xargs, argsp) \
    (*(xprt)->xp_ops->xp_freeargs)((xprt), (xargs), (argsp))
#define svc_destroy(xprt) \
    (*(xprt)->xp_ops->xp_destroy)(xprt)

struct svc_req {
    u_long      rq_prog;
    u_long      rq_vers;
    u_long      rq_proc;
    struct opaque_auth rq_cred;
    caddr_t     rq_clntcred;
    SVCXPRT     *rq_xprt;
};

#define FD_SETSIZE      1024
#define NBBY 8

typedef long    fd_mask;
#define NFDBITS (sizeof(fd_mask) * NBBY)
#define howmany(x, y) (((x)+(y)-1)/(y))

typedef struct fd_set {
    fd_mask fds_bits[howmany(FD_SETSIZE, NFDBITS)];
} fd_set;

extern fd_set svc_fdset;
```

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Figure 6-72: <rpc.h>, Part 16 of 16

```
struct rpcb {
    u_long    r_prog;
    u_long    r_vers;
    char      *r_netid;
    char      *r_addr;
    char      *r_owner;
};
typedef struct rpcb RPCB;

struct rpcblist {
    RPCB      rpcb_map;
    struct rpcblist *rpcb_next;
};
```

Figure 6-73: <rtpriority.h>*

```
typedef struct rtparms {
    short rt_pri;
    ulong rt_tqsecs;
    long  rt_tqnsecs;
} rtparms_t;

typedef struct rtinfo {
    short rt_maxpri;
} rtinfo_t;

#define RT_NOCHANGE -1
#define RT_TQINF    -2
#define RT_TQDEF    -3
```

Figure 6-74: <search.h>

```
typedef enum { FIND, ENTER } ACTION;  
typedef struct entry { char *key; void *data; } ENTRY;  
typedef enum { preorder, postorder, endorder, leaf } VISIT;
```

Figure 6-75: <sys/sem.h>

```
#define SEM_UNDO    010000
#define GETNCNT     3
#define GETPID      4
#define GETVAL      5
#define GETALL      6
#define GETZCNT     7
#define SETVAL      8
#define SETALL      9

struct semid_ds {
    struct ipc_perm    sem_perm;
    struct sem         *sem_base;
    ushort             sem_nsems;
    time_t             sem_otime;
    long               sem_pad1;
    time_t             sem_ctime;
    long               sem_pad2;
    long               sem_pad3[4];
};

struct sem {
    ushort             semval;
    pid_t              sempid;
    ushort             semncnt;
    ushort             semzcnt;
};

struct sembuf {
    ushort             sem_num;
    short              sem_op;
    short              sem_flg;
};
```

Figure 6-76: <setjmp.h>

```
#define _SIGJBLEN      128
#define _JBLEN         10

typedef int jmp_buf[_JBLEN];
typedef int sigjmp_buf[_SIGJBLEN];
```

Figure 6-77: <sys/shm.h>

```
#define SHMLBA          ((1)<<12)

#define SHM_RDONLY      010000
#define SHM_RND         020000

struct shmid_ds {
    struct ipc_perm    shm_perm;
    int                shm_segsz;
    struct anon_map    *shm_amp;
    ushort             shm_lkcnt;
    pid_t              shm_lpid;
    pid_t              shm_cpid;
    ulong              shm_nattch;
    ulong              shm_cnattch;
    time_t             shm_atime;
    long               shm_pad1;
    time_t             shm_dtime;
    long               shm_pad2;
    time_t             shm_ctime;
    long               shm_pad3;
    long               shm_pad4[4];
};
```

Figure 6-78: <signal.h>, Part 1 of 3

#define SIGHUP	1
#define SIGINT	2
#define SIGQUIT	3
#define SIGILL	4
#define SIGTRAP	5
#define SIGIOT	6
#define SIGABRT	6
#define SIGEMT	7
#define SIGFPE	8
#define SIGKILL	9
#define SIGBUS	10
#define SIGSEGV	11
#define SIGSYS	12
#define SIGPIPE	13
#define SIGALRM	14
#define SIGTERM	15
#define SIGUSR1	16
#define SIGUSR2	17
#define SIGCLD	18
#define SIGCHLD	18
#define SIGPWR	19
#define SIGWINCH	20
#define SIGURG	21
#define SIGPOLL	22
#define SIGIO	22
#define SIGSTOP	23
#define SIGTSTP	24
#define SIGCONT	25
#define SIGTTIN	26
#define SIGTTOU	27
#define SIGVTALRM	28
#define SIGPROF	29

Figure 6-79: <signal.h>, Part 2 of 3

```
#define SIGXCPU          30
#define SIGXFSZ          31

#define SIG_DFL          (void(*)())0
#define SIG_ERR          (void(*)())-1
#define SIG_IGN          (void(*)())1
#define SIG_HOLD         (void(*)())2

#define SIG_BLOCK        1
#define SIG_UNBLOCK      2
#define SIG_SETMASK      3

typedef struct {
    unsigned int          sa_sigbits[4];
} sigset_t;

struct sigaction {
    int                   sa_flags;
    void                  (*sa_handler)();
    sigset_t              sa_mask;
    int                   sa_resv[2];
};

#define SA_NOCLDSTOP      0x00020000
#define SA_ONSTACK        0x00000001
#define SA_RESETHAND      0x00000002
#define SA_RESTART        0x00000004
#define SA_SIGINFO        0x00000008
#define SA_NODEFER        0x00000010
#define SA_NOCLDWAIT      0x00010000
```

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Figure 6-80: <signal.h>, Part 3 of 3

```
#define SS_ONSTACK      0x00000001
#define SS_DISABLE      0x00000002

struct sigaltstack {
    char    *ss_sp;
    int     ss_size;
    int     ss_flags;
};

typedef struct sigaltstack    stack_t;
```

Figure 6-81: <sys/signinfo.h>, Part 1 of 5

```
#define ILL_ILLOPC      1
#define ILL_ILLOPN      2
#define ILL_ILLADR      3
#define ILL_ILLTRP      4
#define ILL_PRVOPC      5
#define ILL_PRVREG      6
#define ILL_COPROC      7
#define ILL_BADSTK      8

#define FPE_INTDIV      1
#define FPE_INTOVF      2
#define FPE_FLTDIV      3
#define FPE_FLTOVF      4
#define FPE_FLTUND      5
#define FPE_FLTRES      6
#define FPE_FLTINV      7
#define FPE_FLTSUB      8
```

Figure 6-82: <sys/signinfo.h>, Part 2 of 5

```
#define SEGV_MAPERR      1
#define SEGV_ACCERR      2

#define BUS_ADRALN       1
#define BUS_ADRERR       2
#define BUS_OBJERR       3

#define TRAP_BRKPT       1
#define TRAP_TRACE       2

#define CLD_EXITED        1
#define CLD_KILLED        2
#define CLD_DUMPED        3
#define CLD_TRAPPED       4
#define CLD_STOPPED       5
#define CLD_CONTINUED     6

#define POLL_IN          1
#define POLL_OUT         2
#define POLL_MSG         3
#define POLL_ERR         4
#define POLL_PRI         5
#define POLL_HUP         6

#define SI_MAXSZ          128
#define SI_PAD            ((SI_MAXSZ / sizeof(int)) - 3)
```

Figure 6-83: <sys/signinfo.h>, Part 3 of 5

```
typedef struct siginfo {
    int    si_signo;
    int    si_code;
    int    si_errno;
    union {
        int    _pad[SI_PAD];
        struct {
            pid_t _pid;
            union {
                struct {
                    uid_t _uid;
                } _kill;
                struct {
                    clock_t _utime;
                    int _status;
                    clock_t _stime;
                } _cld;
            } _pdata;
        } _proc;
        struct {
            caddr_t _addr;
        } _fault;
        struct {
            int _fd;
            long _band;
        } _file;
    } _data;
} siginfo_t;
```

Figure 6-84: <sys/siginfo.h>, Part 4 of 5

```
#define si_pid      _data._proc._pid
#define si_status   _data._proc._pdata._cld._status
#define si_stime    _data._proc._pdata._cld._stime
#define si_utime    _data._proc._pdata._cld._utime
#define si_uid      _data._proc._pdata._kill._uid
#define si_addr     _data._fault._addr
#define si_fd       _data._file._fd
#define si_band     _data._file._band
```

Figure 6-85: <sys/siginfo.h>*, Part 5 of 5

```
union sigval {
    int      sival_int;
    void     *sival_ptr;
};

union notifyinfo {
    int      nisigno;
    void     (*nifunc)(union sigval);
};

struct sigevent {
    int      sigev_notify;
    union notifyinfo sigev_notifyinfo;
    union sigval sigev_value;
};

#define SIGEV_NONE      1
#define SIGEV_SIGNAL    2
#define SIGEV_CALLBACK  3
```

Figure 6-86: <sys/stat.h>, Part 1 of 2

```
#define _ST_FSTYPsz      16

struct stat {
    dev_t      st_dev;
    long       st_pad1[3];
    ino_t      st_ino;
    mode_t     st_mode;
    nlink_t    st_nlink;
    uid_t      st_uid;
    gid_t      st_gid;
    dev_t      st_rdev;
    long       st_pad2[2];
    off_t      st_size;
    long       st_pad3;
    time_t     st_atim;
    time_t     st_mtim;
    time_t     st_ctim;
    long       st_blksize;
    long       st_blocks;
    char       st_fstype[_ST_FSTYPsz];
    long       st_pad4[8];
};
#define st_atime    st_atim.tv_sec
#define st_mtime    st_mtim.tv_sec
#define st_ctime    st_ctim.tv_sec
```

Figure 6-87: <sys/stat.h>, Part 2 of 2

```
#define S_IFMT          0xF000
#define S_IFIFO         0x1000
#define S_IFCHR         0x2000
#define S_IFDIR         0x4000
#define S_IFBLK         0x6000
#define S_IFREG         0x8000
#define S_IFLNK         0xA000

#define S_ISUID         0x800
#define S_ISGID         0x400
#define S_ISVTX         0x200

#define S_IRWXU         00700
#define S_IRUSR         00400
#define S_IWUSR         00200
#define S_IXUSR         00100
#define S_IRWXG         00070
#define S_IRGRP         00040
#define S_IWGRP         00020
#define S_IXGRP         00010
#define S_IRWXO         00007
#define S_IROTH         00004
#define S_IWOTH         00002
#define S_IXOTH         00001

#define S_ISFIFO(mode)  ((mode&0xF000) == 0x1000)
#define S_ISCHR(mode)   ((mode&0xF000) == 0x2000)
#define S_ISDIR(mode)   ((mode&0xF000) == 0x4000)
#define S_ISBLK(mode)   ((mode&0xF000) == 0x6000)
#define S_ISREG(mode)   ((mode&0xF000) == 0x8000)
```

Figure 6-88: <sys/statvfs.h>

```
#define FSTYP SZ      16

typedef struct statvfs {
    u_long    f_bsize;
    u_long    f_frsize;
    u_long    f_blocks;
    u_long    f_bfree;
    u_long    f_bavail;
    u_long    f_files;
    u_long    f_ffree;
    u_long    f_favail;
    u_long    f_fsid;
    char      f_basetype[FSTYP SZ];
    u_long    f_flag;
    u_long    f_namemax;
    char      f_fstr[32];
    u_long    f_filler[16];
} statvfs_t;

#define ST_RDONLY    0x01
#define ST_NOSUID    0x02
#define ST_NOTRUNC    0x04
```

Figure 6-89: <stdarg.h>

```
typedef void          *va_list;
extern void           va_end(va_list);
#define va_start(list, name) (void) (list = (void *)((char *)&...))
#define va_arg(list, mode)  ((mode *)((list = (char *)list + \
                                   sizeof(mode)))[-1])
#define va_end(list)       (void)0
```

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NOTE

The construction &... is a syntactic extension to ANSI C and may not be supported by all C compilers. The intended semantics are to set list to the address on the stack of the first incoming argument in the variable part of the argument list. See "Function Calling Sequence" in Chapter 3.

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Figure 6-90: <stddef.h>

```
typedef int           ptrdiff_t;
typedef unsigned int  size_t;
#define NULL          0
typedef long          wchar_t;

#define offsetof(s, m) (size_t)((s *)0->m)
```

Figure 6-91: <stdio.h>, Part 1 of 2

```
typedef unsigned int    size_t;
typedef long            fpos_t;

#define NULL            0

#define BUFSIZ          1024

#define _IOFBF          0000
#define _IOLBF          0100
#define _IONBF          0004
#define _IOEOF          0020
#define _IOERR          0040

#define EOF              (-1)

#define FOPEN_MAX        60
#define FILENAME_MAX     1024

#define L_ctermid         9
#define L_cuserid         9
#define P_tmpdir          "/var/tmp/"
#define L_tmpnam          25

#define stdin            (&__iob[0])
#define stdout            (&__iob[1])
#define stderr            (&__iob[2])
```


Figure 6-92: , Part 2 of 2

```
typedef struct {
    int _cnt;
    unsigned char *_ptr;
    unsigned char *_base;
    unsigned char _flag;
    unsigned char _file; ++
} FILE;

extern FILE __iob[];

#define clearerr(p) ((void)((p)->_flag &= ~(_IOERR | _IOEOF))) +
#define feof(p) ((p)->_flag & _IOEOF)
#define ferror(p) ((p)->_flag & _IOERR)
#define fileno(p) (p)->_file +

/* Non reentrant */
#define getc_unlocked(p) (--(p)->_cnt < 0 ? _ _filbuf(p)
    : (int)*(p)->_ptr++)
#define putc_unlocked(x, p) (--(p)->_cnt < 0 ? _ _flsbuf(x, p)
    : (int)*(p)->_ptr++ = (x)))
#define getchar_unlocked() getc_unlocked(stdin)
#define putchar_unlocked(x) putc_unlocked((x), stdout)

#define getc(p) (--(p)->_cnt < 0 ? _ _filbuf(p)
    : (int)*(p)->_ptr++)
#define putc(x, p) (--(p)->_cnt < 0 ? _ _flsbuf(x, p)
    : (int)*(p)->_ptr++ = (x)))
#define getchar() getc(stdin)
#define putchar(x) putc((x), stdout)

/* Reentrant versions available as functions only */
```

† These macro definitions are moved to Level 2 as of January 1, 1993.

†† The `_file` member of the `FILE` struct is moved to Level 2 as of January 1, 1993.

NOTE

The macros `clearerr`, and `fileno` will be removed as a source interface in a future release supporting multi-processing. Applications should transition to the function equivalents of these macros in `libc`. Binary portability will be supported for existing applications.

CAUTION

The constant `_NFILE` has been removed. It should still appear in `stdio.h`, but may be removed in a future version of the header file. Applications may not be able to depend on `fopen()` failing on an attempt to open more than `_NFILE` files.

Figure 6-93: <stdlib.h>

```
typedef struct {
    int    quot;
    int    rem;
} div_t;

typedef struct {
    long   quot;
    long   rem;
} ldiv_t;

typedef unsigned int    size_t;

#define NULL            0

#define EXIT_FAILURE    1
#define EXIT_SUCCESS    0
#define RAND_MAX        32767

extern unsigned char    __ctype[];
#define MB_CUR_MAX      (int) __ctype[520]
```

Figure 6-94: <stropts.h>, Part 1 of 6

#define SNDZERO	0x001
#define SNDPIPE	0x002
#define RNORM	0x000
#define RMSGD	0x001
#define RMSGN	0x002
#define RMODEMASK	0x003
#define RPROTDAT	0x004
#define RPROTDIS	0x008
#define RPROTNORM	0x010
#define FLUSHR	0x01
#define FLUSHW	0x02
#define FLUSHRW	0x03
#define FLUSHBAND	0x04

Figure 6-95: <stropts.h>, Part 2 of 6

```
#define S_INPUT          0x0001
#define S_HIPRI          0x0002
#define S_OUTPUT         0x0004
#define S_MSG            0x0008
#define S_ERROR          0x0010
#define S_HANGUP         0x0020
#define S_RDNORM         0x0040
#define S_WRNORM         S_OUTPUT
#define S_RDBAND         0x0080
#define S_WRBAND         0x0100
#define S_BANDURG        0x0200

#define RS_HIPRI          0x01

#define MSG_HIPRI         0x01
#define MSG_ANY           0x02
#define MSG_BAND          0x04

#define MORECTL           1
#define MOREDATA          2

#define MUXID_ALL         (-1)

#define ANYMARK           0x01
#define LASTMARK          0x02
```

Figure 6-96: <stropts.h>, Part 3 of 6

```
#define STR                ('S'<<8)
#define I_NREAD            (STR|01)
#define I_PUSH             (STR|02)
#define I_POP              (STR|03)
#define I_LOOK             (STR|04)
#define I_FLUSH            (STR|05)
#define I_SRDOPT           (STR|06)
#define I_GRDOPT           (STR|07)
#define I_STR              (STR|010)
#define I_SETSIG           (STR|011)
#define I_GETSIG           (STR|012)
#define I_FIND             (STR|013)
#define I_LINK             (STR|014)
#define I_UNLINK           (STR|015)
#define I_PEEK             (STR|017)
#define I_FDINSERT         (STR|020)
#define I_SENDFD           (STR|021)
#define I_RECVFD           (STR|016)
#define I_SWROPT           (STR|023)
#define I_GWROPT           (STR|024)
#define I_LIST             (STR|025)
#define I_PLINK            (STR|026)
#define I_PUNLINK          (STR|027)
```

Figure 6-97: <stropts.h>, Part 4 of 6

```
#define I_FLUSHBAND      (STR|034)
#define I_CKBAND        (STR|035)
#define I_GETBAND       (STR|036)
#define I_ATMARK        (STR|037)
#define I_SETCLTIME     (STR|040)
#define I_GETCLTIME     (STR|041)
#define I_CANPUT        (STR|042)
```

Figure 6-98: <stropts.h>, Part 5 of 6

```
struct strioctl {
    int    ic_cmd;
    int    ic_timeout;
    int    ic_len;
    char   *ic_dp;
};

struct strbuf {
    int    maxlen;
    int    len;
    char   *buf;
};

struct strpeek {
    struct strbuf    ctlbuf;
    struct strbuf    databuf;
    long             flags;
};

struct strfdinsert {
    struct strbuf    ctlbuf;
    struct strbuf    databuf;
    long             flags;
    int              fildes;
    int              offset;
};
```

Figure 6-99: <stropts.h>, Part 6 of 6

```
struct strrecvfd {
    int    fd;
    uid_t  uid;
    gid_t  gid;
    char   fill[8];
};

#define FMNAMESZ    8

struct str_mlist {
    char   l_name[FMNAMESZ+1];
};

struct str_list {
    int                sl_nmods;
    struct str_mlist  *sl_modlist;
};

struct bandinfo {
    unsigned char      bi_pri;
    int                bi_flag;
};
```


Figure 6-100: <synch.h>*, Part 1 of 3

```
#define USYNC_THREAD    0
#define USYNC_PROCESS   1

typedef struct thrq_elt thrq_elt_t;

struct thrq_elt {
    thrq_elt_t *thrq_next;
    thrq_elt_t *thrq_prev;
};

typedef volatile struct {
    lwp_mutex_t  m_lmutex;
    long         m_type;
    lwp_mutex_t  m_sync_lock;
    thrq_elt_t   m_sleepq;
    long         filler;
} mutex_t;

typedef volatile struct {
    lwp_cond_t    c_lcond;
    long          c_type;
    thrq_elt_t    *c_syncq;
    lwp_mutex_t   c_sync_lock;
} cond_t;
```

Figure 6-101: <synch.h>*, Part 2 of 3

```
typedef volatile struct {
    mutex_t      s_mutex;
    cond_t       s_cond;
    short        s_count;
    short        s_wakecnt;
    int          s_type;
} sema_t;

typedef volatile struct rwcvt rwcvt_t;

struct rwcvt {
    cond_t rwcvt_cond;
    rwcvt_t *rwcvt_next;
    char    rwcvt_rw;
    char    rwcvt_wakeup;
    short   rwcvt_readerwanted;
} ;

typedef volatile struct rwlock rwlock_t;

struct rwlock {
    mutex_t      rw_mutex;
    lwp_cond_t   rw_lwpcond;
    int          rw_type;
    short        rw_readers;
    char         rw_writer;
    char         rw_wrwakeup;
    short        rw_writerwanted;
    short        rw_rdwakecnt;
    rwcvt_t      *rw_cvqhead;
    rwcvt_t      *rw_cvqtail;
    long         pad[4];
} ;
```

Figure 6-102: <synch.h>*, Part 3 of 3

```
typedef volatile struct {
    mutex_t      rm_mutex;
    pid_t        rm_pid;
    thread_t      rm_owner;
    int          rm_depth;
    long         filler;
} rmutex_t;

typedef volatile struct barrier barrier_t;

struct barrier {
    mutex_t      b_lock;
    int          b_type;
    unsigned int b_count;
    unsigned int b_waiting;
    unsigned int b_generation;
    cond_t       b_cond;
} ;
```

Figure 6-103: <sys/sysi86.h>

```
#define SI86FPHW      40

#define FP_NO         0
#define FP_SW         1
#define FP_HW         2
#define FP_287        2
#define FP_387        3
```

Figure 6-104: <termios.h>, Part 1 of 10

```
#define _POSIX_VDISABLE    0

#define CTRL(c)            ((c)&037)
#define IBSHIFT            16
#define NCC                8
#define NCCS               19

typedef unsigned long      tcflag_t;
typedef unsigned char      cc_t;
typedef unsigned long      speed_t;

struct termios {
    tcflag_t    c_iflag;
    tcflag_t    c_oflag;
    tcflag_t    c_cflag;
    tcflag_t    c_lflag;
    cc_t        c_cc[NCCS];
};
```

Figure 6-105: <termios.h>, Part 2 of 10

#define VINTR	0
#define VQUIT	1
#define VERASE	2
#define VKILL	3
#define VEOF	4
#define VEOL	5
#define VEOL2	6
#define VMIN	4
#define VTIME	5
#define VSWTCH	7
#define VSTART	8
#define VSTOP	9
#define VSUSP	10
#define VDSUSP	11
#define VREPRINT	12
#define VDISCARD	13
#define VWERASE	14
#define VLNEXT	15

Figure 6-106: <termios.h>, Part 3 of 10

```
#define CNUL          0
#define CDEL          0177
#define CESC          '\\\'
#define CINTR         0177
#define CQUIT         034
#define CERASE        '#'
#define CKILL         '@'
#define CEOT          04
#define CEOL          0
#define CEOL2         0
#define CEOF          04
#define CSTART        021
#define CSTOP         023
#define CSWTCH        032
#define CNSWTCH       0
#define CSUSP         CTRL('z')
#define CDSUSP        CTRL('y')
#define CRPRNT        CTRL('r')
#define CFLUSH        CTRL('o')
#define CWERASE        CTRL('w')
#define CLNEXT        CTRL('v')
```

Figure 6-107: <termios.h>, Part 4 of 10

#define IGNBRK	00000001
#define BRKINT	00000002
#define IGNPAR	00000004
#define PARMRK	00000010
#define INPCK	00000020
#define ISTRIP	00000040
#define INLCR	00001000
#define IGNCR	00002000
#define ICRNL	00004000
#define IUCLC	00010000
#define IXON	00020000
#define IXANY	00040000
#define IXOFF	00100000
#define IMAXBEL	00200000
#define DOSMODE	01000000

Figure 6-108: <termios.h>, Part 5 of 10

#define OPOST	0000001
#define OLCUC	0000002
#define ONLCR	0000004
#define OCRNL	0000010
#define ONOCR	0000020
#define ONLRET	0000040
#define OFILL	0000100
#define OFDEL	0000200
#define NLDLY	0000400
#define NL0	0
#define NL1	0000400
#define CRDLY	0003000
#define CR0	0
#define CR1	0001000
#define CR2	0002000
#define CR3	0003000
#define TABDLY	0014000

Figure 6-109: <termios.h>, Part 6 of 10

#define TAB0	0
#define TAB1	0004000
#define TAB2	0010000
#define TAB3	0014000
#define XTABS	0014000
#define BSDLY	0020000
#define BS0	0
#define BS1	0020000
#define VTDLY	0040000
#define VT0	0
#define VT1	0040000
#define FFDLY	0100000
#define FF0	0
#define FF1	0100000
#define PAGEOUT	0200000
#define WRAP	0400000

Figure 6-110: <termios.h>, Part 7 of 10

#define CBAUD	0000017
#define CSIZE	0000060
#define CS5	0
#define CS6	0000020
#define CS7	0000040
#define CS8	0000060
#define CSTOPB	0000100
#define CREAD	0000200
#define PARENB	0000400
#define PARODD	0001000
#define HUPCL	0002000
#define CLOCAL	0004000
#define RCV1EN	0010000
#define XMT1EN	0020000
#define LOBLK	0040000
#define XCLUDE	0100000
#define CIBAUD	03600000
#define PAREXT	04000000

Figure 6-111: <termios.h>, **Part 8 of 10**

#define ISIG	0000001
#define ICANON	0000002
#define XCASE	0000004
#define ECHO	0000010
#define ECHOE	0000020
#define ECHOK	0000040
#define ECHONL	0000100
#define NOFLSH	0000200
#define TOSTOP	0000400
#define ECHOCTL	0001000
#define ECHOPRT	0002000
#define ECHOKE	0004000
#define DEFECHO	0010000
#define FLUSHO	0020000
#define PENDIN	0040000
#define IEXTEN	0100000

Figure 6-112: <termios.h>, Part 9 of 10

```
#define TIOC          ('T'<<8)

#define TCGETA        (TIOC|1)
#define TCSETA        (TIOC|2)
#define TCSETAW       (TIOC|3)
#define TCSETAF       (TIOC|4)
#define TCSBRK        (TIOC|5)
#define TCXONC        (TIOC|6)
#define TCFLSH        (TIOC|7)

#define TIOCGWINSZ    (TIOC|104)
#define TIOCSWINSZ    (TIOC|103)

#define TCGETS        (TIOC|13)
#define TCSETS        (TIOC|14)
#define TCSANOW       (('T'<<8)|14)
#define TCSETSW       (TIOC|15)
#define TCSADRAIN     (('T'<<8)|15)
#define TCSETSF       (TIOC|16)
#define TCSAFLUSH     (('T'<<8)|16)
```

Figure 6-113: <termios.h>, Part 10 of 10

```
#define TCIFLUSH      0
#define TCOFLUSH     1
#define TCIOFLUSH    2

#define TC00FF       0
#define TC00N        1
#define TCIOFF        2
#define TCION         3

#define B0            0
#define B50           1
#define B75           2
#define B110          3
#define B134          4
#define B150          5
#define B200          6
#define B300          7
#define B600          8
#define B1200         9
#define B1800        10
#define B2400        11
#define B4800        12
#define B9600        13
#define B19200       14
#define B38400       15

struct winsize {
    unsigned short    ws_row;
    unsigned short    ws_col;
    unsigned short    ws_xpixel;
    unsigned short    ws_ypixel;
};
```

Figure 6-114: <thread.h>*, Part 1 of 2

```
#define THR_SUSPENDED 0x1
#define THR_BOUND     0x2
#define THR_INCR_CONC 0x4
#define THR_DETACHED  0x8
#define THR_DAEMON     0x10
#define SCHED_TS       1
#define SCHED_OTHER    1
#define SCHED_FIFO     2
#define SCHED_RR       3

typedef      id_t  thread_t;

#define POLICY_PARAM_SZ      PC_CLPARMSZ
```

Figure 6-115: <thread.h>*, Part 2 of 2

```
typedef struct {
    id_t    policy;
    long    policy_params[POLICY_PARAM_SZ];
} sched_param_t;

struct ts_param {
    int     prio;
};

struct fifo_param {
    int     prio;
};

struct rr_param {
    int     prio;
};

typedef unsigned int thread_key_t;
```

Figure 6-116: <sys/ticlts.h>

```
#define TCL_BADADDR      1
#define TCL_BADOPT       2
#define TCL_NOPEER       3
#define TCL_PEERBADSTATE 4
#define TCL_DEFAULTADDRSZ 4
```

Figure 6-117: <sys/ticots.h>

```
#define TCO_NOPEER          ECONNREFUSED
#define TCO_PEERNOROOMONQ   ECONNREFUSED
#define TCO_PEERBADSTATE    ECONNREFUSED
#define TCO_PEERINITIATED   ECONNRESET
#define TCO_PROVIDERINITIATED ECONNABORTED
#define TCO_DEFAULTADDRSZ   4
```

Figure 6-118: <sys/ticotsord.h>

```
#define TC00_NOPEER          1
#define TC00_PEERNOROOMONQ   2
#define TC00_PEERBADSTATE    3
#define TC00_PEERINITIATED   4
#define TC00_PROVIDERINITIATED 5
#define TC00_DEFAULTADDRSZ   4
```

NOTE

The sys/tihdr.h and sys/timod.h headers previously included in this document were unnecessary as they did not contain user level information and have therefore been removed from this document.

Figure 6-119: <time.h>*

```
struct timespec {  
    time_t      tv_sec;  
    long        tv_nsec;  
} ;
```

Figure 6-120: <sys/time.h>

```
typedef long clock_t;
typedef long time_t;
typedef unsigned int size_t;

typedef struct timespec {
    time_t      tv_sec;
    long        tv_nsec;
} timestruc_t;

#define CLOCKS_PER_SEC      1000000

struct tm {
    int    tm_sec;
    int    tm_min;
    int    tm_hour;
    int    tm_mday;
    int    tm_mon;
    int    tm_year;
    int    tm_wday;
    int    tm_yday;
    int    tm_isdst;
};

extern char *tzname[2];

#define CLK_TCK    _sysconf(3)

extern long timezone;
extern int daylight;
```

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Figure 6-121: <sys/times.h>

```
struct tms {  
    clock_t    tms_utime;  
    clock_t    tms_stime;  
    clock_t    tms_cutime;  
    clock_t    tms_cstime;  
};
```

NOTE

This edition introduces the xti.h header which contains the same information as the current tiuser.h. The new xti.h header is a superset of the previous edition's tiuser.h.

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tiuser.h has been moved to Level 2 and will be removed in future editions of the ABI. In the future xti.h should be used as a replacement for tiuser.h.

Figure 6-122: <tiuser.h>, Error Return Values

#define TBADADDR	1	
#define TBADOPT	2	
#define TACCES	3	
#define TBADF	4	
#define TNOADDR	5	
#define TOUTSTATE	6	
#define TBADSEQ	7	
#define TSYSERR	8	
#define TLOOK	9	
#define TBADDATA	10	
#define TBUFOVFLW	11	
#define TFLOW	12	
#define TNODATA	13	
#define TNODIS	14	
#define TNOUDERR	15	
#define TBADFLAG	16	
#define TNOREL	17	
#define TNOTSUPPORT	18	
#define TSTATECHNG	19	
#define TNOSTRUCTYPE	20	M
#define TBADNAME	21	M
#define TBADQLEN	22	M
#define TADDRBUSY	23	M
#define TINDOUT	24	M
#define TPROVMISMATCH	25	M
#define TRESQLEN	26	M
#define TRESADDR	27	M
#define TQFULL	28	M
#define TPROTO	29	M

Figure 6-123: <tiuser.h>, Event Bitmasks

#define T_LISTEN	0x0001	
#define T_CONNECT	0x0002	
#define T_DATA	0x0004	
#define T_EXDATA	0x0008	
#define T_DISCONNECT	0x0010	
#define T_ERROR	0x0020	
#define T_UDERR	0x0040	
#define T_ORDREL	0x0080	
#define T_GODATA	0x0100	M
#define T_GOEXDATA	0x0200	M
#define T_EVENTS	0x03ff	M

Figure 6-124: <tiuser.h>, Flags

#define T_MORE	0x001
#define T_EXPEDITED	0x002
#define T_NEGOTIATE	0x004
#define T_CHECK	0x008
#define T_DEFAULT	0x010
#define T_SUCCESS	0x020
#define T_FAILURE	0x040

Figure 6-125: <tiuser.h>, Service Types

#define T_COTS	01
#define T_COTS_ORD	02
#define T_CLTS	03

Figure 6-126: <tiuser.h>, Values for flags field in t_info structure

#define T_SENDZERO	0x0000001
--------------------	-----------

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Figure 6-127: <tiuser.h>, Transport Interface Data Structures, 1 of 2

```
struct t_info {
    long    addr;
    long    options;
    long    tsdu;
    long    etsdu;
    long    connect;
    long    discon;
    long    servtype;
    long    flags;
};

struct netbuf {
    unsigned int    maxlen;
    unsigned int    len;
    char            *buf;
};

struct t_bind {
    struct netbuf    addr;
    unsigned         qlen;
};

struct t_optmgmt {
    struct netbuf    opt;
    long             flags;
};
```

M

NOTE

Applications invoking TLI binary interfaces to `t_open` or `t_getinfo` will see the `t_info` structure without the `flags` member. Those applications invoking the XTI versions of `t_open` or `t_getinfo` will see the `t_info` structure with the `flags` member.

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Figure 6-128: <tiuser.h>, Transport Interface Data Structures, 2 of 2

```
struct t_discon {
    struct netbuf    udata;
    int              reason;
    int              sequence;
};

struct t_call {
    struct netbuf    addr;
    struct netbuf    opt;
    struct netbuf    udata;
    int              sequence;
};

struct t_unitdata {
    struct netbuf    addr;
    struct netbuf    opt;
    struct netbuf    udata;
};

struct t_uderr {
    struct netbuf    addr;
    struct netbuf    opt;
    long             error;
};
```


Figure 6-129: <tiuser.h>, Structure Types

#define T_BIND	1
#define T_OPTMGMT	2
#define T_CALL	3
#define T_DIS	4
#define T_UNITDATA	5
#define T_UDERROR	6
#define T_INFO	7

Figure 6-130: <tiuser.h>, Fields of Structures

#define T_ADDR	0x01
#define T_OPT	0x02
#define T_UDATA	0x04
#define T_ALL	0xffff

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NOTE

Differences between XTI and TLI have forced the value of T_ALL to change. The previous edition's T_ALL value will not produce the same results as the new T_ALL.

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Figure 6-131: <tiuser.h>, Transport Interface States

#define T_UNINIT	0
#define T_UNBND	1
#define T_IDLE	2
#define T_OUTCON	3
#define T_INCON	4
#define T_DATAXFER	5
#define T_OUTREL	6
#define T_INREL	7
#define T_FAKE	8
#define T_NOSTATES	9

Figure 6-132: <tiuser.h>, User-level Events

#define T_OPEN	0
#define T_BIND	1
#define T_OPTMGMT	2
#define T_UNBIND	3
#define T_CLOSE	4
#define T_SNDUDATA	5
#define T_RCVUDATA	6
#define T_RCVUDERR	7
#define T_CONNECT1	8
#define T_CONNECT2	9
#define T_RCVCONNECT	10
#define T_LISTN	11
#define T_ACCEPT1	12
#define T_ACCEPT2	13
#define T_ACCEPT3	14
#define T_SND	15
#define T_RCV	16
#define T_SNDDIS1	17
#define T_SNDDIS2	18
#define T_RCVDIS1	19
#define T_RCVDIS2	20
#define T_RCVDIS3	21
#define T_SNDREL	22
#define T_RCVREL	23
#define T_PASSCON	24
#define T_NOEVENTS	25

Figure 6-133: <tsprioctl.h>*

```
typedef struct tsparms {
    short ts_uprilm;
    short ts_upri;
} tsparms_t;

typedef struct tsinfo {
    short ts_maxupri;
} tsinfo_t;

#define      TS_NOCHANGE    -32768
```

Figure 6-134: <sys/types.h>

```
typedef unsigned char    uchar_t;
typedef unsigned short   ushort_t;
typedef unsigned int     uint_t;
typedef unsigned long    ulong_t;

typedef char *           caddr_t;
typedef long             daddr_t;
typedef long             off_t;
typedef long             id_t;
typedef int              key_t;
typedef ulong_t          mode_t;
typedef long             uid_t;
typedef uid_t            gid_t;
typedef ulong_t          nlink_t;
typedef ulong_t          dev_t;
typedef ulong_t          ino_t;
typedef long             pid_t;
typedef uint_t           size_t;
typedef long             time_t;
typedef long             clock_t;

typedef unsigned short   ushort;
typedef unsigned long    ulong;

typedef unsigned char    u_char;
typedef unsigned short   u_short;
typedef unsigned int     u_int;
typedef unsigned long    u_long;
```

Figure 6-135: <ucontext.h>, Part 1 of 2

```
typedef int      greg_t;
#define NGREG    19
typedef greg_t   gregset_t[NGREG];

#define SS       18
#define UESP     17
#define EFL      16
#define CS       15
#define EIP      14
#define ERR      13
#define TRAPNO   12
#define EAX      11
#define ECX      10
#define EDX      9
#define EBX      8
#define ESP      7
#define EBP      6
#define ESI      5
#define EDI      4
#define DS       3
#define ES       2
#define FS       1
#define GS       0
```

Figure 6-136: <ucontext.h>, Part 2 of 2

```
typedef struct fpregset {
    union {
        struct fpchip_state {
            int    state[27];
            int    status;
        } fpchip_state;
        struct fp_emul_space {
            char    fp_emul[246];
            char    fp_epad[2];
        } fp_emul_space;
        int    f_fpregs[62];
    } fp_reg_set;
    long    f_wregs[33];
} fpregset_t;

typedef struct {
    gregset_t    gregs;
    fpregset_t    fpregs;
} mcontext_t;

typedef struct ucontext {
    u_long        uc_flags;
    struct ucontext    *uc_link;
    sigset_t        uc_sigmask;
    stack_t        uc_stack;
    mcontext_t        uc_mcontext;
    long        uc_filler[5];
} ucontext_t;
```

Figure 6-137: <sys/uio.h>

```
typedef struct iovec {
    caddr_t    iov_base;
    int        iov_len;
} iovec_t;
```

Figure 6-138: <ulimit.h>

```
#define UL_GETFSIZE    1
#define UL_SETFSIZE    2
```

Figure 6-139: <unistd.h>, Part 1 of 2

```
#define R_OK            4
#define W_OK            2
#define X_OK            1
#define F_OK            0

#define F_ULOCK          0
#define F_LOCK          1
#define F_TLOCK         2
#define F_TEST          3

#define SEEK_SET         0
#define SEEK_CUR         1
#define SEEK_END         2

#define _SC_ARG_MAX      1
#define _SC_CHILD_MAX    2
#define _SC_CLK_TCK      3
```


Figure 6-139: <unistd.h>, Part 1 of 2 (continued)

#define	_SC_NGROUPS_MAX	4	
#define	_SC_OPEN_MAX	5	
#define	_SC_JOB_CONTROL	6	
#define	_SC_SAVED_IDS	7	
#define	_SC_VERSION	8	
#define	_SC_PASS_MAX	9	
#define	_SC_LOGNAME_MAX	10	
#define	_SC_PAGESIZE	11	
#define	_SC_XOPEN_VERSION	12	
#define	_CS_PATH	1	M
#define	_CS_HOSTNAME	2	M
#define	_CS_RELEASE	3	M
#define	_CS_VERSION	4	M
#define	_CS_MACHINE	5	M
#define	_CS_ARCHITECTURE	6	M
#define	_CS_HW_SERIAL	7	M
#define	_CS_HW_PROVIDER	8	M
#define	_CS_SRPC_DOMAIN	9	M
#define	_CS_SYSNAME	11	M

Figure 6-140: <unistd.h>, Part 2 of 2

```
#define _PC_LINK_MAX          1
#define _PC_MAX_CANON        2
#define _PC_MAX_INPUT        3
#define _PC_NAME_MAX         4
#define _PC_PATH_MAX         5
#define _PC_PIPE_BUF         6
#define _PC_NO_TRUNC         7
#define _PC_VDISABLE         8
#define _PC_CHOWN_RESTRICTED 9

#define _POSIX_JOB_CONTROL    1
#define _POSIX_SAVED_IDS     1
#define _POSIX_VDISABLE      0

#define _POSIX_VERSION        *
#define _XOPEN_VERSION        *

/* starred values vary and should be
   retrieved using sysconf() or pathconf() */

#define STDIN_FILENO          0
#define STDOUT_FILENO         1
#define STDERR_FILENO         2
```

Figure 6-141: <utime.h>

```
struct utimbuf {
    time_t      actime;
    time_t      modtime;
};
```

Figure 6-142: <sys/utsname.h>

```
#define SYS_NMLN    257

struct utsname {
    char    sysname[SYS_NMLN];
    char    nodename[SYS_NMLN];
    char    release[SYS_NMLN];
    char    version[SYS_NMLN];
    char    machine[SYS_NMLN];
};
```

Figure 6-143: <wait.h>

```
#define WEXITED          0001
#define WTRAPPED         0002
#define WSTOPPED         0004
#define WCONTINUED       0010
#define WUNTRACED        0004
#define WNOHANG          0100
#define WNOWAIT          0200

#define WCONTFLG         0177777
#define WCOREFLG         0200

#define WWORD(stat)      ((int)((stat)&0177777))

#define WSTOPFLG         0177
#define WSIGMASK         0177
#define WLOBYTE(stat)    ((int)((stat)&0377))
#define WHIBYTE(stat)    ((int)(((stat)>>8)&0377))

#define WIFEXITED(stat)  (WLOBYTE(stat)==0)
#define WIFSIGNALED(stat) (WLOBYTE(stat)>0&&WHIBYTE(stat)==0)
#define WIFSTOPPED(stat) (WLOBYTE(stat)==WSTOPFLG&&WHIBYTE(stat)!=0)

#define WIFCONTINUED(stat) (WWORD(stat)==WCONTFLG)

#define WEXITSTATUS(stat) WHIBYTE(stat)
#define WTERMSIG(stat)    (WLOBYTE(stat)&WSIGMASK)
#define WSTOPSIG(stat)    WHIBYTE(stat)

#define WCOREDUMP(stat)  ((stat)&WCOREFLG)
```

Figure 6-144: <wchar.h>

```
typedef long wchar_t;
typedef unsigned int    size_t;
typedef long wint_t;

typedef struct
{
    wchar_t    ;
    wchar_t    ;
} mbstate_t;

#define NULL 0
#define WEOF (-1)

#define WCHAR_MAX  2147483647
#define WCHAR_MIN  (-2147483647-1)

#define mbrlen(x, n, p)  mbrtowc((wchar_t *)0, x, n, p)
```

Figure 6-145: <wctype.h>*, Part 1 of 3

```
typedef long      wint_t;
typedef unsigned long  wctype_t;

#define WEOF (-1)

#define _U      01
#define _L      02
#define _N      04
#define _S      010
#define _P      020
#define _C      040
#define _B      0100
#define _X      0200

#define _E1      0x00000100
#define _E2      0x00000200
#define _E3      0x00000400
#define _E4      0x00000800
#define _E5      0x00001000
#define _E6      0x00002000
#define _E7      0x00004000
#define _E8      0x00008000
#define _E9      0x00010000
#define _E10     0x00020000
#define _E11     0x00040000
#define _E12     0x00080000
#define _E13     0x00100000
#define _E14     0x00200000
#define _E15     0x00400000
#define _E16     0x00800000
#define _E17     0x01000000
#define _E18     0x02000000
#define _E19     0x04000000
#define _E20     0x08000000
#define _E21     0x10000000
```

Figure 6-146: <wctype.h>*, Part 2 of 3

```
#define _PD_ALNUM    (_U | _L | _N)
#define _PD_ALPHA    (_U | _L)
#define _PD_BLANK    (_B)
#define _PD_CNTRL    (_C)
#define _PD_DIGIT    (_N)
#define _PD_GRAPH    (_P | _U | _L | _N | _E1 |
                     _E2 | _E5 | _E6)
#define _PD_LOWER    (_L)
#define _PD_PRINT    (_P | _U | _L | _N | _B |
                     _E1 | _E2 | _E5 | _E6)
#define _PD_PUNCT    (_P)
#define _PD_SPACE    (_S)
#define _PD_UPPER    (_U)
#define _PD_XDIGIT   (_X)

#define iswalnum(c)    __isw(c, _PD_ALNUM)
#define iswalpha(c)    __isw(c, _PD_ALPHA)
#define iswcntrl(c)    __isw(c, _PD_CNTRL)
#define iswdigit(c)    __isw(c, _PD_DIGIT)
#define iswgraph(c)    __isw(c, _PD_GRAPH)
#define iswlower(c)    __isw(c, _PD_LOWER)
#define iswprint(c)    __isw(c, _PD_PRINT)
#define iswpunct(c)    __isw(c, _PD_PUNCT)
#define iswspace(c)    __isw(c, _PD_SPACE)
#define iswupper(c)    __isw(c, _PD_UPPER)
#define iswxdigit(c)    __isw(c, _PD_XDIGIT)
#define tolower(c)      __tow(c, _PD_UPPER)
#define toupper(c)      __tow(c, _PD_LOWER)
#define isphonogram(c)  __isx(c, _E1)
#define isideogram(c)   __isx(c, _E2)
#define isenglish(c)    __isx(c, _E3)
#define isnumber(c)     __isx(c, _E4)
#define isspecial(c)    __isx(c, _E5)
```

Figure 6-147: <wctype.h>*, Part 3 of 3

```
#define iscodeset0(c)    (((c) & ~(wchar_t)0xff) == 0)
#define iscodeset1(c)    (((c) >> 28) == 0x3)
#define iscodeset2(c)    (((c) >> 28) == 0x1)
#define iscodeset3(c)    (((c) >> 28) == 0x2)

inline int __isw(wint_t c, wctype_t t){
    if (c > 255)
        return (__iswctype(c, t));
    return (1 + __ctype)[c] & t ;
}

inline int __isx(wint_t c, wctype_t t){
    return (c > 255 && __iswctype(c, t));
}

inline wint_t __tow(wint_t c, wctype_t t){
    if (c > 255)
        return (__trwctype(c, t));
    if ((1 + __ctype)[c] & t )
        return (258 + __ctype)[c];
    return (c);
}
```

NOTE

The construction inline is a syntactic extension to ANSI C and may not be supported by all C compilers. The intended semantics are to behave like regular preprocessor function like macros except parameter names are local and expressions giving their initial values are evaluated exactly once.

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Figure 6-148: <wordexp.h>*

#define WRDE_APPEND	0001	M
#define WRDE_DOOFFS	0002	M
#define WRDE_NOCMD	0004	M
#define WRDE_REUSE	0010	M
#define WRDE_SHOWERR	0020	M
#define WRDE_UNDEF	0040	M
#define WRDE_NOSYS	(-1)	M
#define WRDE_BADCHAR	(-2)	M
#define WRDE_BADVAL	(-3)	M
#define WRDE_CMDSUB	(-4)	M
#define WRDE_NOSPACE	(-5)	M
#define WRDE_SYNTAX	(-6)	M
typedef struct		M
{		M
size_t we_wordc;		M
char **we_wordv;		M
size_t we_offs;		M
} wordexp_t;		M

X Window Data Definitions

NOTE

This section is new to the Third Edition of this document, but will not be marked with the "G" diff-mark.

This section contains standard data definitions that describe system data for the optional X Window System libraries listed in the Generic ABI. These data definitions are referred to by their names in angle brackets: `<name.h>` and `<sys/name.h>`. Included in these data definitions are macro definitions and structure definitions. While an ABI-conforming system may provide X11 and X Toolkit Intrinsics interfaces, it need not contain the actual data definitions referenced here. Programmers should observe that the sources of the structures defined in these data definitions are defined in SVID or the appropriate X Consortium documentation (see chapter 10 in the Generic ABI).

Figure 6-149: <X11/Atom.h>, Part 1 of 3

```
#define XA_PRIMARY ((Atom) 1)
#define XA_SECONDARY ((Atom) 2)
#define XA_ARC ((Atom) 3)
#define XA_ATOM ((Atom) 4)
#define XA_BITMAP ((Atom) 5)
#define XA_CARDINAL ((Atom) 6)
#define XA_COLORMAP ((Atom) 7)
#define XA_CURSOR ((Atom) 8)
#define XA_CUT_BUFFER0 ((Atom) 9)
#define XA_CUT_BUFFER1 ((Atom) 10)
#define XA_CUT_BUFFER2 ((Atom) 11)
#define XA_CUT_BUFFER3 ((Atom) 12)
#define XA_CUT_BUFFER4 ((Atom) 13)
#define XA_CUT_BUFFER5 ((Atom) 14)
#define XA_CUT_BUFFER6 ((Atom) 15)
#define XA_CUT_BUFFER7 ((Atom) 16)
#define XA_DRAWABLE ((Atom) 17)
#define XA_FONT ((Atom) 18)
#define XA_INTEGER ((Atom) 19)
#define XA_PIXMAP ((Atom) 20)
#define XA_POINT ((Atom) 21)
#define XA_RECTANGLE ((Atom) 22)
#define XA_RESOURCE_MANAGER ((Atom) 23)
#define XA_RGB_COLOR_MAP ((Atom) 24)
#define XA_RGB_BEST_MAP ((Atom) 25)
#define XA_RGB_BLUE_MAP ((Atom) 26)
#define XA_RGB_DEFAULT_MAP ((Atom) 27)
#define XA_RGB_GRAY_MAP ((Atom) 28)
#define XA_RGB_GREEN_MAP ((Atom) 29)
#define XA_RGB_RED_MAP ((Atom) 30)
#define XA_STRING ((Atom) 31)
#define XA_VISUALID ((Atom) 32)
```

Figure 6-150: <X11/Atom.h>, Part 2 of 3

```
#define XA_WINDOW                ((Atom) 33)
#define XA_WM_COMMAND            ((Atom) 34)
#define XA_WM_HINTS              ((Atom) 35)
#define XA_WM_CLIENT_MACHINE     ((Atom) 36)
#define XA_WM_ICON_NAME         ((Atom) 37)
#define XA_WM_ICON_SIZE         ((Atom) 38)
#define XA_WM_NAME               ((Atom) 39)
#define XA_WM_NORMAL_HINTS      ((Atom) 40)
#define XA_WM_SIZE_HINTS        ((Atom) 41)
#define XA_WM_ZOOM_HINTS        ((Atom) 42)
#define XA_MIN_SPACE             ((Atom) 43)
#define XA_NORM_SPACE            ((Atom) 44)
#define XA_MAX_SPACE             ((Atom) 45)
#define XA_END_SPACE             ((Atom) 46)
#define XA_SUPERSCRIPT_X        ((Atom) 47)
#define XA_SUPERSCRIPT_Y        ((Atom) 48)
#define XA_SUBSCRIPT_X          ((Atom) 49)
#define XA_SUBSCRIPT_Y          ((Atom) 50)
#define XA_UNDERLINE_POSITION    ((Atom) 51)
#define XA_UNDERLINE_THICKNESS  ((Atom) 52)
#define XA_STRIKEOUT_ASCENT      ((Atom) 53)
#define XA_STRIKEOUT_DESCENT     ((Atom) 54)
#define XA_ITALIC_ANGLE          ((Atom) 55)
#define XA_X_HEIGHT              ((Atom) 56)
#define XA_QUAD_WIDTH            ((Atom) 57)
#define XA_WEIGHT                ((Atom) 58)
#define XA_POINT_SIZE            ((Atom) 59)
#define XA_RESOLUTION            ((Atom) 60)
#define XA_COPYRIGHT             ((Atom) 61)
#define XA_NOTICE                ((Atom) 62)
#define XA_FONT_NAME             ((Atom) 63)
#define XA_FAMILY_NAME           ((Atom) 64)
```

Figure 6-151: <X11/Atom.h>, Part 3 of 3

```
#define XA_FULL_NAME          ((Atom) 65)
#define XA_CAP_HEIGHT        ((Atom) 66)
#define XA_WM_CLASS          ((Atom) 67)
#define XA_WM_TRANSIENT_FOR  ((Atom) 68)
#define XA_LAST_PREDEFINED   ((Atom) 68)
```

Figure 6-152: <X11/Composite.h>

```
extern WidgetClass compositeWidgetClass;
```

Figure 6-153: <X11/Constraint.h>

```
extern WidgetClass constraintWidgetClass;
```

Figure 6-154: <X11/Core.h>

```
extern WidgetClass coreWidgetClass;
```

Figure 6-155: <X11/cursorfont.h>, Part 1 of 3

```
#define XC_num_glyphs      154
#define XC_X_cursor        0
#define XC_arrow           2
#define XC_based_arrow_down 4
#define XC_based_arrow_up  6
#define XC_boat            8
#define XC_bogosity        10
#define XC_bottom_left_corner 12
#define XC_bottom_right_corner 14
#define XC_bottom_side     16
#define XC_bottom_tee      18
#define XC_box_spiral      20
#define XC_center_ptr      22
#define XC_circle          24
#define XC_clock            26
#define XC_coffee_mug      28
#define XC_cross            30
#define XC_cross_reverse   32
#define XC_crosshair       34
#define XC_diamond_cross   36
#define XC_dot             38
#define XC_dotbox          40
#define XC_double_arrow    42
#define XC_draft_large     44
#define XC_draft_small     46
#define XC_draped_box      48
#define XC_exchange        50
#define XC_fleur           52
#define XC_gobbler         54
#define XC_gumby           56
#define XC_hand1           58
#define XC_hand2           60
```

Figure 6-156: <X11/cursorfont.h>, Part 2 of 3

```
#define XC_heart          62
#define XC_icon           64
#define XC_iron_cross     66
#define XC_left_ptr       68
#define XC_left_side      70
#define XC_left_tee        72
#define XC_leftbutton     74
#define XC_ll_angle       76
#define XC_lr_angle       78
#define XC_man            80
#define XC_middlebutton   82
#define XC_mouse          84
#define XC_pencil         86
#define XC_pirate         88
#define XC_plus           90
#define XC_question_arrow 92
#define XC_right_ptr      94
#define XC_right_side     96
#define XC_right_tee      98
#define XC_rightbutton    100
#define XC_rtl_logo       102
#define XC_sailboat       104
#define XC_sb_down_arrow  106
#define XC_sb_h_double_arrow 108
#define XC_sb_left_arrow  110
#define XC_sb_right_arrow 112
#define XC_sb_up_arrow    114
#define XC_sb_v_double_arrow 116
#define XC_shuttle        118
#define XC_sizing         120
#define XC_spider         122
#define XC_spraycan       124
```

Figure 6-157: <X11/cursorfont.h>, Part 3 of 3

#define XC_star	126
#define XC_target	128
#define XC_tcross	130
#define XC_top_left_arrow	132
#define XC_top_left_corner	134
#define XC_top_right_corner	136
#define XC_top_side	138
#define XC_top_tee	140
#define XC_trek	142
#define XC_ul_angle	144
#define XC_umbrella	146
#define XC_ur_angle	148
#define XC_watch	150
#define XC_xterm	152

Figure 6-158: <X11/Intrinsic.h>, Part 1 of 6

```
typedef char *String;

#define XtNumber(arr)\
    ((Cardinal) (sizeof(arr) / sizeof(arr[0])))

typedef void          *Widget;
typedef Widget        *WidgetList;
typedef void          *CompositeWidget;
typedef void          *WidgetClass;
typedef XtActionsRec  *XtActionList;

typedef void          *XtAppContext;
typedef unsigned long  XtValueMask;
typedef unsigned long  XtIntervalId;
typedef unsigned long  XtInputId;
typedef unsigned long  XtWorkProcId;
typedef unsigned int   XtGeometryMask;
typedef unsigned long  XtGCMask;
typedef unsigned long  Pixel;
typedef int            XtCacheType;
#define XtCacheNone      0x001
#define XtCacheAll       0x002
#define XtCacheByDisplay 0x003
#define XtCacheRefCount   0x100

typedef char           Boolean;
typedef long           XtArgVal;
typedef unsigned char  XtEnum;

typedef unsigned int    Cardinal;
typedef unsigned short  Dimension;
typedef short           Position;

typedef void           *XtPointer;
```

Figure 6-159: <X11/Intrinsic.h>, Part 2 of 6

```
typedef void          *XtTranslations;
typedef void          *XtAccelerators;
typedef unsigned int  Modifiers;

#define XtCWQueryOnly    (1 << 7)
#define XtSMDontChange  5

typedef void *XtCacheRef;
typedef void *XtActionHookId;
typedef unsigned long EventMask;
typedef enum {XtListHead, XtListTail } XtListPosition;
typedef unsigned long XtInputMask;

typedef struct {
    String      string;
    XtActionProc proc;
} XtActionsRec;

typedef enum {
    XtAddress,
    XtBaseOffset,
    XtImmediate,
    XtResourceString,
    XtResourceQuark,
    XtWidgetBaseOffset,
    XtProcedureArg
} XtAddressMode;

typedef struct {
    XtAddressMode      address_mode;
    XtPointer          address_id;
    Cardinal            size;
} XtConvertArgRec, *XtConvertArgList;
```

Figure 6-160: <Xt11/Intrinsic.h>, Part 3 of 6

```
#define XtInputNoneMask      0L
#define XtInputReadMask     (1L<<0)
#define XtInputWriteMask    (1L<<1)
#define XtInputExceptMask   (1L<<2)

typedef struct {
    XtGeometryMask request_mode;
    Position      x, y;
    Dimension     width, height, border_width;
    Widget sibling;
} XtWidgetGeometry;

typedef struct {
    String      name;
    XtArgVal    value;
} Arg, *ArgList;

typedef XtPointer  XtVarArgsList;

typedef struct {
    XtCallbackProc  callback;
    XtPointer       closure;
} XtCallbackRec, *XtCallbackList;

typedef enum {
    XtCallbackNoList,
    XtCallbackHasNone,
    XtCallbackHasSome
} XtCallbackStatus;

typedef struct {
    Widget      shell_widget;
    Widget      enable_widget;
} XtPopdownIDRec, *XtPopdownID;
```

Figure 6-161: <X11/Intrinsic.h>, Part 4 of 6

```
typedef enum {
    XtGeometryYes,
    XtGeometryNo,
    XtGeometryAlmost,
    XtGeometryDone
} XtGeometryResult;

typedef enum {
    XtGrabNone,
    XtGrabNonexclusive,
    XtGrabExclusive
} XtGrabKind;

typedef struct {
    String      resource_name;
    String      resource_class;
    String      resource_type;
    Cardinal    resource_size;
    Cardinal    resource_offset;
    String      default_type;
    XtPointer   default_addr;
} XtResource, *XtResourceList;

typedef struct {
    char        match;
    String      substitution;
} SubstitutionRec, *Substitution;

typedef Boolean      (*XtFilePredicate);
typedef XtPointer    XtRequestId;

extern XtConvertArgRec const colorConvertArgs[];
extern XtConvertArgRec const screenConvertArg[];
```

Figure 6-162: <X11/Intrinsic.h>, Part 5 of 6

```
#define XtAllEvents ((EventMask) -1L)
#define XtIMXEvent      1
#define XtIMTimer       2
#define XtIMAlternateInput 4
#define XtIMAll (XtIMXEvent | XtIMTimer | XtIMAlternateInput)

#define XtOffsetOf(s_type,field) XtOffset(s_type*,field)
#define XtNew(type) ((type *) XtMalloc((unsigned) sizeof(type)))
#define XT_CONVERT_FAIL      (Atom)0x80000001

#define XtIsRectObj(object) \
    (_XtCheckSubclassFlag(object, (XtEnum)0x02))
#define XtIsWidget(object) \
    (_XtCheckSubclassFlag(object, (XtEnum)0x04))
#define XtIsComposite(widget) \
    (_XtCheckSubclassFlag(widget, (XtEnum)0x08))
#define XtIsConstraint(widget) \
    (_XtCheckSubclassFlag(widget, (XtEnum)0x10))
#define XtIsShell(widget) \
    (_XtCheckSubclassFlag(widget, (XtEnum)0x20))
#define XtIsOverrideShell(widget) \
    (_XtIsSubclassOf(widget, (WidgetClass)overrideShellWidgetClass, \
    (WidgetClass)shellWidgetClass, (XtEnum)0x20))
#define XtIsWMShell(widget) \
    (_XtCheckSubclassFlag(widget, (XtEnum)0x40))
#define XtIsVendorShell(widget) \
    (_XtIsSubclassOf(widget, (WidgetClass)vendorShellWidgetClass, \
    (WidgetClass)wmShellWidgetClass, (XtEnum)0x40))
#define XtIsTransientShell(widget) \
    (_XtIsSubclassOf(widget, (WidgetClass)transientShellWidgetClass, \
    (WidgetClass)wmShellWidgetClass, (XtEnum)0x40))
#define XtIsTopLevelShell(widget) \
    (_XtCheckSubclassFlag(widget, (XtEnum)0x80))
#define XtIsApplicationShell(widget) \
    (_XtIsSubclassOf(widget, (WidgetClass)applicationShellWidgetClass, \
    (WidgetClass)topLevelShellWidgetClass, (XtEnum)0x80))
```

Figure 6-163: <X11/Intrinsic.h>, Part 6 of 6

```
#define XtSetArg(arg,n,d)\
    ((void)( (arg).name = (n), (arg).value = (XtArgVal)(d) ))
#define XtOffset(p_type,field)\
    ((Cardinal) (((char *) (&((p_type)NULL)->field))) - ((char *) NULL)))

#define XtVaNestedList          "XtVaNestedList"
#define XtVaTypedArg            "XtVaTypedArg"
#define XtUnspecifiedPixmap    ((Pixmap)2)
#define XtUnspecifiedShellInt   (-1)
#define XtUnspecifiedWindow     ((Window)2)
#define XtUnspecifiedWindowGroup ((Window)3)
#define XtDefaultForeground     "XtDefaultForeground"
#define XtDefaultBackground     "XtDefaultBackground"
#define XtDefaultFont           "XtDefaultFont"
#define XtDefaultFontSet        "XtDefaultFontSet"
```

Figure 6-164: <X11/Object.h>

```
extern WidgetClass objectClass;
```

Figure 6-165: <X11/RectObj.h>

```
extern WidgetClass rectObjClass;
```

Figure 6-166: <X11/extensions/shape.h>*

```
#define ShapeSet          0
#define ShapeUnion        1
#define ShapeIntersect    2
#define ShapeSubtract     3
#define ShapeInvert       4

#define ShapeBounding      0
#define ShapeClip         1

#define ShapeNotifyMask    (1L << 0)
#define ShapeNotify       0
```

Figure 6-167: <X11/Shell.h>

```
extern WidgetClass shellWidgetClass;
extern WidgetClass overrideShellWidgetClass;
extern WidgetClass wmShellWidgetClass;
extern WidgetClass transientShellWidgetClass;
extern WidgetClass topLevelShellWidgetClass;
extern WidgetClass applicationShellWidgetClass;
```

Figure 6-168: <X11/Vendor.h>

```
extern WidgetClass vendorShellWidgetClass;
```


Figure 6-169: <X11/X.h>, Part 1 of 12

```
typedef unsigned long XID;

typedef XID Window;
typedef XID Drawable;
typedef XID Font;
typedef XID Pixmap;
typedef XID Cursor;
typedef XID Colormap;
typedef XID GCcontext;
typedef XID KeySym;

typedef unsigned long Atom;
typedef unsigned long VisualID;
typedef unsigned long Time;
typedef unsigned char KeyCode;

#define AllTemporary      0L
#define AnyButton         0L
#define AnyKey            0L
#define AnyPropertyType  0L
#define CopyFromParent    0L
#define CurrentTime       0L
#define InputFocus        1L
#define NoEventMask       0L
#define None              0L
#define NoSymbol          0L
#define ParentRelative    1L
#define PointerWindow     0L
#define PointerRoot       1L
```

Figure 6-170: <X11/X.h>, Part 2 of 12

```
#define KeyPressMask          (1L<<0)
#define KeyReleaseMask       (1L<<1)
#define ButtonPressMask      (1L<<2)
#define ButtonReleaseMask    (1L<<3)
#define EnterWindowMask      (1L<<4)
#define LeaveWindowMask      (1L<<5)
#define PointerMotionMask    (1L<<6)
#define PointerMotionHintMask (1L<<7)
#define Button1MotionMask    (1L<<8)
#define Button2MotionMask    (1L<<9)
#define Button3MotionMask    (1L<<10)
#define Button4MotionMask    (1L<<11)
#define Button5MotionMask    (1L<<12)
#define ButtonMotionMask     (1L<<13)
#define KeymapStateMask      (1L<<14)
#define ExposureMask         (1L<<15)
#define VisibilityChangeMask (1L<<16)
#define StructureNotifyMask  (1L<<17)
#define ResizeRedirectMask   (1L<<18)
#define SubstructureNotifyMask (1L<<19)
#define SubstructureRedirectMask (1L<<20)
#define FocusChangeMask      (1L<<21)
#define PropertyChangeMask   (1L<<22)
#define ColormapChangeMask   (1L<<23)
#define OwnerGrabButtonMask  (1L<<24)
```

Figure 6-171: <X11/X.h>, Part 3 of 12

```
#define KeyPress          2
#define KeyRelease        3
#define ButtonPress       4
#define ButtonRelease     5
#define MotionNotify      6
#define EnterNotify       7
#define LeaveNotify       8
#define FocusIn           9
#define FocusOut          10
#define KeymapNotify      11
#define Expose            12
#define GraphicsExpose    13
#define NoExpose           14
#define VisibilityNotify   15
#define CreateNotify      16
#define DestroyNotify     17
#define UnmapNotify       18
#define MapNotify         19
#define MapRequest        20
#define ReparentNotify    21
#define ConfigureNotify   22
#define ConfigureRequest  23
#define GravityNotify     24
#define ResizeRequest     25
#define CirculateNotify   26
#define CirculateRequest  27
#define PropertyNotify    28
#define SelectionClear    29
#define SelectionRequest  30
#define SelectionNotify   31
#define ColormapNotify    32
#define ClientMessage     33
#define MappingNotify     34
#define LASTEvent         35
                        /* must be bigger than any event # */
```

Figure 6-172: <X11/X.h>, Part 4 of 12

```
#define ShiftMask      (1<<0)
#define LockMask      (1<<1)
#define ControlMask   (1<<2)
#define Mod1Mask      (1<<3)
#define Mod2Mask      (1<<4)
#define Mod3Mask      (1<<5)
#define Mod4Mask      (1<<6)
#define Mod5Mask      (1<<7)

#define Button1Mask    (1<<8)
#define Button2Mask    (1<<9)
#define Button3Mask    (1<<10)
#define Button4Mask    (1<<11)
#define Button5Mask    (1<<12)
#define AnyModifier    (1<<15)

#define Button1        1
#define Button2        2
#define Button3        3
#define Button4        4
#define Button5        5

#define NotifyNormal    0
#define NotifyGrab      1
#define NotifyUngrab    2
#define NotifyWhileGrabbed 3
#define NotifyHint      1
#define NotifyAncestor  0
#define NotifyVirtual    1
#define NotifyInferior   2
#define NotifyNonlinear  3
#define NotifyNonlinearVirtual 4
#define NotifyPointer    5
#define NotifyPointerRoot 6
#define NotifyDetailNone 7
```

Figure 6-173: <X11/X.h>, Part 5 of 12

```
#define VisibilityUnobscured      0
#define VisibilityPartiallyObscured  1
#define VisibilityFullyObscured    2

#define PlaceOnTop                0
#define PlaceOnBottom            1

#define PropertyNewValue          0
#define PropertyDelete            1

#define ColormapUninstalled       0
#define ColormapInstalled        1

#define GrabModeSync              0
#define GrabModeAsync            1

#define GrabSuccess                0
#define AlreadyGrabbed            1
#define GrabInvalidTime          2
#define GrabNotViewable          3
#define GrabFrozen                4

#define AsyncPointer              0
#define SyncPointer               1
#define ReplayPointer             2
#define AsyncKeyboard             3
#define SyncKeyboard             4
#define ReplayKeyboard            5
#define AsyncBoth                 6
#define SyncBoth                  7

#define RevertToNone              (int)None
#define RevertToPointerRoot       (int)PointerRoot
#define RevertToParent            2
```

Figure 6-174: <X11/X.h>, Part 6 of 12

```
#define Success          0
#define BadRequest       1
#define BadValue         2
#define BadWindow        3
#define BadPixmap        4
#define BadAtom           5
#define BadCursor        6
#define BadFont           7
#define BadMatch          8
#define BadDrawable       9
#define BadAccess        10
#define BadAlloc          11
#define BadColor          12
#define BadGC             13
#define BadIDChoice       14
#define BadName           15
#define BadLength         16
#define BadImplementation 17

#define InputOutput       1
#define InputOnly         2

#define CWBackPixmap      (1L<<0)
#define CWBackPixel       (1L<<1)
#define CWBorderPixmap    (1L<<2)
#define CWBorderPixel     (1L<<3)
#define CWBitGravity      (1L<<4)
#define CWWinGravity      (1L<<5)
#define CWBackingStore    (1L<<6)
#define CWBackingPlanes  (1L<<7)
#define CWBackingPixel    (1L<<8)
#define CWOverrideRedirect (1L<<9)
#define CWSaveUnder       (1L<<10)
#define CWEventMask       (1L<<11)
#define CWDontPropagate   (1L<<12)
#define CWColormap        (1L<<13)
#define CWCursor          (1L<<14)
```

Figure 6-175: <X11/X.h>, Part 7 of 12

```
#define CWX                (1<<0)
#define CWY                (1<<1)
#define CWWidth            (1<<2)
#define CWHeight           (1<<3)
#define CWBorderWidth      (1<<4)
#define CWSibling           (1<<5)
#define CWStackMode        (1<<6)

#define ForgetGravity       0
#define NorthWestGravity   1
#define NorthGravity       2
#define NorthEastGravity   3
#define WestGravity        4
#define CenterGravity       5
#define EastGravity         6
#define SouthWestGravity    7
#define SouthGravity        8
#define SouthEastGravity    9
#define StaticGravity       10
#define UnmapGravity        0

#define NotUseful           0
#define WhenMapped          1
#define Always              2

#define IsUnmapped          0
#define IsUnviewable        1
#define IsViewable          2

#define SetModeInsert       0
#define SetModeDelete       1

#define DestroyAll          0
#define RetainPermanent     1
#define RetainTemporary     2
```

Figure 6-176: <X11/X.h>, Part 8 of 12

```
#define Above          0
#define Below          1
#define TopIf          2
#define BottomIf       3
#define Opposite       4
#define RaiseLowest    0
#define LowerHighest   1
#define PropModeReplace 0
#define PropModePrepend 1
#define PropModeAppend 2

#define GXclear         0x0
#define GXand           0x1
#define GXandReverse    0x2
#define GXcopy          0x3
#define GXandInverted   0x4
#define GXnoop          0x5
#define GXxor           0x6
#define GXor            0x7
#define GXnor           0x8
#define GXequiv         0x9
#define GXinvert        0xa
#define GXorReverse     0xb
#define GXcopyInverted  0xc
#define GXorInverted    0xd
#define GXnand          0xe
#define GXset           0xf

#define LineSolid       0
#define LineOnOffDash   1
#define LineDoubleDash  2
#define CapNotLast      0
#define CapButt         1
#define CapRound        2
#define CapProjecting   3
```

Figure 6-177: <X11/X.h>, Part 9 of 12

```
#define JoinMiter      0
#define JoinRound      1
#define JoinBevel      2

#define FillSolid       0
#define FillTiled       1
#define FillStippled    2
#define FillOpaqueStippled3

#define EvenOddRule     0
#define WindingRule     1

#define ClipByChildren  0
#define IncludeInferiors 1

#define Unsorted        0
#define YSorted         1
#define YXSorted        2
#define YXBanded        3

#define CoordModeOrigin 0
#define CoordModePrevious 1

#define Complex         0
#define Nonconvex       1
#define Convex          2

#define ArcChord        0
#define ArcPieSlice     1
```

Figure 6-178: <X11/X.h>, Part 10 of 12

```
#define GCFunction          (1L<<0)
#define GCPlaneMask        (1L<<1)
#define GCForeground        (1L<<2)
#define GCBackground        (1L<<3)
#define GCLineWidth         (1L<<4)
#define GCLineStyle         (1L<<5)
#define GCCapStyle          (1L<<6)
#define GCJoinStyle         (1L<<7)
#define GCFillStyle         (1L<<8)
#define GCFillRule          (1L<<9)
#define GCTile              (1L<<10)
#define GCStipple           (1L<<11)
#define GCTileStipXOrigin   (1L<<12)
#define GCTileStipYOrigin   (1L<<13)
#define GCFont              (1L<<14)
#define GCSubwindowMode     (1L<<15)
#define GCGraphicsExposures (1L<<16)
#define GCClipXOrigin       (1L<<17)
#define GCClipYOrigin       (1L<<18)
#define GCClipMask          (1L<<19)
#define GCDashOffset        (1L<<20)
#define GCDashList          (1L<<21)
#define GCArcMode           (1L<<22)

#define FontLeftToRight     0
#define FontRightToLeft     1

#define XYBitmap            0
#define XYPixmap            1
#define ZPixmap             2

#define AllocNone           0
#define AllocAll            1

#define DoRed               (1<<0)
#define DoGreen             (1<<1)
#define DoBlue              (1<<2)
```

Figure 6-179: <X11/X.h>, Part 11 of 12

```
#define CursorShape          0
#define TileShape            1
#define StippleShape         2

#define AutoRepeatModeOff    0
#define AutoRepeatModeOn    1
#define AutoRepeatModeDefault 2

#define LedModeOff           0
#define LedModeOn            1

#define KBKeyClickPercent    (1L<<0)
#define KBBellPercent        (1L<<1)
#define KBBellPitch          (1L<<2)
#define KBBellDuration       (1L<<3)
#define KBLed                (1L<<4)
#define KBLedMode            (1L<<5)
#define KBKey                (1L<<6)
#define KBAutoRepeatMode     (1L<<7)

#define MappingSuccess        0
#define MappingBusy           1
#define MappingFailed         2

#define MappingModifier      0
#define MappingKeyboard       1
#define MappingPointer        2
#define DontPreferBlanking    0
#define PreferBlanking        1
#define DefaultBlanking       2

#define DontAllowExposures    0
#define AllowExposures        1
#define DefaultExposures      2
```

Figure 6-180: <X11/X.h>, Part 12 of 12

```
#define ScreenSaverReset 0
#define ScreenSaverActive 1

#define EnableAccess 1
#define DisableAccess 0
#define StaticGray 0
#define GrayScale 1

#define StaticColor 2
#define PseudoColor 3
#define TrueColor 4
#define DirectColor 5

#define LSBFirst 0
#define MSBFirst 1
```

Figure 6-181: <X11/Xcms.h>, Part 1 of 5

```
#define XcmsFailure          0
#define XcmsSuccess         1
#define XcmsSuccessWithCompression 2

#define XcmsUndefinedFormat (XcmsColorFormat)0x00000000
#define XcmsCIEXYZFormat    (XcmsColorFormat)0x00000001
#define XcmsCIEuvYFormat    (XcmsColorFormat)0x00000002
#define XcmsCIExyYFormat    (XcmsColorFormat)0x00000003
#define XcmsCIELabFormat    (XcmsColorFormat)0x00000004
#define XcmsCIELuvFormat    (XcmsColorFormat)0x00000005
#define XcmsTekHVCFormat    (XcmsColorFormat)0x00000006
#define XcmsRGBFormat       (XcmsColorFormat)0x80000000
#define XcmsRGBiFormat      (XcmsColorFormat)0x80000001

#define XcmsInitNone        0x00
#define XcmsInitSuccess     0x01

typedef unsigned int XcmsColorFormat;

typedef double XcmsFloat;

typedef struct {
    unsigned short red;
    unsigned short green;
    unsigned short blue;
} XcmsRGB;
```

Figure 6-182: <X11/Xcms.h>, Part 2 of 5

```
typedef struct {
    XcmsFloat red;
    XcmsFloat green;
    XcmsFloat blue;
} XcmsRGBi;

typedef struct {
    XcmsFloat X;
    XcmsFloat Y;
    XcmsFloat Z;
} XcmsCIEXYZ;

typedef struct {
    XcmsFloat u_prime;
    XcmsFloat v_prime;
    XcmsFloat Y;
} XcmsCIEuvY;

typedef struct {
    XcmsFloat x;
    XcmsFloat y;
    XcmsFloat Y;
} XcmsCIExyY;

typedef struct {
    XcmsFloat L_star;
    XcmsFloat a_star;
    XcmsFloat b_star;
} XcmsCIELab;
```

Figure 6-183: <X11/Xcms.h>, Part 3 of 5

```
typedef struct {
    XcmsFloat L_star;
    XcmsFloat u_star;
    XcmsFloat v_star;
} XcmsCIELuv;

typedef struct {
    XcmsFloat H;
    XcmsFloat V;
    XcmsFloat C;
} XcmsTekHVC;

typedef struct {
    XcmsFloat pad0;
    XcmsFloat pad1;
    XcmsFloat pad2;
    XcmsFloat pad3;
} XcmsPad;
```

Figure 6-184: <X11/Xcms.h>, Part 4 of 5

```
typedef struct {
    union {
        XcmsRGB      RGB;
        XcmsRGBi     RGBi;
        XcmsCIEXYZ   CIEXYZ;
        XcmsCIEuvY   CIEuvY;
        XcmsCIExyY   CIExyY;
        XcmsCIELab    CIELab;
        XcmsCIELuv    CIELuv;
        XcmsTekHVC    TekHVC;
        XcmsPad       Pad;
    } spec;
    unsigned longpixel;
    XcmsColorFormat  format;
} XcmsColor;

typedef struct {
    XcmsColor      screenWhitePt;
    XPointer       functionSet;
    XPointer       screenData;
    unsigned char  state;
    char           pad[3];
} XcmsPerScrnInfo;

typedef void *XcmsCCC;

typedef Status (*XcmsConversionProc)();
typedef XcmsConversionProc *XcmsFuncListPtr;
```

Figure 6-185: <X11/Xcms.h>, Part 5 of 5

```
typedef struct {
    char                *prefix;
    XcmsColorFormat     id;
    XcmsParseStringProc parseString;
    XcmsFuncListPtr     to_CIEXYZ;
    XcmsFuncListPtr     from_CIEXYZ;
    int                 inverse_flag;
} XcmsColorSpace;

typedef struct {
    XcmsColorSpace      **DDColorSpaces;
    XcmsScreenInitProc  screenInitProc;
    XcmsScreenFreeProc  screenFreeProc;
} XcmsFunctionSet;
```

Figure 6-186: <X11/Xlib.h>, Part 1 of 27

```
typedef void *XPointer;

#define Bool                int
#define Status              int
#define True                1
#define False               0
#define QueuedAlready       0
#define QueuedAfterReading  1
#define QueuedAfterFlush    2

#define AllPlanes            ((unsigned long)~0L)
```

Figure 6-187: <X11/Xlib.h>, Part 2 of 27

```
typedef void XExtData;

typedef void XExtCodes;

typedef struct {
    int depth;
    int bits_per_pixel;
    int scanline_pad;
} XPixmapFormatValues;
```

Figure 6-188: <X11/Xlib.h>, Part 3 of 27

```
typedef struct {
    int function;
    unsigned long plane_mask;
    unsigned long foreground;
    unsigned long background;
    int line_width;
    int line_style;
    int cap_style;
    int join_style;
    int fill_style;
    int fill_rule;
    int arc_mode;
    Pixmap tile;
    Pixmap stipple;
    int ts_x_origin;
    int ts_y_origin;
    Font font;
    int subwindow_mode;
    Bool graphics_exposures;
    int clip_x_origin;
    int clip_y_origin;
    Pixmap clip_mask;
    int dash_offset;
    char dashes;
} XGCValues;

typedef void *GC;

typedef struct _dummy Visual;
```

Figure 6-189: <X11/Xlib.h>, Part 4 of 27

```
typedef struct _dummy Screen;

typedef struct {
    Pixmap background_pixmap;
    unsigned long background_pixel;
    Pixmap border_pixmap;
    unsigned long border_pixel;
    int bit_gravity;
    int win_gravity;
    int backing_store;
    unsigned long backing_planes;
    unsigned long backing_pixel;
    Bool save_under;
    long event_mask;
    long do_not_propagate_mask;
    Bool override_redirect;
    Colormap colormap;
    Cursor cursor;
} XSetWindowAttributes;
```

Figure 6-190: <X11/Xlib.h>, Part 5 of 27

```
typedef struct _dummy ScreenFormat;

typedef struct {
    int x, y;
    int width, height;
    int border_width;
    int depth;
    Visual *visual;
    Window root;
    int class;
    int bit_gravity;
    int win_gravity;
    int backing_store;
    unsigned long backing_planes;
    unsigned long backing_pixel;
    Bool save_under;
    Colormap colormap;
    Bool map_installed;
    int map_state;
    long all_event_masks;
    long your_event_mask;
    long do_not_propagate_mask;
    Bool override_redirect;
    Screen *screen;
} XWindowAttributes;
```

Figure 6-191: <X11/Xlib.h>, Part 6 of 27

```
typedef struct {
    int family;
    int length;
    char *address;
} XHostAddress;

typedef struct _XImage {
    int width, height;
    int xoffset;
    int format;
    char *data;
    int byte_order;
    int bitmap_unit;
    int bitmap_bit_order;
    int bitmap_pad;
    int depth;
    int bytes_per_line;
    int bits_per_pixel;
    unsigned long red_mask;
    unsigned long green_mask;
    unsigned long blue_mask;
    XPointer obdata;
    struct funcs {
        struct _XImage *(*create_image)();
        int (*destroy_image)();
        unsigned long (*get_pixel)();
        int (*put_pixel)();
        struct _XImage *(*sub_image)();
        int (*add_pixel)();
    } f;
} XImage;
```

Figure 6-192: <X11/Xlib.h>, Part 7 of 27

```
typedef struct {
    int x, y;
    int width, height;
    int border_width;
    Window sibling;
    int stack_mode;
} XWindowChanges;

typedef struct {
    unsigned long pixel;
    unsigned short red, green, blue;
    char flags;
    char pad;
} XColor;

typedef struct {
    short x1, y1, x2, y2;
} XSegment;

typedef struct {
    short x, y;
} XPoint;

typedef struct {
    short x, y;
    unsigned short width, height;
} XRectangle;

typedef struct {
    short x, y;
    unsigned short width, height;
    short angle1, angle2;
} XArc;
```

Figure 6-193: <X11/Xlib.h>, Part 8 of 27

```
typedef struct {
    int key_click_percent;
    int bell_percent;
    int bell_pitch;
    int bell_duration;
    int led;
    int led_mode;
    int key;
    int auto_repeat_mode;
} XKeyboardControl;

typedef struct {
    int key_click_percent;
    int bell_percent;
    unsigned int bell_pitch, bell_duration;
    unsigned long led_mask;
    int global_auto_repeat;
    char auto_repeats[32];
} XKeyboardState;

typedef struct {
    Time time;
    short x, y;
} XTimeCoord;

typedef struct {
    int          max_keypermod;
    KeyCode      *modifiermap;
} XModifierKeymap;

typedef struct _dummy Display;
```


Figure 6-194: <X11/Xlib.h>, Part 9 of 27

```
typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window window;
    Window root;
    Window subwindow;
    Time time;
    int x, y;
    int x_root, y_root;
    unsigned int state;
    unsigned int keycode;
    Bool same_screen;
} XKeyEvent;
typedef XKeyEvent XKeyPressedEvent;
typedef XKeyEvent XKeyReleasedEvent;

typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window window;
    Window root;
    Window subwindow;
    Time time;
    int x, y;
    int x_root, y_root;
    unsigned int state;
    unsigned int button;
    Bool same_screen;
} XButtonEvent;
typedef XButtonEvent XButtonPressedEvent;
typedef XButtonEvent XButtonReleasedEvent;
```

Figure 6-195: <X11/Xlib.h>, Part 10 of 27

```
typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window window;
    Window root;
    Window subwindow;
    Time time;
    int x, y;
    int x_root, y_root;
    unsigned int state;
    char is_hint;
    Bool same_screen;
} XMotionEvent;
typedef XMotionEvent XPointerMovedEvent;

typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window window;
    Window root;
    Window subwindow;
    Time time;
    int x, y;
    int x_root, y_root;
    int mode;
    int detail;
    Bool same_screen;
    Bool focus;
    unsigned int state;
} XCrossingEvent;
```

Figure 6-196: <X11/Xlib.h>, Part 11 of 27

```
typedef XCrossingEvent XEnterWindowEvent;
typedef XCrossingEvent XLeaveWindowEvent;
typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window window;
    int mode;
    int detail;
} XFocusChangeEvent;
typedef XFocusChangeEvent XFocusInEvent;
typedef XFocusChangeEvent XFocusOutEvent;

typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window window;
    char key_vector[32];
} XKeymapEvent;

typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window window;
    int x, y;
    int width, height;
    int count;
} XExposeEvent;
```

Figure 6-197: <X11/Xlib.h>, Part 12 of 27

```
typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Drawable drawable;
    int x, y;
    int width, height;
    int count;
    int major_code;
    int minor_code;
} XGraphicsExposeEvent;

typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Drawable drawable;
    int major_code;
    int minor_code;
} XNoExposeEvent;

typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window window;
    int state;
} XVisibilityEvent;
```

Figure 6-198: <X11/Xlib.h>, Part 13 of 27

```
typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window parent;
    Window window;
    int x, y;
    int width, height;
    int border_width;
    Bool override_redirect;
} XCreateWindowEvent;

typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window event;
    Window window;
} XDestroyWindowEvent;

typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window event;
    Window window;
    Bool from_configure;
} XUnmapEvent;
```

Figure 6-199: <X11/Xlib.h>, Part 14 of 27

```
typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window event;
    Window window;
    Bool override_redirect;
} XMapEvent;

typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window parent;
    Window window;
} XMapRequestEvent;

typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window event;
    Window window;
    Window parent;
    int x, y;
    Bool override_redirect;
} XReparentEvent;
```

Figure 6-200: <X11/Xlib.h>, Part 15 of 27

```
typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window event;
    Window window;
    int x, y;
    int width, height;
    int border_width;
    Window above;
    Bool override_redirect;
} XConfigureEvent;

typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window event;
    Window window;
    int x, y;
} XGravityEvent;

typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window window;
    int width, height;
} XResizeRequestEvent;
```

Figure 6-201: <X11/Xlib.h>, Part 16 of 27

```
typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window parent;
    Window window;
    int x, y;
    int width, height;
    int border_width;
    Window above;
    int detail;
    unsigned long value_mask;
} XConfigureRequestEvent;

typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window event;
    Window window;
    int place;
} XCirculateEvent;

typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window parent;
    Window window;
    int place;
} XCirculateRequestEvent;
```


Figure 6-202: <X11/Xlib.h>, Part 17 of 27

```
typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window window;
    Atom atom;
    Time time;
    int state;
} XPropertyEvent;

typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window window;
    Atom selection;
    Time time;
} XSelectionClearEvent;

typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window owner;
    Window requestor;
    Atom selection;
    Atom target;
    Atom property;
    Time time;
} XSelectionRequestEvent;
```

Figure 6-203: <X11/Xlib.h>, Part 18 of 27

```
typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window requestor;
    Atom selection;
    Atom target;
    Atom property;
    Time time;
} XSelectionEvent;

typedef struct {
    int type;
    Display *display;
    XID resourceid;
    unsigned long serial;
    unsigned char error_code;
    unsigned char request_code;
    unsigned char minor_code;
} XErrorEvent;

typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window window;
    Atom message_type;
    int format;
    union {
        char b[20];
        short s[10];
        long l[5];
    } data;
} XClientMessageEvent;
```

Figure 6-204: <X11/Xlib.h>, Part 19 of 27

```
typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window window;
    Colormap colormap;
    Bool new;
    int state;
} XColormapEvent;

typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window window;
    int request;
    int first_keycode;
    int count;
} XMappingEvent;

typedef struct {
    int type;
    unsigned long serial;
    Bool send_event;
    Display *display;
    Window window;
} XAnyEvent;
```

Figure 6-205: <X11/Xlib.h>, Part 20 of 27

```
typedef union _XEvent {
    int                type;
    XAnyEvent          xany;
    XKeyEvent          xkey;
    XButtonEvent       xbutton;
    XMotionEvent       xmotion;
    XCrossingEvent     xcrossing;
    XFocusChangeEvent  xfocus;
    XExposeEvent        xexpose;
    XGraphicsExposeEvent xgraphicsexpose;
    XNoExposeEvent      xnoexpose;
    XVisibilityEvent   xvisibility;
    XCreateWindowEvent xcreatewindow;
    XDestroyWindowEvent xdestroywindow;
    XUnmapEvent        xunmap;
    XMapEvent          xmap;
    XMapRequestEvent   xmaprequest;
    XReparentEvent     xreparent;
    XConfigureEvent    xconfigure;
    XGravityEvent      xgravity;
    XResizeRequestEvent xresizerequest;
    XConfigureRequestEvent xconfigurerequest;
    XCirculateEvent    xcirculate;
    XCirculateRequestEvent xcirculaterequest;
    XPropertyEvent     xproperty;
    XSelectionClearEvent xselectionclear;
    XSelectionRequestEvent xselectionrequest;
    XSelectionEvent     xselection;
    XColormapEvent      xcolormap;
    XClientMessageEvent xclient;
    XMappingEvent       xmapping;
    XErrorEvent         xerror;
    XKeymapEvent        xkeymap;
    long               pad[24];
} XEvent;
```

Figure 6-206: <X11/Xlib.h>, Part 21 of 27

```
typedef struct {
    short lbearing;
    short rbearing;
    short width;
    short ascent;
    short descent;
    unsigned short attributes;
} XCharStruct;

typedef struct {
    Atom name;
    unsigned long card32;
} XFontProp;

typedef struct {
    XExtData      *ext_data;
    Font          fid;
    unsigned      direction;
    unsigned      min_char_or_byte2;
    unsigned      max_char_or_byte2;
    unsigned      min_byte1;
    unsigned      max_byte1;
    Bool          all_chars_exist;
    unsigned      default_char;
    int           n_properties;
    XFontProp     *properties;
    XCharStruct    min_bounds;
    XCharStruct    max_bounds;
    XCharStruct    *per_char;
    int           ascent;
    int           descent;
} XFontStruct;
```

Figure 6-207: <X11/Xlib.h>, Part 22 of 27

```
typedef struct {
    char *chars;
    int nchars;
    int delta;
    Font font;
} XTextItem;

typedef struct {
    unsigned char byte1;
    unsigned char byte2;
} XChar2b;

typedef struct {
    XChar2b *chars;
    int nchars;
    int delta;
    Font font;
} XTextItem16;

typedef union {
    Display *display;
    GC gc;
    Visual *visual;
    Screen *screen;
    ScreenFormat *pixmap_format;
    XFontStruct *font;
} XDataObject;

typedef struct {
    XRectangle    max_ink_extent;
    XRectangle    max_logical_extent;
} XFontSetExtents;

typedef struct _dummy XFontSet;
```

Figure 6-208: <X11/Xlib.h>, Part 23 of 27

```
typedef struct {
    char        *chars;
    int         nchars;
    int         delta;
    XFontSet    *font_set;
} XmbTextItem;

typedef struct {
    wchar_t     *chars;
    int         nchars;
    int         delta;
    XFontSet    font_set;
} XwcTextItem;

typedef void (*XIMProc)();

typedef void *XIM;
typedef void *XIC;

typedef unsigned long XIMStyle;

typedef struct {
    unsigned short count_styles;
    XIMStyle *supported_styles;
} XIMStyles;

#define XIMPreeditArea          0x0001L
#define XIMPreeditCallbacks     0x0002L
#define XIMPreeditPosition      0x0004L
#define XIMPreeditNothing       0x0008L
#define XIMPreeditNone          0x0010L
#define XIMStatusArea           0x0100L
#define XIMStatusCallbacks      0x0200L
#define XIMStatusNothing        0x0400L
#define XIMStatusNone           0x0800L
```

Figure 6-209: <X11/Xlib.h>, Part 24 of 27

```
#define XNVaNestedList      "XNVaNestedList"
#define XNQueryInputStyle  "queryInputStyle"
#define XNClientWindow     "clientWindow"
#define XNInputStyle       "inputStyle"
#define XNFocusWindow      "focusWindow"
#define XNResourceName     "resourceName"
#define XNResourceClass    "resourceClass"
#define XNGeometryCallback "geometryCallback"
#define XNFilterEvents     "filterEvents"
#define XNPreeditStartCallback "preeditStartCallback"
#define XNPreeditDoneCallback "preeditDoneCallback"
#define XNPreeditDrawCallback "preeditDrawCallback"
#define XNPreeditCaretCallback "preeditCaretCallback"
#define XNPreeditAttributes "preeditAttributes"
#define XNStatusStartCallback "statusStartCallback"
#define XNStatusDoneCallback "statusDoneCallback"
#define XNStatusDrawCallback "statusDrawCallback"
#define XNStatusAttributes  "statusAttributes"
#define XNArea             "area"
#define XNAreaNeeded       "areaNeeded"
#define XNSpotLocation     "spotLocation"
#define XNColormap         "colorMap"
#define XNStdColormap      "stdColorMap"
#define XNForeground       "foreground"
#define XNBackground       "background"
#define XNBackgroundPixmap "backgroundPixmap"
#define XNFontSet          "fontSet"
#define XNLineSpace        "lineSpace"
#define XNCursor           "cursor"
```


Figure 6-210: <X11/Xlib.h>, Part 25 of 27

```
#define XBufferOverflow    -1
#define XLookupNone       1
#define XLookupChars      2
#define XLookupKeySym     3
#define XLookupBoth       4

typedef XPointer XVaNestedList;

typedef struct {
    XPointer client_data;
    XIMProc callback;
} XIMCallback;

typedef unsigned long XIMFeedback;

#define XIMReverse        1
#define XIMUnderline      (1<<1)
#define XIMHighlight      (1<<2)
#define XIMPrimary        (1<<5)
#define XIMSecondary      (1<<6)
#define XIMTertiary       (1<<7)

typedef struct _XIMText {
    unsigned short length;
    XIMFeedback *feedback;
    Bool encoding_is_wchar;
    union {
        char *multi_byte;
        wchar_t *wide_char;
    } string;
} XIMText;
```

Figure 6-211: <X11/Xlib.h>, Part 26 of 27

```
typedef struct _XIMPreeditDrawCallbackStruct {
    int caret;
    int chg_first;
    int chg_length;
    XIMText *text;
} XIMPreeditDrawCallbackStruct;

typedef enum {
    XIMForwardChar, XIMBackwardChar,
    XIMForwardWord, XIMBackwardWord,
    XIMCaretUp, XIMCaretDown,
    XIMNextLine, XIMPreviousLine,
    XIMLineStart, XIMLineEnd,
    XIMAbsolutePosition,
    XIMDontChange
} XIMCaretDirection;

typedef enum {
    XIMIsInvisible,
    XIMIsPrimary,
    XIMIsSecondary
} XIMCaretStyle;

typedef struct _XIMPreeditCaretCallbackStruct {
    int position;
    XIMCaretDirection direction;
    XIMCaretStyle style;
} XIMPreeditCaretCallbackStruct;
```

Figure 6-212: <X11/Xlib.h>, Part 27 of 27

```
typedef enum {
    XIMTextType,
    XIMBitmapType
} XIMStatusDataType;

typedef struct _XIMStatusDrawCallbackStruct {
    XIMStatusDataType type;
    union {
        XIMText *text;
        Pixmap bitmap;
    } data;
} XIMStatusDrawCallbackStruct;
```

Figure 6-213: <X11/Xresource.h>, Part 1 of 2

```
typedef int          XrmQuark, *XrmQuarkList;
#define NULLQUARK    ((XrmQuark) 0)

typedef enum {XrmBindTightly, XrmBindLoosely} \
            XrmBinding, *XrmBindingList;

typedef XrmQuark      XrmName;
typedef XrmQuarkList  XrmNameList;
typedef XrmQuark      XrmClass;
typedef XrmQuarkList  XrmClassList;
typedef XrmQuark      XrmRepresentation;

#define XrmStringToName(string)      XrmStringToQuark(string)
#define XrmStringToNameList(str, name) XrmStringToQuarkList(str, name)
#define XrmStringToClass(class)      XrmStringToQuark(class)
#define XrmStringToClassList(str, class) XrmStringToQuarkList(str, class)
#define XrmStringToRepresentation(string) XrmStringToQuark(string)

typedef struct {
    unsigned int      size;
    XPointer          addr;
} XrmValue, *XrmValuePtr;

typedef void          *XrmHashBucket;
typedef XrmHashBucket *XrmHashTable;
typedef XrmHashTable  XrmSearchList[];
typedef void          *XrmDatabase;

#define XrmEnumAllLevels      0
#define XrmEnumOneLevel      1
```

Figure 6-214: <X11/Xresource.h>, Part 2 of 2

```
typedef enum {
    XrmoptionNoArg,
    XrmoptionIsArg,
    XrmoptionStickyArg,
    XrmoptionSepArg,
    XrmoptionResArg,
    XrmoptionSkipArg,
    XrmoptionSkipLine,
    XrmoptionSkipNArgs
} XrmOptionKind;

typedef struct {
    char          *option;
    char          *specifier;
    XrmOptionKind argKind;
    XPointer      value;
} XrmOptionDescRec, *XrmOptionDescList;
```

Figure 6-215: <X11/Xutil.h>, Part 1 of 5

```
#define NoValue      0x0000
#define XValue 0x0001
#define YValue 0x0002
#define WidthValue  0x0004
#define HeightValue 0x0008
#define AllValues    0x000F
#define XNegative    0x0010
#define YNegative    0x0020

typedef struct {
    long flags;
    int x, y;
    int width, height;
    int min_width, min_height;
    int max_width, max_height;
    int width_inc, height_inc;
    struct {
        int x;
        int y;
    } min_aspect, max_aspect;
    int base_width, base_height;
    int win_gravity;
} XSizeHints;

#define USPosition    (1L << 0)
#define USSize (1L << 1)
#define PPosition    (1L << 2)
#define PSize         (1L << 3)
#define PMinSize      (1L << 4)
#define PMaxSize      (1L << 5)
#define PResizeInc    (1L << 6)
#define PAspect       (1L << 7)
#define PBaseSize     (1L << 8)
#define PWinGravity   (1L << 9)
#define PAllHints (PPosition|PSize|PMinSize|PMaxSize|PResizeInc|PAspect)
```

Figure 6-216: <X11/Xutil.h>, Part 2 of 5

```
typedef struct {
    long    flags;
    Bool    input;
    int     initial_state;
    Pixmap  icon_pixmap;
    Window  icon_window;
    int     icon_x, icon_y;
    Pixmap  icon_mask;
    XID     window_group;
} XWMHints;

#define InputHint      (1L << 0)
#define StateHint      (1L << 1)
#define IconPixmapHint (1L << 2)
#define IconWindowHint (1L << 3)
#define IconPositionHint (1L << 4)
#define IconMaskHint  (1L << 5)
#define WindowGroupHint (1L << 6)
#define AllHints (InputHint|StateHint|
                  IconPixmapHint|IconWindowHint|
                  IconPositionHint|IconMaskHint|WindowGroupHint)

#define WithdrawnState 0
#define NormalState    1
#define IconicState    3

typedef struct {
    unsigned char    *value;
    Atom             encoding;
    int              format;
    unsigned long     nitems;
} XTextProperty;

#define XNoMemory          -1
#define XLocaleNotSupported -2
#define XConverterNotFound -3
```

Figure 6-217: <X11/Xutil.h>, Part 3 of 5

```
typedef int XContext;

typedef enum {
    XStringStyle,
    XCompoundTextStyle,
    XTextStyle,
    XStdICCTextStyle
} XICCEncodingStyle;

typedef struct {
    int min_width, min_height;
    int max_width, max_height;
    int width_inc, height_inc;
} XIconSize;

typedef struct {
    char *res_name;
    char *res_class;
} XClassHint;

#define XDestroyImage(ximage)
    ((*((ximage)->f.destroy_image))((ximage)))
#define XGetPixel(ximage, x, y)
    ((*((ximage)->f.get_pixel))((ximage), (x), (y)))
#define XPutPixel(ximage, x, y, pixel)
    ((*((ximage)->f.put_pixel))((ximage), (x), (y), (pixel)))
#define XSubImage(ximage, x, y, width, height)
    ((*((ximage)->f.sub_image))((ximage), (x), (y), (width), (height)))
#define XAddPixel(ximage, value)
    ((*((ximage)->f.add_pixel))((ximage), (value)))

typedef struct _XComposeStatus {
    XPointer compose_ptr;
    int chars_matched;
} XComposeStatus;
```


Figure 6-218: <X11/Xutil.h>, Part 4 of 5

```
#define IsKeypadKey(keysym)
    (((unsigned)(keysym) >= XK_KP_Space) && \
     ((unsigned)(keysym) <= XK_KP_Equal))
#define IsCursorKey(keysym)
    (((unsigned)(keysym) >= XK_Home) && \
     ((unsigned)(keysym) < XK_Select))
#define IsPFKey(keysym)
    (((unsigned)(keysym) >= XK_KP_F1) && \
     ((unsigned)(keysym) <= XK_KP_F4))
#define IsFunctionKey(keysym)
    (((unsigned)(keysym) >= XK_F1) && ((unsigned)(keysym) <= XK_F35))
#define IsMiscFunctionKey(keysym)
    (((unsigned)(keysym) >= XK_Select) && \
     ((unsigned)(keysym) <= XK_Break))
#define IsModifierKey(keysym)
    (((unsigned)(keysym) >= XK_Shift_L) && \
     ((unsigned)(keysym) <= XK_Hyper_R))
    || ((unsigned)(keysym) == XK_Mode_switch)
    || ((unsigned)(keysym) == XK_Num_Lock))

typedef void *Region;

#define RectangleOut 0
#define RectangleIn 1
#define RectanglePart 2

typedef struct {
    Visual *visual;
    VisualID visualid;
    int screen;
    int depth;
    int class;
    unsigned long red_mask;
    unsigned long green_mask;
    unsigned long blue_mask;
    int colormap_size;
    int bits_per_rgb;
} XVisualInfo;
```

Figure 6-219: <X11/Xutil.h>, Part 5 of 5

```
#define VisualNoMask      0x0
#define VisualIDMask      0x1
#define VisualScreenMask      0x2
#define VisualDepthMask      0x4
#define VisualClassMask      0x8
#define VisualRedMaskMask      0x10
#define VisualGreenMaskMask      0x20
#define VisualBlueMaskMask      0x40
#define VisualColormapSizeMask      0x80
#define VisualBitsPerRGBMask      0x100
#define VisualAllMask      0x1FF

typedef struct {
    Colormap      colormap;
    unsigned long red_max;
    unsigned long red_mult;
    unsigned long green_max;
    unsigned long green_mult;
    unsigned long blue_max;
    unsigned long blue_mult;
    unsigned long base_pixel;
    VisualID      visualid;
    XID           killid;
} XStandardColormap;

#define ReleaseByFreeingColormap ((XID) 1L)
#define BitmapSuccess      0
#define BitmapOpenFailed      1
#define BitmapFileInvalid      2
#define BitmapNoMemory      3
#define XCSUCCESS      0
#define XCNOMEM      1
#define XCNOENT      2
#define XUniqueContext()      ((XContext) XrmUniqueQuark())
```

M

Motif 1.2 Data Definitions

This section contains standard data definitions that describe system data for the optional Motif 1.2 libraries. These data definitions are referred to by their names in angle brackets: `<name.h>` and `<sys/name.h>`. Included in these data definitions are macro definitions and structure definitions. While an ABI-conforming system may provide Motif 1.2 interfaces, it need not contain the actual data definitions referenced here. Programmers should observe that the sources of the structures defined in these data definitions are defined in SVID or the appropriate Motif documentation (see chapter 10 in the Generic ABI).

6-192

LIBRARIES

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File: abi_386/chap6 (Delta 44.21)
386:adm.book:sum

Figure 6-220: <Xm/ArrowB.h>*

```
typedef struct _XmArrowButtonClassRec * XmArrowButtonWidgetClass;  
typedef struct _XmArrowButtonRec      * XmArrowButtonWidget;
```

Figure 6-221: <Xm/ArrowBG.h>*

```
typedef struct _XmArrowButtonGadgetClassRec * XmArrowButtonGadgetClass;  
typedef struct _XmArrowButtonGadgetRec     * XmArrowButtonGadget;
```

Figure 6-222: <Xm/BulletinB.h>*

```
typedef struct _XmBulletinBoardClassRec * XmBulletinBoardWidgetClass;  
typedef struct _XmBulletinBoardRec     * XmBulletinBoardWidget;
```

Figure 6-223: <Xm/CascadeB.h>*

```
typedef struct _XmCascadeButtonRec      * XmCascadeButtonWidget;  
typedef struct _XmCascadeButtonClassRec * XmCascadeButtonWidgetClass;
```

Figure 6-224: <Xm/CascadeBG.h>*

```
typedef struct _XmCascadeButtonGadgetClassRec    * XmCascadeButtonGadgetClass;  
typedef struct _XmCascadeButtonGadgetRec        * XmCascadeButtonGadget;  
typedef struct _XmCascadeButtonGCacheObjRec     * XmCascadeButtonGCacheObject;
```

Figure 6-225: <Xm/Command.h>*

```
typedef struct _XmCommandClassRec    * XmCommandWidgetClass;  
typedef struct _XmCommandRec        * XmCommandWidget;
```

Figure 6-226: <Xm/CutPaste.h>*

```
#define XmClipboardFail      0
#define XmClipboardSuccess  1
#define XmClipboardTruncate 2
#define XmClipboardLocked   4
#define XmClipboardBadFormat 5
#define XmClipboardNoData   6
#define ClipboardFail      0

#define ClipboardSuccess    1
#define ClipboardTruncate  2
#define ClipboardLocked     4
#define ClipboardBadFormat  5
#define ClipboardNoData     6

typedef struct {
    long DataId;
    long PrivateId;
} XmClipboardPendingRec, *XmClipboardPendingList;
```

Figure 6-227: <Xm/DialogS.h>*

```
typedef struct _XmDialogShellClassRec    * XmDialogShellWidgetClass;
typedef struct _XmDialogShellRec         * XmDialogShellWidget;
```

Figure 6-228: <Xm/Display.h>*

```
enum {
    XmDRAG_NONE,
    XmDRAG_DROP_ONLY,
    XmDRAG_PREFER_PREREGISTER,
    XmDRAG_PREREGISTER,
    XmDRAG_PREFER_DYNAMIC,
    XmDRAG_DYNAMIC,
    XmDRAG_PREFER_RECEIVER
};

typedef struct _XmDisplayRec *XmDisplay;
typedef struct _XmDisplayClassRec *XmDisplayClass;
```

Figure 6-229: <Xm/DragC.h>*, Part 1 of 4

```
#define XmDROP_MOVE      (1L << 0)
#define XmDROP_COPY      (1L << 1)
#define XmDROP_LINK      (1L << 2)

#define XmHELP            2
typedef unsigned int      XmID;

#define _XA_MOTIF_DROP    "_MOTIF_DROP"
#define _XA_DRAG_FAILURE  "_MOTIF_DRAG_FAILURE"
#define _XA_DRAG_SUCCESS  "_MOTIF_DRAG_SUCCESS"

enum{
    XmTOP_LEVEL_ENTER,          XmTOP_LEVEL_LEAVE,
    XmDRAG_MOTION,              XmDROP_SITE_ENTER,
    XmDROP_SITE_LEAVE,         XmDROP_START,
    XmDROP_FINISH,             XmDRAG_DROP_FINISH,
    XmOPERATION_CHANGED
} ;

enum{
    XmDROP,                    XmDROP_HELP,
    XmDROP_CANCEL,             XmDROP_INTERRUPT
} ;
```


Figure 6-230: <Xm/DragC.h>*, Part 2 of 4

```
#define XmDROP_NOOP      0L

enum{   XmBLEND_ALL,                XmBLEND_STATE_SOURCE,
        XmBLEND_JUST_SOURCE,       XmBLEND_NONE
        } ;

enum{   XmDROP_FAILURE,             XmDROP_SUCCESS
        } ;

enum{   XmCR_TOP_LEVEL_ENTER,       XmCR_TOP_LEVEL_LEAVE,
        XmCR_DRAG_MOTION,           XmCR_DROP_SITE_ENTER,
        XmCR_DROP_SITE_LEAVE,       XmCR_DROP_START,
        XmCR_DROP_FINISH,           XmCR_DRAG_DROP_FINISH,
        XmCR_OPERATION_CHANGED,
        _XmNUMBER_DND_CB_REASONS
        } ;

typedef struct _XmDragContextClassRec *XmDragContextClass;
typedef struct _XmDragContextRec      *XmDragContext;

typedef struct _XmAnyICCCallbackStruct{
    int          reason;
    XEvent       *event;
    Time         timeStamp;
}XmAnyICCCallbackStruct, *XmAnyICCCallback;

typedef struct _XmTopLevelEnterCallbackStruct{
    int          reason;
    XEvent       *event;
    Time         timeStamp;
    Screen       *screen;
    Window       window;
    Position     x, y;
    unsigned char dragProtocolStyle;
    Atom         iccHandle;
}XmTopLevelEnterCallbackStruct, *XmTopLevelEnterCallback;
```

Figure 6-231: <Xm/DragC.h>*, Part 3 of 4

```
typedef struct _XmTopLevelLeaveCallbackStruct{
    int            reason;
    XEvent         *event;
    Time          timeStamp;
    Screen         *screen;
    Window         window;
}XmTopLevelLeaveCallbackStruct, *XmTopLevelLeaveCallback;

typedef struct _XmDropSiteEnterCallbackStruct{
    int            reason;
    XEvent         *event;
    Time          timeStamp;
    unsigned char  operation;
    unsigned char  operations;
    unsigned char  dropSiteStatus;
    Position       x, y;
}XmDropSiteEnterCallbackStruct, *XmDropSiteEnterCallback;

typedef struct _XmDropSiteLeaveCallbackStruct{
    int            reason;
    XEvent         *event;
    Time          timeStamp;
}XmDropSiteLeaveCallbackStruct, *XmDropSiteLeaveCallback;

typedef struct _XmDragMotionCallbackStruct{
    int            reason;
    XEvent         *event;
    Time          timeStamp;
    unsigned char  operation;
    unsigned char  operations;
    unsigned char  dropSiteStatus;
    Position       x, y;
}XmDragMotionCallbackStruct, *XmDragMotionCallback;
```

Figure 6-232: <Xm/DragC.h>*, Part 4 of 4

```
typedef struct _XmOperationChangedCallbackStruct{
    int                reason;
    XEvent             *event;
    Time               timeStamp;
    unsigned char       operation;
    unsigned char       operations;
    unsigned char       dropSiteStatus;
}XmOperationChangedCallbackStruct, *XmOperationChangedCallback;

typedef struct _XmDropStartCallbackStruct{
    int                reason;
    XEvent             *event;
    Time               timeStamp;
    unsigned char       operation;
    unsigned char       operations;
    unsigned char       dropSiteStatus;
    unsigned char       dropAction;
    Position            x, y;
    Window              window;
    Atom                iccHandle;
}XmDropStartCallbackStruct, *XmDropStartCallback;

typedef struct _XmDropFinishCallbackStruct{
    int                reason;
    XEvent             *event;
    Time               timeStamp;
    unsigned char       operation;
    unsigned char       operations;
    unsigned char       dropSiteStatus;
    unsigned char       dropAction;
    unsigned char       completionStatus;
}XmDropFinishCallbackStruct, *XmDropFinishCallback;
typedef struct _XmDragDropFinishCallbackStruct{
    int                reason;
    XEvent             *event;
    Time               timeStamp;
}XmDragDropFinishCallbackStruct, *XmDragDropFinishCallback;
```

Figure 6-233: <Xm/DragIcon.h>*

```
enum {
    XmATTACH_NORTH_WEST,
    XmATTACH_NORTH,
    XmATTACH_NORTH_EAST,
    XmATTACH_EAST,
    XmATTACH_SOUTH_EAST,
    XmATTACH_SOUTH,
    XmATTACH_SOUTH_WEST,
    XmATTACH_WEST,
    XmATTACH_CENTER,
    XmATTACH_HOT
};

typedef struct _XmDragIconRec *XmDragIconObject;
typedef struct _XmDragIconClassRec *XmDragIconObjectClass;
```

Figure 6-234: <Xm/DragOverS.h>*

```
typedef struct _XmDragOverShellRec          *XmDragOverShellWidget;
typedef struct _XmDragOverShellClassRec      *XmDragOverShellWidgetClass;
```

Figure 6-235: <Xm/DrawingA.h>*

```
typedef struct _XmDrawingAreaClassRec * XmDrawingAreaWidgetClass;
typedef struct _XmDrawingAreaRec      * XmDrawingAreaWidget;
```

Figure 6-236: <Xm/DrawnB.h>*

```
typedef struct _XmDrawnButtonClassRec *XmDrawnButtonWidgetClass;
typedef struct _XmDrawnButtonRec      *XmDrawnButtonWidget;
```

Figure 6-237: <Xm/DropSMgr.h>*, Part 1 of 2

```
#define XmCR_DROP_SITE_LEAVE_MESSAGE 1
#define XmCR_DROP_SITE_ENTER_MESSAGE 2
#define XmCR_DROP_SITE_MOTION_MESSAGE 3
#define XmCR_DROP_MESSAGE 4

#define XmNO_DROP_SITE 1
#define XmINVALID_DROP_SITE 2
#define XmVALID_DROP_SITE 3

enum { XmDRAG_UNDER_NONE, XmDRAG_UNDER_PIXMAP,
       XmDRAG_UNDER_SHADOW_IN, XmDRAG_UNDER_SHADOW_OUT,
       XmDRAG_UNDER_HIGHLIGHT };

enum { XmDROP_SITE_SIMPLE, XmDROP_SITE_COMPOSITE,
       XmDROP_SITE_SIMPLE_CLIP_ONLY = 128,
       XmDROP_SITE_COMPOSITE_CLIP_ONLY };

enum { XmABOVE, XmBELOW };

enum { XmDROP_SITE_ACTIVE, XmDROP_SITE_INACTIVE };

typedef struct _XmDragProcCallbackStruct {
    int          reason;
    XEvent *     event;
    Time         timeStamp;
    Widget       dragContext;
    Position     x, y;
    unsigned char dropSiteStatus;
    unsigned char operation;
    unsigned char operations;
    Boolean      animate;
} XmDragProcCallbackStruct, * XmDragProcCallback;
```

Figure 6-238: <Xm/DropSMgr.h>*, **Part 2 of 2**

```
typedef struct _XmDropProcCallbackStruct {
    int                reason;
    XEvent *           event;
    Time               timeStamp;
    Widget             dragContext;
    Position           x, y;
    unsigned char      dropSiteStatus;
    unsigned char      operation;
    unsigned char      operations;
    unsigned char      dropAction;
} XmDropProcCallbackStruct, * XmDropProcCallback;

typedef struct _XmDropSiteVisualsRec {
    Pixel              background;
    Pixel              foreground;
    Pixel              topShadowColor;
    Pixmap             topShadowPixmap;
    Pixel              bottomShadowColor;
    Pixmap             bottomShadowPixmap;
    Dimension          shadowThickness;
    Pixel              highlightColor;
    Pixmap             highlightPixmap;
    Dimension          highlightThickness;
    Dimension          borderWidth;
} XmDropSiteVisualsRec, * XmDropSiteVisuals;

typedef struct _XmDropSiteManagerClassRec *XmDropSiteManagerObjectClass;
typedef struct _XmDropSiteManagerRec *XmDropSiteManagerObject;
```

Figure 6-239: <Xm/DropTrans.h>*

```
#define XmTRANSFER_FAILURE 0
#define XmTRANSFER_SUCCESS 1

typedef struct _XmDropTransferClassRec * XmDropTransferObjectClass;
typedef struct _XmDropTransferRec      * XmDropTransferObject;

typedef struct _XmDropTransferEntryRec {
    XtPointer    client_data;
    Atom         target;
} XmDropTransferEntryRec, * XmDropTransferEntry;
```

Figure 6-240: <Xm/FileSB.h>*

```
typedef struct _XmFileSelectionBoxClassRec * XmFileSelectionBoxWidgetClass;
typedef struct _XmFileSelectionBoxRec      * XmFileSelectionBoxWidget;
```

Figure 6-241: <Xm/Form.h>*

```
typedef struct _XmFormClassRec * XmFormWidgetClass;
typedef struct _XmFormRec      * XmFormWidget;
```

Figure 6-242: <Xm/Frame.h>*

```
typedef struct _XmFrameClassRec * XmFrameWidgetClass;
typedef struct _XmFrameRec      * XmFrameWidget;
```

Figure 6-243: <Xm/Label.h>*

```
typedef struct _XmLabelClassRec * XmLabelWidgetClass;
typedef struct _XmLabelRec      * XmLabelWidget;
```

Figure 6-244: <Xm/LabelG.h>*

```
typedef struct _XmLabelGadgetClassRec * XmLabelGadgetClass;
typedef struct _XmLabelGadgetRec      * XmLabelGadget;
typedef struct _XmLabelGCacheObjRec   * XmLabelGCacheObject;
```

Figure 6-245: <Xm/List.h>*

```
#define XmINITIAL      0
#define XmADDITION     1
#define XmMODIFICATION 2

typedef struct _XmListClassRec * XmListWidgetClass;
typedef struct _XmListRec      * XmListWidget;
```

Figure 6-246: <Xm/MainW.h>*

```
typedef struct _XmMainWindowClassRec * XmMainWindowWidgetClass;
typedef struct _XmMainWindowRec      * XmMainWindowWidget;
```

Figure 6-247: <Xm/MenuShell.h>*

```
typedef struct _XmMenuShellClassRec      * XmMenuShellWidgetClass;
typedef struct _XmMenuShellWidgetRec     * XmMenuShellWidget;
```

Figure 6-248: <Xm/MessageB.h>*

```
typedef struct _XmMessageBoxClassRec * XmMessageBoxWidgetClass;  
typedef struct _XmMessageBoxRec      * XmMessageBoxWidget;
```

Figure 6-249: <Mrm/MrmPublic.h>*, **Part 1 of 3**

```
#define MrmSUCCESS          1
#define MrmCREATE_NEW      3
#define MrmINDEX_RETRY     5
#define MrmINDEX_GT        7
#define MrmINDEX_LT        9
#define MrmPARTIAL_SUCCESS 11

#define MrmFAILURE          0
#define MrmNOT_FOUND        2
#define MrmEXISTS           4
#define MrmNUL_GROUP        6
#define MrmNUL_TYPE         8
#define MrmWRONG_GROUP     10
#define MrmWRONG_TYPE       12
#define MrmOUT_OF_RANGE     14
#define MrmBAD_RECORD       16
#define MrmNULL_DATA        18
#define MrmBAD_DATA_INDEX   20
#define MrmBAD_ORDER        22
#define MrmBAD_CONTEXT      24
#define MrmNOT_VALID        26
#define MrmBAD_BTREE        28
#define MrmBAD_WIDGET_REC   30
#define MrmBAD_CLASS_TYPE   32
#define MrmNO_CLASS_NAME    34
#define MrmTOO_MANY         36
#define MrmBAD_IF_MODULE    38
#define MrmNULL_DESC        40
#define MrmOUT_OF_BOUNDS    42
#define MrmBAD_COMPRESS     44
#define MrmBAD_ARG_TYPE     46
#define MrmNOT_IMP          48
#define MrmNULL_INDEX       50
#define MrmBAD_KEY_TYPE     52
#define MrmBAD_CALLBACK     54
```

Figure 6-250: <Mrm/MrmPublic.h>*, Part 2 of 3

```
#define MrmNULL_ROUTINE          56
#define MrmVEC_TOO_BIG          58
#define MrmBAD_HIERARCHY        60
#define MrmBAD_CLASS_CODE       62
#define MrmDISPLAY_NOT_OPENED   63
#define MrmEOF                   64
#define MrmUNRESOLVED_REFS      65
#define MrmNcreateCallback      "createCallback"
#define MrmCR_CREATE            XmCR_CREATE

#define MrmwcUnknown            1

#define MrmRtypeMin              1
#define MrmRtypeInteger          1
#define MrmRtypeBoolean          2
#define MrmRtypeChar8            3
#define MrmRtypeChar8Vector      4
#define MrmRtypeCString          5
#define MrmRtypeCStringVector    6
#define MrmRtypeFloat            7
#define MrmRtypeCallback         9
#define MrmRtypePixmapImage     10
#define MrmRtypePixmapDDIF      11
#define MrmRtypeResource         12
#define MrmRtypeNull             13
#define MrmRtypeAddrName         14
#define MrmRtypeIconImage        15
#define MrmRtypeFont             16
#define MrmRtypeFontList         17
#define MrmRtypeColor            18
#define MrmRtypeColorTable       19
#define MrmRtypeAny              20
#define MrmRtypeTransTable       21
#define MrmRtypeClassRecName     22
#define MrmRtypeIntegerVector    23
```

Figure 6-251: <Mrm/MrmPublic.h>*, Part 3 of 3

```
#define MrmRtypeXBitmapFile      24
#define MrmRtypeCountedVector    25
#define MrmRtypeKeysym           26
#define MrmRtypeSingleFloat      27
#define MrmRtypeWideCharacter     28
#define MrmRtypeFontSet          29
#define MrmRtypeMax               30
typedef short int                MrmCode ;
typedef unsigned char            MrmSCode ;
typedef unsigned short int       MrmOffset ;
typedef short int                MrmType ;
typedef unsigned short int       MrmSize ;
typedef short int                MrmCount ;
typedef unsigned char            MrmFlag ;
typedef long int                 MrmResource_id ;
typedef short int                MrmGroup ;

#define MrmMaxResourceSize       65535
#define MrmOsOpenParamVersion    1
typedef struct {
    Cardinal                version;
    char                    *default_fname;
    union {
        unsigned long       related_nam;
        Boolean              clobber_flg;
    } nam_flg;
    Display                 *display;
} MrmOsOpenParam, *MrmOsOpenParamPtr ;

typedef struct MrmHierarchyDescStruct *MrmHierarchy;
typedef struct {
    String                  name ;
    XtPointer               value ;
} MRMRegisterArg, MrmRegisterArg, *MrmRegisterArglist ;

#define URMwcUnknown           1
```

Figure 6-252: <Xm/MwmUtil.h>*, Part 1 of 3

```
typedef struct
{
    long      flags;
    long      functions;
    long      decorations;
    int       input_mode;
    long      status;
} MotifWmHints;

typedef MotifWmHints    MwmHints;

#define MWM_HINTS_FUNCTIONS      (1L << 0)
#define MWM_HINTS_DECORATIONS   (1L << 1)
#define MWM_HINTS_INPUT_MODE    (1L << 2)
#define MWM_HINTS_STATUS        (1L << 3)

#define MWM_FUNC_ALL             (1L << 0)
#define MWM_FUNC_RESIZE         (1L << 1)
#define MWM_FUNC_MOVE           (1L << 2)
#define MWM_FUNC_MINIMIZE       (1L << 3)
#define MWM_FUNC_MAXIMIZE       (1L << 4)
#define MWM_FUNC_CLOSE          (1L << 5)

#define MWM_DECOR_ALL           (1L << 0)
#define MWM_DECOR_BORDER        (1L << 1)
#define MWM_DECOR_RESIZEH       (1L << 2)
#define MWM_DECOR_TITLE         (1L << 3)
#define MWM_DECOR_MENU          (1L << 4)
#define MWM_DECOR_MINIMIZE      (1L << 5)
#define MWM_DECOR_MAXIMIZE      (1L << 6)

#define MWM_INPUT_MODELESS      0
#define MWM_INPUT_PRIMARY_APPLICATION_MODAL 1
#define MWM_INPUT_SYSTEM_MODAL 2
#define MWM_INPUT_FULL_APPLICATION_MODAL 3
```

Figure 6-253: <Xm/MwmUtil.h>*, Part 2 of 3

```
#define MWM_TEAROFF_WINDOW      (1L << 0)
#define MWM_INPUT_APPLICATION_MODAL    MWM_INPUT_PRIMARY_APPLICATION_MODAL

typedef struct
{
    long        flags;
    Window      wm_window;
} MotifWmInfo;

typedef MotifWmInfo    MwmInfo;

#define MWM_INFO_STARTUP_STANDARD    (1L << 0)
#define MWM_INFO_STARTUP_CUSTOM     (1L << 1)

typedef struct
{
    CARD32      flags;
    CARD32      functions;
    CARD32      decorations;
    INT32       inputMode;
    CARD32      status;
} PropMotifWmHints;

typedef PropMotifWmHints    PropMwmHints;

#define PROP_MOTIF_WM_HINTS_ELEMENTS    5
#define PROP_MWM_HINTS_ELEMENTS        PROP_MOTIF_WM_HINTS_ELEMENTS

#define _XA_MOTIF_WM_HINTS              "_MOTIF_WM_HINTS"
#define _XA_MWM_HINTS                   _XA_MOTIF_WM_HINTS

#define _XA_MOTIF_WM_MESSAGES            "_MOTIF_WM_MESSAGES"
#define _XA_MWM_MESSAGES                 _XA_MOTIF_WM_MESSAGES

#define _XA_MOTIF_WM_OFFSET              "_MOTIF_WM_OFFSET"
```

Figure 6-254: <Xm/MwmUtil.h>*, Part 3 of 3

```
#define _XA_MOTIF_WM_MENU      "_MOTIF_WM_MENU"
#define _XA_MWM_MENU          _XA_MOTIF_WM_MENU

typedef struct
{
    CARD32 flags;
    CARD32 wmWindow;
} PropMotifWmInfo;

typedef PropMotifWmInfo PropMwmInfo;

#define PROP_MOTIF_WM_INFO_ELEMENTS      2
#define PROP_MWM_INFO_ELEMENTS          PROP_MOTIF_WM_INFO_ELEMENTS

#define _XA_MOTIF_WM_INFO      "_MOTIF_WM_INFO"
#define _XA_MWM_INFO          _XA_MOTIF_WM_INFO

#define _XA_MOTIF_BINDINGS    "_MOTIF_BINDINGS"
```

Figure 6-255: <Xm/PanedW.h>*

```
typedef struct _XmPanedWindowClassRec    *XmPanedWindowWidgetClass;
typedef struct _XmPanedWindowRec         *XmPanedWindowWidget;
```

Figure 6-256: <Xm/PushB.h>*

```
typedef struct _XmPushButtonClassRec *XmPushButtonWidgetClass;
typedef struct _XmPushButtonRec      *XmPushButtonWidget;
```

Figure 6-257: <Xm/PushBG.h>*

```
typedef struct _XmPushButtonGadgetClassRec *XmPushButtonGadgetClass;
typedef struct _XmPushButtonGadgetRec     *XmPushButtonGadget;
typedef struct _XmPushButtonGCacheObjRec  *XmPushButtonGCacheObject;
```

Figure 6-258: <Xm/RepType.h>*

```
#define XmREP_TYPE_INVALID          0x1FFF

typedef unsigned short XmRepTypeId ;

typedef struct
{
    String rep_type_name ;
    String *value_names ;
    unsigned char *values ;
    unsigned char num_values ;
    Boolean reverse_installed ;
    XmRepTypeId rep_type_id ;
}XmRepTypeEntryRec, *XmRepTypeEntry, XmRepTypeListRec, *XmRepTypeList ;
```

Figure 6-259: <Xm/RowColumn.h>*

```
typedef struct _XmRowColumnClassRec * XmRowColumnWidgetClass;  
typedef struct _XmRowColumnRec      * XmRowColumnWidget;
```

Figure 6-260: <Xm/Scale.h>*

```
typedef struct _XmScaleClassRec * XmScaleWidgetClass;  
typedef struct _XmScaleRec      * XmScaleWidget;
```

Figure 6-261: <Xm/Screen.h>*

```
typedef struct _XmScreenRec      *XmScreen;  
typedef struct _XmScreenClassRec *XmScreenClass;
```

Figure 6-262: <Xm/ScrollBar.h>*

```
typedef struct _XmScrollBarClassRec * XmScrollBarWidgetClass;  
typedef struct _XmScrollBarRec      * XmScrollBarWidget;
```

Figure 6-263: <Xm/ScrolledW.h>*

```
typedef struct _XmScrolledWindowClassRec * XmScrolledWindowWidgetClass;
typedef struct _XmScrolledWindowRec      * XmScrolledWindowWidget;
```

Figure 6-264: <Xm/SelectioB.h>*

```
typedef struct _XmSelectionBoxClassRec * XmSelectionBoxWidgetClass;
typedef struct _XmSelectionBoxRec      * XmSelectionBoxWidget;
```

Figure 6-265: <Xm/SeparatoG.h>*

```
typedef struct _XmSeparatorGadgetClassRec * XmSeparatorGadgetClass;
typedef struct _XmSeparatorGadgetRec      * XmSeparatorGadget;
typedef struct _XmSeparatorGCacheObjRec   * XmSeparatorGCacheObject;
```

Figure 6-266: <Xm/Separator.h>*

```
typedef struct _XmSeparatorClassRec * XmSeparatorWidgetClass;
typedef struct _XmSeparatorRec      * XmSeparatorWidget;
```

Figure 6-267: <Xm/Text.h>*

```
typedef struct _XmTextSourceRec *XmTextSource;  
typedef struct _XmTextClassRec *XmTextWidgetClass;  
typedef struct _XmTextRec *XmTextWidget;
```

Figure 6-268: <Xm/TextF.h>*

```
typedef struct _XmTextFieldClassRec *XmTextFieldWidgetClass;  
typedef struct _XmTextFieldRec *XmTextFieldWidget;
```

Figure 6-269: <Xm/ToggleB.h>*

```
typedef struct _XmToggleButtonClassRec *XmToggleButtonWidgetClass;  
typedef struct _XmToggleButtonRec *XmToggleButtonWidget;
```

Figure 6-270: <Xm/ToggleBG.h>*

```
typedef struct _XmToggleButtonGadgetClassRec *XmToggleButtonGadgetClass;  
typedef struct _XmToggleButtonGadgetRec *XmToggleButtonGadget;  
typedef struct _XmToggleButtonGCacheObjRec *XmToggleButtonGCacheObject;
```

Figure 6-271: <Xm/VendorS.h>*

```
typedef struct _XmVendorShellRec      *XmVendorShellWidget;  
typedef struct _XmVendorShellClassRec *XmVendorShellWidgetClass;
```

Figure 6-272: <Xm/VirtKeys.h>*, **Part 1 of 2**

```
#define _OSF_Keysyms  
  
#define osfXK_BackSpace      0x1004FF08  
#define osfXK_Insert        0x1004FF63  
#define osfXK_Delete        0x1004FFFF  
#define osfXK_Copy          0x1004FF02  
#define osfXK_Cut           0x1004FF03  
#define osfXK_Paste         0x1004FF04  
  
#define osfXK_AddMode        0x1004FF31  
#define osfXK_PrimaryPaste   0x1004FF32  
#define osfXK_QuickPaste     0x1004FF33  
  
#define osfXK_PageLeft      0x1004FF40  
#define osfXK_PageUp        0x1004FF41  
#define osfXK_PageDown      0x1004FF42  
#define osfXK_PageRight     0x1004FF43
```

Figure 6-273: <Xm/VirtKeys.h>*, **Part 2 of 2**

#define osfXK_EndLine	0x1004FF57
#define osfXK_BeginLine	0x1004FF58
#define osfXK_Activate	0x1004FF44
#define osfXK_MenuBar	0x1004FF45
#define osfXK_Clear	0x1004FF0B
#define osfXK_Cancel	0x1004FF69
#define osfXK_Help	0x1004FF6A
#define osfXK_Menu	0x1004FF67
#define osfXK_Select	0x1004FF60
#define osfXK_Undo	0x1004FF65
#define osfXK_Left	0x1004FF51
#define osfXK_Up	0x1004FF52
#define osfXK_Right	0x1004FF53
#define osfXK_Down	0x1004FF54

Figure 6-274: <Xm/Xm.h>*, Part 1 of 14

```
#define XmUNSPECIFIED_PIXMAP          2

#define XmSTRING_OS_CHARSET           XmSTRING_ISO8859_1
#define XmFALLBACK_CHARSET           XmSTRING_ISO8859_1

#define XmDEFAULT_FONT                _XmSDEFAULT_FONT
#define XmDEFAULT_BACKGROUND          _XmSDEFAULT_BACKGROUND
#define XmDEFAULT_DARK_THRESHOLD      20
#define XmDEFAULT_LIGHT_THRESHOLD     90
#define XmDEFAULT_FOREGROUND_THRESHOLD 70

typedef enum{ XmFONT_IS_FONT, XmFONT_IS_FONTSET } XmFontType;

enum{    XmSTRING_DIRECTION_L_TO_R,      XmSTRING_DIRECTION_R_TO_L
        } ;
#define XmSTRING_DIRECTION_DEFAULT ((XmStringDirection) 255)

typedef unsigned char * XmString;
typedef XmString *      XmStringTable;
typedef char *          XmStringCharSet;
typedef unsigned char   XmStringComponentType;
typedef unsigned char   XmStringDirection;

typedef struct _XmFontListRec        *XmFontListEntry;
typedef struct _XmFontListRec        *XmFontList;
typedef struct __XmStringContextRec  *_XmStringContext;
typedef struct __XmStringRec         *_XmString;
typedef struct _XmtStringContextRec  *XmStringContext;
typedef struct _XmFontListContextRec *XmFontContext;

enum{    XmSTRING_COMPONENT_UNKNOWN,      XmSTRING_COMPONENT_CHARSET,
        XmSTRING_COMPONENT_TEXT,         XmSTRING_COMPONENT_DIRECTION,
        XmSTRING_COMPONENT_SEPARATOR,    XmSTRING_COMPONENT_LOCALE_TEXT
        } ;
```


Figure 6-275: <Xm/Xm.h>*, Part 2 of 14

```
#define XmSTRING_COMPONENT_END      ((XmStringComponentType) 126)
#define XmSTRING_COMPONENT_USER_BEGIN ((XmStringComponentType) 128)
#define XmSTRING_COMPONENT_USER_END ((XmStringComponentType) 255)

typedef struct _XmPrimitiveClassRec * XmPrimitiveWidgetClass;
typedef struct _XmPrimitiveRec      * XmPrimitiveWidget;

typedef struct _XmGadgetClassRec * XmGadgetClass;
typedef struct _XmGadgetRec      * XmGadget;

typedef struct _XmManagerClassRec * XmManagerWidgetClass;
typedef struct _XmManagerRec      * XmManagerWidget;

enum{
    XmCHANGE_ALL,                XmCHANGE_NONE,
    XmCHANGE_WIDTH,              XmCHANGE_HEIGHT
} ;

enum{
    XmPIXELS,                    Xm100TH_MILLIMETERS,
    Xm1000TH_INCHES,             Xm100TH_POINTS,
    Xm100TH_FONT_UNITS
} ;

enum{
    XmDESTROY,                   XmUNMAP,
    XmDO_NOHING
} ;

enum{
    XmEXPLICIT,                  XmPOINTER
} ;

enum{
    XmNONE,                      XmTAB_GROUP,
    XmSTICKY_TAB_GROUP,          XmEXCLUSIVE_TAB_GROUP
} ;

#define XmDYNAMIC_DEFAULT_TAB_GROUP ((XmNavigationType) 255)
```

Figure 6-276: <Xm/Xm.h>*, Part 3 of 14

```
enum{                                XmBELL = 1
    } ;

enum{  XmNO_ORIENTATION,            XmVERTICAL,
      XmHORIZONTAL
    } ;

enum{  XmWORK_AREA,                XmMENU_BAR,
      XmMENU_PULLDOWN,            XmMENU_POPUP,
      XmMENU_OPTION
    } ;

enum{  XmNO_PACKING,                XmPACK_TIGHT,
      XmPACK_COLUMN,              XmPACK_NONE
    } ;

enum{  XmALIGNMENT_CONTENTS_TOP = 3,
      XmALIGNMENT_CONTENTS_BOTTOM
    } ;

enum{  XmTEAR_OFF_ENABLED,          XmTEAR_OFF_DISABLED
    } ;

enum{  XmUNPOST,                    XmUNPOST_AND_REPLAY
    } ;

enum{  XmLAST_POSITION = -1,        XmFIRST_POSITION
    } ;

enum{  XmALIGNMENT_BEGINNING,        XmALIGNMENT_CENTER,
      XmALIGNMENT_END
    } ;
```

Figure 6-277: <Xm/Xm.h>*, Part 4 of 14

```
enum{    XmALIGNMENT_BASELINE_TOP,
        XmALIGNMENT_BASELINE_BOTTOM = 2, XmALIGNMENT_WIDGET_TOP,
        XmALIGNMENT_WIDGET_BOTTOM
    } ;

enum{    XmFRAME_GENERIC_CHILD,          XmFRAME_WORKAREA_CHILD,
        XmFRAME_TITLE_CHILD
    } ;

enum{    XmN_OF_MANY = 1,                XmONE_OF_MANY
    } ;

enum{    XmATTACH_NONE,                  XmATTACH_FORM,
        XmATTACH_OPPOSITE_FORM,        XmATTACH_WIDGET,
        XmATTACH_OPPOSITE_WIDGET,      XmATTACH_POSITION,
        XmATTACH_SELF
    } ;

enum{    XmRESIZE_NONE,                  XmRESIZE_GROW,
        XmRESIZE_ANY
    } ;
```

Figure 6-278: <Xm/Xm.h>*, Part 5 of 15

```
enum{    XmCR_NONE,                XmCR_HELP,
        XmCR_VALUE_CHANGED,      XmCR_INCREMENT,
        XmCR_DECREMENT,         XmCR_PAGE_INCREMENT,
        XmCR_PAGE_DECREMENT,    XmCR_TO_TOP,
        XmCR_TO_BOTTOM,         XmCR_DRAG,
        XmCR_ACTIVATE,          XmCR_ARM,
        XmCR_DISARM,            XmCR_MAP = 16,
        XmCR_UNMAP,             XmCR_FOCUS,
        XmCR_LOSING_FOCUS,       XmCR_MODIFYING_TEXT_VALUE,
        XmCR_MOVING_INSERT_CURSOR, XmCR_EXECUTE,
        XmCR_SINGLE_SELECT,      XmCR_MULTIPLE_SELECT,
        XmCR_EXTENDED_SELECT,    XmCR_BROWSE_SELECT,
        XmCR_DEFAULT_ACTION,     XmCR_CLIPBOARD_DATA_REQUEST,
        XmCR_CLIPBOARD_DATA_DELETE, XmCR_CASCADING,
        XmCR_OK,                 XmCR_CANCEL,
        XmCR_APPLY = 34,         XmCR_NO_MATCH,
        XmCR_COMMAND_ENTERED,    XmCR_COMMAND_CHANGED,
        XmCR_EXPOSE,             XmCR_RESIZE,
        XmCR_INPUT,              XmCR_GAIN_PRIMARY,
        XmCR_LOSE_PRIMARY,       XmCR_CREATE,
        XmCR_TEAR_OFF_ACTIVATE,  XmCR_TEAR_OFF_DEACTIVATE,
        XmCR_OBSCURED_TRAVERSAL
    } ;

typedef struct
{
    int      reason;
    XEvent   *event;
} XmAnyCallbackStruct;

typedef struct
{
    int      reason;
    XEvent   *event;
    int      click_count;
} XmArrowButtonCallbackStruct;
```

Figure 6-279: <Xm/Xm.h>*, Part 6 of 14

```
typedef struct
{
    int      reason;
    XEvent   *event;
    Window   window;
} XmDrawingAreaCallbackStruct;

typedef struct
{
    int      reason;
    XEvent   *event;
    Window   window;
    int      click_count;
} XmDrawnButtonCallbackStruct;

typedef struct
{
    int      reason;
    XEvent   *event;
    int      click_count;
} XmPushButtonCallbackStruct;

typedef struct
{
    int      reason;
    XEvent   *event;
    Widget   widget;
    char     *data;
    char     *callbackstruct;
} XmRowColumnCallbackStruct;
```

Figure 6-280: <Xm/Xm.h>*, Part 7 of 14

```
typedef struct
{
    int reason;
    XEvent * event;
    int value;
    int pixel;
} XmScrollBarCallbackStruct;

typedef struct
{
    int reason;
    XEvent * event;
    int set;
} XmToggleButtonCallbackStruct;

typedef struct
{
    int      reason;
    XEvent   *event;
    XmString item;
    int      item_length;
    int      item_position;
    XmString *selected_items;
    int      selected_item_count;
    int      *selected_item_positions;
    char     selection_type;
} XmListCallbackStruct;

typedef struct
{
    int reason;
    XEvent *event;
    XmString value;
    int length;
} XmSelectionBoxCallbackStruct;
```

Figure 6-281: <Xm/Xm.h>*, Part 8 of 14

```
typedef struct
{
    int reason;
    XEvent      *event;
    XmString     value;
    int          length;
} XmCommandCallbackStruct;

typedef struct
{
    int          reason;
    XEvent      *event;
    XmString     value;
    int          length;
    XmString     mask;
    int          mask_length;
    XmString     dir ;
    int          dir_length ;
    XmString     pattern ;
    int          pattern_length ;
} XmFileSelectionBoxCallbackStruct;

typedef struct
{
    int reason;
    XEvent * event;
    int value;
} XmScaleCallbackStruct;

enum{   XmMULTICLICK_DISCARD,           XmMULTICLICK_KEEP
        } ;

enum{   XmSHADOW_IN = 7,               XmSHADOW_OUT
        } ;
```

Figure 6-282: <Xm/Xm.h>*, Part 9 of 14

enum{	XmARROW_UP, XmARROW_LEFT, } ;	XmARROW_DOWN, XmARROW_RIGHT
enum{	XmNO_LINE, XmDOUBLE_LINE, XmDOUBLE_DASHED_LINE, XmSHADOW_ETCHED_OUT, XmSHADOW_ETCHED_OUT_DASH, } ;	XmSINGLE_LINE, XmSINGLE_DASHED_LINE, XmSHADOW_ETCHED_IN, XmSHADOW_ETCHED_IN_DASH, XmINVALID_SEPARATOR_TYPE
enum{	XmPIXMAP = 1, } ;	XmSTRING
enum{	XmWINDOW, XmCURSOR = 2 } ;	
enum{	XmMAX_ON_TOP, XmMAX_ON_LEFT, } ;	XmMAX_ON_BOTTOM, XmMAX_ON_RIGHT
enum{	XmSINGLE_SELECT, XmEXTENDED_SELECT, } ;	XmMULTIPLE_SELECT, XmBROWSE_SELECT
enum{	XmSTATIC, } ;	XmDYNAMIC
enum{	XmVARIABLE, XmRESIZE_IF_POSSIBLE } ;	XmCONSTANT,
enum{	XmAUTOMATIC, } ;	XmAPPLICATION_DEFINED

Figure 6-283: <Xm/Xm.h>*, Part 10 of 14

```
enum{ XmAS_NEEDED = 1
      } ;

#define SW_TOP          1
#define SW_BOTTOM      0
#define SW_LEFT        2
#define SW_RIGHT       0

#define XmTOP_LEFT      (SW_TOP | SW_LEFT)
#define XmBOTTOM_LEFT  (SW_BOTTOM | SW_LEFT)
#define XmTOP_RIGHT     (SW_TOP | SW_RIGHT)
#define XmBOTTOM_RIGHT (SW_BOTTOM | SW_RIGHT)

enum{ XmCOMMAND_ABOVE_WORKSPACE,      XmCOMMAND_BELOW_WORKSPACE
      } ;

enum{ XmMULTI_LINE_EDIT,              XmSINGLE_LINE_EDIT
      } ;

typedef enum{
    XmTEXT_FORWARD,
    XmTEXT_BACKWARD
} XmTextDirection;

typedef long XmTextPosition;
typedef Atom XmTextFormat;

#define XmFMT_8_BIT      ((XmTextFormat) XA_STRING)
#define XmFMT_16_BIT     ((XmTextFormat) 2)

#define FMT8BIT          XmFMT_8_BIT
#define FMT16BIT         XmFMT_16_BIT
```

Figure 6-284: <Xm/Xm.h>*, Part 11 of 14

```
typedef enum{
    XmSELECT_POSITION,          XmSELECT_WHITESPACE,
    XmSELECT_WORD,              XmSELECT_LINE,
    XmSELECT_ALL,               XmSELECT_PARAGRAPH
} XmTextScanType ;

typedef enum{
    XmHIGHLIGHT_NORMAL,          XmHIGHLIGHT_SELECTED,
    XmHIGHLIGHT_SECONDARY_SELECTED
} XmHighlightMode ;

typedef struct {
    char *ptr;
    int length;
    XmTextFormat format;
} XmTextBlockRec, *XmTextBlock;

typedef struct
{
    int reason;
    XEvent *event;
    Boolean doit;
    long currInsert, newInsert;
    long startPos, endPos;
    XmTextBlock text;
} XmTextVerifyCallbackStruct, *XmTextVerifyPtr;

typedef struct {
    wchar_t *wcsptr;
    int length;
} XmTextBlockRecWcs, *XmTextBlockWcs;
```

Figure 6-285: <Xm/Xm.h>*, Part 12 of 14

```
typedef struct
{
    int reason;
    XEvent *event;
    Boolean doit;
    long currInsert, newInsert;
    long startPos, endPos;
    XmTextBlockWcs text;
} XmTextVerifyCallbackStructWcs, *XmTextVerifyPtrWcs;

#define XmTextGetTopPosition      XmTextGetTopCharacter
#define XmTextSetTopPosition      XmTextSetTopCharacter

#define XmCOPY_FAILED            0
#define XmCOPY_SUCCEEDED        1
#define XmCOPY_TRUNCATED        2

enum{
    XmDIALOG_NONE,                XmDIALOG_APPLY_BUTTON,
    XmDIALOG_CANCEL_BUTTON,       XmDIALOG_DEFAULT_BUTTON,
    XmDIALOG_OK_BUTTON,           XmDIALOG_FILTER_LABEL,
    XmDIALOG_FILTER_TEXT,         XmDIALOG_HELP_BUTTON,
    XmDIALOG_LIST,                XmDIALOG_LIST_LABEL,
    XmDIALOG_MESSAGE_LABEL,       XmDIALOG_SELECTION_LABEL,
    XmDIALOG_SYMBOL_LABEL,        XmDIALOG_TEXT,
    XmDIALOG_SEPARATOR,           XmDIALOG_DIR_LIST,
    XmDIALOG_DIR_LIST_LABEL
} ;

#define XmDIALOG_HISTORY_LIST     XmDIALOG_LIST
#define XmDIALOG_PROMPT_LABEL     XmDIALOG_SELECTION_LABEL
#define XmDIALOG_VALUE_TEXT       XmDIALOG_TEXT
#define XmDIALOG_COMMAND_TEXT     XmDIALOG_TEXT
#define XmDIALOG_FILE_LIST        XmDIALOG_LIST
#define XmDIALOG_FILE_LIST_LABEL  XmDIALOG_LIST_LABEL
```

Figure 6-286: <Xm/Xm.h>*, Part 13 of 14

```
enum{    XmDIALOG_MODELESS,                XmDIALOG_PRIMARY_APPLICATION_MODAL,
        XmDIALOG_FULL_APPLICATION_MODAL, XmDIALOG_SYSTEM_MODAL
    } ;

#define XmDIALOG_APPLICATION_MODAL    XmDIALOG_PRIMARY_APPLICATION_MODAL

enum{    XmPLACE_TOP,                        XmPLACE_ABOVE_SELECTION,
        XmPLACE_BELOW_SELECTION
    } ;

#define XmFILE_DIRECTORY (1 << 0)
#define XmFILE_REGULAR   (1 << 1)
#define XmFILE_ANY_TYPE  (XmFILE_DIRECTORY | XmFILE_REGULAR)

enum{    XmDIALOG_WORK_AREA,                XmDIALOG_PROMPT,
        XmDIALOG_SELECTION,                XmDIALOG_COMMAND,
        XmDIALOG_FILE_SELECTION
    } ;

enum{    XmDIALOG_TEMPLATE,                XmDIALOG_ERROR,
        XmDIALOG_INFORMATION,             XmDIALOG_MESSAGE,
        XmDIALOG_QUESTION,                XmDIALOG_WARNING,
        XmDIALOG_WORKING
    } ;

typedef enum{
    XmVISIBILITY_UNOBSURED,                XmVISIBILITY_PARTIALLY_OBSCURED,
    XmVISIBILITY_FULLY_OBSCURED
} XmVisibility ;

typedef enum{
    XmTRAVERSE_CURRENT,                    XmTRAVERSE_NEXT,
    XmTRAVERSE_PREV,                       XmTRAVERSE_HOME,
    XmTRAVERSE_NEXT_TAB_GROUP,             XmTRAVERSE_PREV_TAB_GROUP,
    XmTRAVERSE_UP,                         XmTRAVERSE_DOWN,
    XmTRAVERSE_LEFT,                       XmTRAVERSE_RIGHT
} XmTraversalDirection ;
```

Figure 6-287: <Xm/Xm.h>*, Part 14 of 14

```
typedef struct _XmTraverseObscuredCallbackStruct
{
    int                reason ;
    XEvent *           event ;
    Widget             traversal_destination ;
    XmTraversalDirection direction ;
} XmTraverseObscuredCallbackStruct ;

typedef unsigned char  XmNavigationType;
typedef unsigned char  XmButtonType;
typedef XmButtonType * XmButtonTypeTable;
typedef KeySym *       XmKeySymTable;
typedef XmStringCharSet * XmStringCharSetTable;

enum{  XmPUSHBUTTON = 1,                XmTOGGLEBUTTON,
       XmRADIOBUTTON,                XmCASCADEBUTTON,
       XmSEPARATOR,                  XmDOUBLE_SEPARATOR,
       XmTITLE
    } ;
#define XmCHECKBUTTON                XmTOGGLEBUTTON

typedef struct _XmSecondaryResourceDataRec{
    XmResourceBaseProc  base_proc;
    XtPointer           client_data;
    String              name;
    String              res_class;
    XtResourceList      resources;
    Cardinal            num_resources;
}XmSecondaryResourceDataRec, *XmSecondaryResourceData;
typedef long XmOffset;
typedef XmOffset *XmOffsetPtr;
```

Figure 6-288: <Xm/XmStrDefs.h>*, Part 1 of 34

```
#define XmS ""
#define XmCAccelerator "Accelerator"
#define XmCAcceleratorText "AcceleratorText"
#define XmCAdjustLast "AdjustLast"
#define XmCAdjustMargin "AdjustMargin"
#define XmCAlignment "Alignment"
#define XmCAllowOverlap "AllowOverlap"
#define XmCAnimationMask "AnimationMask"
#define XmCAnimationPixmap "AnimationPixmap"
#define XmCAnimationPixmapDepth "AnimationPixmapDepth"
#define XmCAnimationStyle "AnimationStyle"
#define XmCApplyLabelString "ApplyLabelString"
#define XmCArmCallback "ArmCallback"
#define XmCArmColor "ArmColor"
#define XmCArmPixmap "ArmPixmap"
#define XmCArrowDirection "ArrowDirection"
#define XmCAttachment "Attachment"
#define XmCAudibleWarning "AudibleWarning"
#define XmCAutoShowCursorPosition "AutoShowCursorPosition"
#define XmCAutoUnmanage "AutoUnmanage"
#define XmCAutomaticSelection "AutomaticSelection"
#define XmCAvailability "Availability"
#define XmCBackgroundPixmap "BackgroundPixmap"
#define XmCBlendModel "BlendModel"
#define XmCBlinkRate "BlinkRate"
#define XmCBottomShadowColor "BottomShadowColor"
#define XmCBottomShadowPixmap "BottomShadowPixmap"
#define XmCButtonAcceleratorText "ButtonAcceleratorText"
#define XmCButtonAccelerators "ButtonAccelerators"
#define XmCButtonCount "ButtonCount"
#define XmCButtonFontList "ButtonFontList"
#define XmCButtonMnemonicCharSets "ButtonMnemonicCharSets"
#define XmCButtonMnemonics "ButtonMnemonics"
#define XmCButtonSet "ButtonSet"
#define XmCButtonType "ButtonType"
```

Figure 6-289: <Xm/XmStrDefs.h>*, Part 2 of 34

```
#define XmCButtons "Buttons"
#define XmCCancelLabelString "CancelLabelString"
#define XmCChildHorizontalAlignment "ChildHorizontalAlignment"
#define XmCChildHorizontalSpacing "ChildHorizontalSpacing"
#define XmCChildPlacement "ChildPlacement"
#define XmCChildType "ChildType"
#define XmCChildVerticalAlignment "ChildVerticalAlignment"
#define XmCChildren "Children"
#define XmCClientData "ClientData"
#define XmCClipWindow "ClipWindow"
#define XmCColumns "Columns"
#define XmCCommandWindow "CommandWindow"
#define XmCCommandWindowLocation "CommandWindowLocation"
#define XmCConvertProc "ConvertProc"
#define XmCCursorBackground "CursorBackground"
#define XmCCursorForeground "CursorForeground"
#define XmCCursorPosition "CursorPosition"
#define XmCCursorPositionVisible "CursorPositionVisible"
#define XmCDarkThreshold "DarkThreshold"
#define XmCDecimalPoints "DecimalPoints"
#define XmCDefaultButtonShadowThickness "DefaultButtonShadowThickness"
#define XmCDefaultButtonType "DefaultButtonType"
#define XmCDefaultCopyCursorIcon "DefaultCopyCursorIcon"
#define XmCDefaultFontList "DefaultFontList"
#define XmCDefaultInvalidCursorIcon "DefaultInvalidCursorIcon"
#define XmCDefaultLinkCursorIcon "DefaultLinkCursorIcon"
#define XmCDefaultMoveCursorIcon "DefaultMoveCursorIcon"
#define XmCDefaultNoneCursorIcon "DefaultNoneCursorIcon"
#define XmCDefaultPosition "DefaultPosition"
#define XmCDefaultSourceCursorIcon "DefaultSourceCursorIcon"
#define XmCDefaultValidCursorIcon "DefaultValidCursorIcon"
#define XmCDeleteResponse "DeleteResponse"
#define XmCDesktopParent "DesktopParent"
#define XmCDialogStyle "DialogStyle"
#define XmCDialogTitle "DialogTitle"
```

Figure 6-290: <Xm/XmStrDefs.h>*, Part 3 of 34

```
#define XmCDialogType "DialogType"
#define XmCDirListItemCount "DirListItemCount"
#define XmCDirListItems "DirListItems"
#define XmCDirListLabelString "DirListLabelString"
#define XmCDirMask "DirMask"
#define XmCDirSearchProc "DirSearchProc"
#define XmCDirSpec "DirSpec"
#define XmCDirectory "Directory"
#define XmCDirectoryValid "DirectoryValid"
#define XmCDisarmCallback "DisarmCallback"
#define XmCDoubleClickInterval "DoubleClickInterval"
#define XmCDragContextClass "DragContextClass"
#define XmCDragDropFinishCallback "DragDropFinishCallback"
#define XmCDragIconClass "DragIconClass"
#define XmCDragInitiatorProtocolStyle "DragInitiatorProtocolStyle"
#define XmCDragMotionCallback "DragMotionCallback"
#define XmCDragOperations "DragOperations"
#define XmCDragOverMode "DragOverMode"
#define XmCDragProc "DragProc"
#define XmCDragReceiverProtocolStyle "DragReceiverProtocolStyle"
#define XmCDropProc "DropProc"
#define XmCDropRectangles "DropRectangles"
#define XmCDropSiteActivity "DropSiteActivity"
#define XmCDropSiteEnterCallback "DropSiteEnterCallback"
#define XmCDropSiteLeaveCallback "DropSiteLeaveCallback"
#define XmCDropSiteManagerClass "DropSiteManagerClass"
#define XmCDropSiteOperations "DropSiteOperations"
#define XmCDropSiteType "DropSiteType"
#define XmCDropStartCallback "DropStartCallback"
#define XmCDropTransferClass "DropTransferClass"
#define XmCDropTransfers "DropTransfers"
#define XmCEditable "Editable"
#define XmCEntryBorder "EntryBorder"
#define XmCEntryClass "EntryClass"
#define XmCExportTargets "ExportTargets"
```


Figure 6-291: <Xm/XmStrDefs.h>*, Part 4 of 34

```
#define XmCExposeCallback "ExposeCallback"
#define XmCExtensionType "ExtensionType"
#define XmCFileListItemCount "FileListItemCount"
#define XmCFileListItems "FileListItems"
#define XmCFileListLabelString "FileListLabelString"
#define XmCFileSearchProc "FileSearchProc"
#define XmCFileTypeMask "FileTypeMask"
#define XmCFillOnArm "FillOnArm"
#define XmCFillOnSelect "FillOnSelect"
#define XmCFilterLabelString "FilterLabelString"
#define XmCFontList "FontList"
#define XmCForegroundThreshold "ForegroundThreshold"
#define XmCHelpLabelString "HelpLabelString"
#define XmCHighlightColor "HighlightColor"
#define XmCHighlightOnEnter "HighlightOnEnter"
#define XmCHighlightPixmap "HighlightPixmap"
#define XmCHighlightThickness "HighlightThickness"
#define XmCHorizontalFontUnit "HorizontalFontUnit"
#define XmCHorizontalScrollBar "HorizontalScrollBar"
#define XmCHot "Hot"
#define XmCICCHandle "ICCHandle"
#define XmCImportTargets "ImportTargets"
#define XmCIncrement "Increment"
#define XmCIncremental "Incremental"
#define XmCIndicatorOn "IndicatorOn"
#define XmCIndicatorSize "IndicatorSize"
#define XmCIndicatorType "IndicatorType"
#define XmCInitialDelay "InitialDelay"
#define XmCInitialFocus "InitialFocus"
#define XmCInputCreate "InputCreate"
#define XmCInputMethod "InputMethod"
#define XmCInvalidCursorForeground "InvalidCursorForeground"
#define XmCIsAligned "IsAligned"
#define XmCIsHomogeneous "IsHomogeneous"
#define XmCItemCount "ItemCount"
```

Figure 6-292: <Xm/XmStrDefs.h>*, Part 5 of 34

```
#define XmCItems "Items"
#define XmCKeyboardFocusPolicy "KeyboardFocusPolicy"
#define XmCLabelFontList "LabelFontList"
#define XmCLabelInsensitivePixmap "LabelInsensitivePixmap"
#define XmCLabelPixmap "LabelPixmap"
#define XmCLabelString "LabelString"
#define XmCLabelType "LabelType"
#define XmCLightThreshold "LightThreshold"
#define XmCListLabelString "ListLabelString"
#define XmCListMarginHeight "ListMarginHeight"
#define XmCListMarginWidth "ListMarginWidth"
#define XmCListSizePolicy "ListSizePolicy"
#define XmCListSpacing "ListSpacing"
#define XmCListUpdated "ListUpdated"
#define XmCLogicalParent "LogicalParent"
#define XmCMainWindowMarginHeight "MainWindowMarginHeight"
#define XmCMainWindowMarginWidth "MainWindowMarginWidth"
#define XmCMappingDelay "MappingDelay"
#define XmCMarginBottom "MarginBottom"
#define XmCMarginHeight "MarginHeight"
#define XmCMarginLeft "MarginLeft"
#define XmCMarginRight "MarginRight"
#define XmCMarginTop "MarginTop"
#define XmCMarginWidth "MarginWidth"
#define XmCMask "Mask"
#define XmCMaxItems "MaxItems"
#define XmCMaxLength "MaxLength"
#define XmCMaxValue "MaxValue"
#define XmCMaximum "Maximum"
#define XmCMenuBar "MenuBar"
#define XmCMenuPost "MenuPost"
#define XmCMenuWidget "MenuWidget"
#define XmCMessageProc "MessageProc"
#define XmCMessageWindow "MessageWindow"
#define XmCMinimizeButtons "MinimizeButtons"
```

Figure 6-293: <Xm/XmStrDefs.h>*, Part 6 of 34

```
#define XmCMinimum "Minimum"
#define XmCMnemonic "Mnemonic"
#define XmCMnemonicCharSet "MnemonicCharSet"
#define XmCMoveOpaque "MoveOpaque"
#define XmCMultiClick "MultiClick"
#define XmCMustMatch "MustMatch"
#define XmCMwmDecorations "MwmDecorations"
#define XmCMwmFunctions "MwmFunctions"
#define XmCMwmInputMode "MwmInputMode"
#define XmCMwmMenu "MwmMenu"
#define XmCMwmMessages "MwmMessages"
#define XmCNavigationType "NavigationType"
#define XmCNeedsMotion "NeedsMotion"
#define XmCNoMatchString "NoMatchString"
#define XmCNoResize "NoResize"
#define XmCNoneCursorForeground "NoneCursorForeground"
#define XmCNotifyProc "NotifyProc"
#define XmCNumChildren "NumChildren"
#define XmCNumColumns "NumColumns"
#define XmCNumDropRectangles "NumDropRectangles"
#define XmCNumDropTransfers "NumDropTransfers"
#define XmCNumExportTargets "NumExportTargets"
#define XmCNumImportTargets "NumImportTargets"
#define XmCOffset "Offset"
#define XmCOkLabelString "OkLabelString"
#define XmCOperationChangedCallback "OperationChangedCallback"
#define XmCOperationCursorIcon "OperationCursorIcon"
#define XmCOptionLabel "OptionLabel"
#define XmCOptionMnemonic "OptionMnemonic"
#define XmCOutputCreate "OutputCreate"
#define XmCPacking "Packing"
#define XmCPageIncrement "PageIncrement"
#define XmCPaneMaximum "PaneMaximum"
#define XmCPaneMinimum "PaneMinimum"
#define XmCPattern "Pattern"
```

Figure 6-294: <Xm/XmStrDefs.h>*, Part 7 of 34

```
#define XmCPendingDelete "PendingDelete"
#define XmCPopupEnabled "PopupEnabled"
#define XmCPositionIndex "PositionIndex"
#define XmCPostFromButton "PostFromButton"
#define XmCPostFromCount "PostFromCount"
#define XmCPostFromList "PostFromList"
#define XmCPreeditType "PreeditType"
#define XmCProcessingDirection "ProcessingDirection"
#define XmCPromptString "PromptString"
#define XmCProtocolCallback "ProtocolCallback"
#define XmCPushButtonEnabled "PushButtonEnabled"
#define XmCQualifySearchDataProc "QualifySearchDataProc"
#define XmCRadioAlwaysOne "RadioAlwaysOne"
#define XmCRadioBehavior "RadioBehavior"
#define XmCRecomputeSize "RecomputeSize"
#define XmCRectangles "Rectangles"
#define XmCRepeatDelay "RepeatDelay"
#define XmCResizeCallback "ResizeCallback"
#define XmCResizeHeight "ResizeHeight"
#define XmCResizePolicy "ResizePolicy"
#define XmCResizeWidth "ResizeWidth"
#define XmCRowColumnType "RowColumnType"
#define XmCRows "Rows"
#define XmCRubberPositioning "RubberPositioning"
#define XmCSashHeight "SashHeight"
#define XmCSashIndent "SashIndent"
#define XmCSashWidth "SashWidth"
#define XmCScaleHeight "ScaleHeight"
#define XmCScaleMultiple "ScaleMultiple"
#define XmCScaleWidth "ScaleWidth"
#define XmCScroll "Scroll"
#define XmCScrollBarDisplayPolicy "ScrollBarDisplayPolicy"
#define XmCScrollBarPlacement "ScrollBarPlacement"
#define XmCScrollSide "ScrollSide"
#define XmCScrolledWindowMarginHeight "ScrolledWindowMarginHeight"
```

Figure 6-295: <Xm/XmStrDefs.h>*, Part 8 of 34

```
#define XmCScrolledWindowMarginWidth "ScrolledWindowMarginWidth"
#define XmCScrollingPolicy "ScrollingPolicy"
#define XmCSelectColor "SelectColor"
#define XmCSelectInsensitivePixmap "SelectInsensitivePixmap"
#define XmCSelectPixmap "SelectPixmap"
#define XmCSelectThreshold "SelectThreshold"
#define XmCSelectedItemCount "SelectedItemCount"
#define XmCSelectedItems "SelectedItems"
#define XmCSelectionArrayCount "SelectionArrayCount"
#define XmCSelectionLabelString "SelectionLabelString"
#define XmCSelectionPolicy "SelectionPolicy"
#define XmCSeparatorOn "SeparatorOn"
#define XmCSeparatorType "SeparatorType"
#define XmCSet "Set"
#define XmCShadowThickness "ShadowThickness"
#define XmCShadowType "ShadowType"
#define XmCShellUnitType "ShellUnitType"
#define XmCShowArrows "ShowArrows"
#define XmCShowAsDefault "ShowAsDefault"
#define XmCShowSeparator "ShowSeparator"
#define XmCShowValue "ShowValue"
#define XmCSimpleCheckBox "SimpleCheckBox"
#define XmCSimpleMenuBar "SimpleMenuBar"
#define XmCSimpleOptionMenu "SimpleOptionMenu"
#define XmCSimplePopupMenu "SimplePopupMenu"
#define XmCSimplePulldownMenu "SimplePulldownMenu"
#define XmCSimpleRadioBox "SimpleRadioBox"
#define XmCSizePolicy "SizePolicy"
#define XmCSliderSize "SliderSize"
#define XmCSource "Source"
#define XmCSourceCursorIcon "SourceCursorIcon"
#define XmCSourceIsExternal "SourceIsExternal"
#define XmCSourcePixmapIcon "SourcePixmapIcon"
#define XmCSourceWidget "SourceWidget"
#define XmCSourceWindow "SourceWindow"
```

Figure 6-296: <Xm/XmStrDefs.h>*, Part 9 of 34

```
#define XmCSpacing "Spacing"
#define XmCStartTime "StartTime"
#define XmCStateCursorIcon "StateCursorIcon"
#define XmCStringDirection "StringDirection"
#define XmCTearOffModel "TearOffModel"
#define XmCTextFontList "TextFontList"
#define XmCTextString "TextString"
#define XmCTextValue "TextValue"
#define XmCTitleString "TitleString"
#define XmCTopCharacter "TopCharacter"
#define XmCTopItemPosition "TopItemPosition"
#define XmCTopLevelEnterCallback "TopLevelEnterCallback"
#define XmCTopLevelLeaveCallback "TopLevelLeaveCallback"
#define XmCTopShadowColor "TopShadowColor"
#define XmCTopShadowPixmap "TopShadowPixmap"
#define XmCTransferProc "TransferProc"
#define XmCTransferStatus "TransferStatus"
#define XmCTraversalOn "TraversalOn"
#define XmCTraversalType "TraversalType"
#define XmCTreeUpdateProc "TreeUpdateProc"
#define XmCTroughColor "TroughColor"
#define XmCUnitType "UnitType"
#define XmCUnpostBehavior "UnpostBehavior"
#define XmCUnselectPixmap "UnselectPixmap"
#define XmCUpdateSliderSize "UpdateSliderSize"
#define XmCUseAsyncGeometry "UseAsyncGeometry"
#define XmCUserData "UserData"
#define XmCValidCursorForeground "ValidCursorForeground"
#define XmCValueChangedCallback "ValueChangedCallback"
#define XmCValueWcs "ValueWcs"
#define XmCVerifyBell "VerifyBell"
#define XmCVerticalAlignment "VerticalAlignment"
#define XmCVerticalFontUnit "VerticalFontUnit"
#define XmCVerticalScrollBar "VerticalScrollBar"
```

Figure 6-297: <Xm/XmStrDefs.h>*, Part 10 of 34

```
#define XmCVisibleItemCount "VisibleItemCount"
#define XmCVisibleWhenOff "VisibleWhenOff"
#define XmCVisualPolicy "VisualPolicy"
#define XmCWhichButton "WhichButton"
#define XmCWordWrap "WordWrap"
#define XmCWorkWindow "WorkWindow"
#define XmCXmString "XmString"
#define XmNaccelerator "accelerator"
#define XmNacceleratorText "acceleratorText"
#define XmNactivateCallback "activateCallback"
#define XmNadjustLast "adjustLast"
#define XmNadjustMargin "adjustMargin"
#define XmNalignment "alignment"
#define XmNallowOverlap "allowOverlap"
#define XmNallowResize "allowResize"
#define XmNanimationMask "animationMask"
#define XmNanimationPixmap "animationPixmap"
#define XmNanimationPixmapDepth "animationPixmapDepth"
#define XmNanimationStyle "animationStyle"
#define XmNapplyCallback "applyCallback"
#define XmNapplyLabelString "applyLabelString"
#define XmNarmCallback "armCallback"
#define XmNarmColor "armColor"
#define XmNarmPixmap "armPixmap"
#define XmNarrowDirection "arrowDirection"
#define XmNattachment "attachment"
#define XmNaudibleWarning "audibleWarning"
#define XmNautoShowCursorPosition "autoShowCursorPosition"
#define XmNautoUnmanage "autoUnmanage"
#define XmNautomaticSelection "automaticSelection"
#define XmNavailability "availability"
#define XmNblendModel "blendModel"
#define XmNblinkRate "blinkRate"
#define XmNbottomAttachment "bottomAttachment"
#define XmNbottomOffset "bottomOffset"
```

Figure 6-298: <Xm/XmStrDefs.h>*, Part 11 of 34

```
#define XmNbottomPosition "bottomPosition"
#define XmNbottomShadowColor "bottomShadowColor"
#define XmNbottomShadowPixmap "bottomShadowPixmap"
#define XmNbottomWidget "bottomWidget"
#define XmNbrowseSelectionCallback "browseSelectionCallback"
#define XmNbuttonAcceleratorText "buttonAcceleratorText"
#define XmNbuttonAccelerators "buttonAccelerators"
#define XmNbuttonCount "buttonCount"
#define XmNbuttonFontList "buttonFontList"
#define XmNbuttonMnemonicCharSets "buttonMnemonicCharSets"
#define XmNbuttonMnemonics "buttonMnemonics"
#define XmNbuttonSet "buttonSet"
#define XmNbuttonType "buttonType"
#define XmNbuttons "buttons"
#define XmNcancelButton "cancelButton"
#define XmNcancelCallback "cancelCallback"
#define XmNcancelLabelString "cancelLabelString"
#define XmNcascadePixmap "cascadePixmap"
#define XmNcascadingCallback "cascadingCallback"
#define XmNchildHorizontalAlignment "childHorizontalAlignment"
#define XmNchildHorizontalSpacing "childHorizontalSpacing"
#define XmNchildPlacement "childPlacement"
#define XmNchildPosition "childPosition"
#define XmNchildType "childType"
#define XmNchildVerticalAlignment "childVerticalAlignment"
#define XmNclientData "clientData"
#define XmNclipWindow "clipWindow"
#define XmNcolumns "columns"
#define XmNcommand "command"
#define XmNcommandChangedCallback "commandChangedCallback"
#define XmNcommandEnteredCallback "commandEnteredCallback"
#define XmNcommandWindow "commandWindow"
#define XmNcommandWindowLocation "commandWindowLocation"
#define XmNconvertProc "convertProc"
#define XmNcursorBackground "cursorBackground"
```


Figure 6-299: <Xm/XmStrDefs.h>*, Part 12 of 34

```
#define XmNcursorForeground "cursorForeground"
#define XmNcursorPosition "cursorPosition"
#define XmNcursorPositionVisible "cursorPositionVisible"
#define XmNdarkThreshold "darkThreshold"
#define XmNdecimalPoints "decimalPoints"
#define XmNdecrementCallback "decrementCallback"
#define XmNdefaultActionCallback "defaultActionCallback"
#define XmNdefaultButton "defaultButton"
#define XmNdefaultButtonShadowThickness "defaultButtonShadowThickness"
#define XmNdefaultButtonType "defaultButtonType"
#define XmNdefaultCopyCursorIcon "defaultCopyCursorIcon"
#define XmNdefaultFontList "defaultFontList"
#define XmNdefaultInvalidCursorIcon "defaultInvalidCursorIcon"
#define XmNdefaultLinkCursorIcon "defaultLinkCursorIcon"
#define XmNdefaultMoveCursorIcon "defaultMoveCursorIcon"
#define XmNdefaultNoneCursorIcon "defaultNoneCursorIcon"
#define XmNdefaultPosition "defaultPosition"
#define XmNdefaultSourceCursorIcon "defaultSourceCursorIcon"
#define XmNdefaultValidCursorIcon "defaultValidCursorIcon"
#define XmNdeleteResponse "deleteResponse"
#define XmNdesktopParent "desktopParent"
#define XmNdialogStyle "dialogStyle"
#define XmNdialogTitle "dialogTitle"
#define XmNdialogType "dialogType"
#define XmNdirListItemCount "dirListItemCount"
#define XmNdirListItems "dirListItems"
#define XmNdirListLabelString "dirListLabelString"
#define XmNdirMask "dirMask"
#define XmNdirSearchProc "dirSearchProc"
#define XmNdirSpec "dirSpec"
#define XmNdirectory "directory"
#define XmNdirectoryValid "directoryValid"
#define XmNdisarmCallback "disarmCallback"
#define XmNdoubleClickInterval "doubleClickInterval"
#define XmNdragCallback "dragCallback"
```

Figure 6-300: <Xm/XmStrDefs.h>*, Part 13 of 34

```
#define XmNdragContextClass "dragContextClass"
#define XmNdragDropFinishCallback "dragDropFinishCallback"
#define XmNdragIconClass "dragIconClass"
#define XmNdragInitiatorProtocolStyle "dragInitiatorProtocolStyle"
#define XmNdragMotionCallback "dragMotionCallback"
#define XmNdragOperations "dragOperations"
#define XmNdragOverMode "dragOverMode"
#define XmNdragProc "dragProc"
#define XmNdragReceiverProtocolStyle "dragReceiverProtocolStyle"
#define XmNdropFinishCallback "dropFinishCallback"
#define XmNdropProc "dropProc"
#define XmNdropRectangles "dropRectangles"
#define XmNdropSiteActivity "dropSiteActivity"
#define XmNdropSiteEnterCallback "dropSiteEnterCallback"
#define XmNdropSiteLeaveCallback "dropSiteLeaveCallback"
#define XmNdropSiteManagerClass "dropSiteManagerClass"
#define XmNdropSiteOperations "dropSiteOperations"
#define XmNdropSiteType "dropSiteType"
#define XmNdropStartCallback "dropStartCallback"
#define XmNdropTransferClass "dropTransferClass"
#define XmNdropTransfers "dropTransfers"
#define XmNeditMode "editMode"
#define XmNeditable "editable"
#define XmNentryAlignment "entryAlignment"
#define XmNentryBorder "entryBorder"
#define XmNentryCallback "entryCallback"
#define XmNentryClass "entryClass"
#define XmNentryVerticalAlignment "entryVerticalAlignment"
#define XmNexportTargets "exportTargets"
#define XmNexposeCallback "exposeCallback"
#define XmNextendedSelectionCallback "extendedSelectionCallback"
#define XmNextensionType "extensionType"
#define XmNfileListItemCount "fileListItemCount"
#define XmNfileListItems "fileListItems"
#define XmNfileListLabelString "fileListLabelString"
```

Figure 6-301: <Xm/XmStrDefs.h>*, Part 14 of 34

```
#define XmNfileSearchProc "fileSearchProc"
#define XmNfileTypeMask "fileTypeMask"
#define XmNfillOnArm "fillOnArm"
#define XmNfillOnSelect "fillOnSelect"
#define XmNfilterLabelString "filterLabelString"
#define XmNfocusCallback "focusCallback"
#define XmNfocusMovedCallback "focusMovedCallback"
#define XmNfocusPolicyChanged "focusPolicyChanged"
#define XmNfontList "fontList"
#define XmNforegroundThreshold "foregroundThreshold"
#define XmNfractionBase "fractionBase"
#define XmNgainPrimaryCallback "gainPrimaryCallback"
#define XmNhelpCallback "helpCallback"
#define XmNhelpLabelString "helpLabelString"
#define XmNhighlightColor "highlightColor"
#define XmNhighlightOnEnter "highlightOnEnter"
#define XmNhighlightPixmap "highlightPixmap"
#define XmNhighlightThickness "highlightThickness"
#define XmNhistoryItemCount "historyItemCount"
#define XmNhistoryItems "historyItems"
#define XmNhistoryMaxItems "historyMaxItems"
#define XmNhistoryVisibleItemCount "historyVisibleItemCount"
#define XmNhorizontalFontUnit "horizontalFontUnit"
#define XmNhorizontalScrollBar "horizontalScrollBar"
#define XmNhorizontalSpacing "horizontalSpacing"
#define XmNhotX "hotX"
#define XmNhotY "hotY"
#define XmNiccHandle "iccHandle"
#define XmNimportTargets "importTargets"
#define XmNincrement "increment"
#define XmNincrementCallback "incrementCallback"
#define XmNincremental "incremental"
#define XmNindicatorOn "indicatorOn"
#define XmNindicatorSize "indicatorSize"
#define XmNindicatorType "indicatorType"
```

Figure 6-302: <Xm/XmStrDefs.h>*, Part 15 of 34

```
#define XmNinitialDelay "initialDelay"
#define XmNinitialFocus "initialFocus"
#define XmNinputCallback "inputCallback"
#define XmNinputCreate "inputCreate"
#define XmNinputMethod "inputMethod"
#define XmNinvalidCursorForeground "invalidCursorForeground"
#define XmNisAligned "isAligned"
#define XmNisHomogeneous "isHomogeneous"
#define XmNitemCount "itemCount"
#define XmNitems "items"
#define XmNkeyboardFocusPolicy "keyboardFocusPolicy"
#define XmNlabelFontList "labelFontList"
#define XmNlabelInsensitivePixmap "labelInsensitivePixmap"
#define XmNlabelPixmap "labelPixmap"
#define XmNlabelString "labelString"
#define XmNlabelType "labelType"
#define XmNleftAttachment "leftAttachment"
#define XmNleftOffset "leftOffset"
#define XmNleftPosition "leftPosition"
#define XmNleftWidget "leftWidget"
#define XmNlightThreshold "lightThreshold"
#define XmNlineSpace "lineSpace"
#define XmNlistItemCount "listItemCount"
#define XmNlistItems "listItems"
#define XmNlistLabelString "listLabelString"
#define XmNlistMarginHeight "listMarginHeight"
#define XmNlistMarginWidth "listMarginWidth"
#define XmNlistSizePolicy "listSizePolicy"
#define XmNlistSpacing "listSpacing"
#define XmNlistUpdated "listUpdated"
#define XmNlistVisibleItemCount "listVisibleItemCount"
#define XmNlogicalParent "logicalParent"
#define XmNlosePrimaryCallback "losePrimaryCallback"
#define XmNlosingFocusCallback "losingFocusCallback"
#define XmNmainWindowMarginHeight "mainWindowMarginHeight"
```

Figure 6-303: <Xm/XmStrDefs.h>*, Part 16 of 34

```
#define XmNmainWindowMarginWidth "mainWindowMarginWidth"
#define XmNmapCallback "mapCallback"
#define XmNmappingDelay "mappingDelay"
#define XmNmargin "margin"
#define XmNmarginBottom "marginBottom"
#define XmNmarginHeight "marginHeight"
#define XmNmarginLeft "marginLeft"
#define XmNmarginRight "marginRight"
#define XmNmarginTop "marginTop"
#define XmNmarginWidth "marginWidth"
#define XmNmask "mask"
#define XmNmaxLength "maxLength"
#define XmNmaximum "maximum"
#define XmNmenuAccelerator "menuAccelerator"
#define XmNmenuBar "menuBar"
#define XmNmenuCursor "menuCursor"
#define XmNmenuHelpWidget "menuHelpWidget"
#define XmNmenuHistory "menuHistory"
#define XmNmenuPost "menuPost"
#define XmNmessageAlignment "messageAlignment"
#define XmNmessageProc "messageProc"
#define XmNmessageString "messageString"
#define XmNmessageWindow "messageWindow"
#define XmNminimizeButtons "minimizeButtons"
#define XmNminimum "minimum"
#define XmNmnemonic "mnemonic"
#define XmNmnemonicCharSet "mnemonicCharSet"
#define XmNmodifyVerifyCallback "modifyVerifyCallback"
#define XmNmodifyVerifyCallbackWcs "modifyVerifyCallbackWcs"
#define XmNmotionVerifyCallback "motionVerifyCallback"
#define XmNmoveOpaque "moveOpaque"
#define XmNmultiClick "multiClick"
#define XmNmultipleSelectionCallback "multipleSelectionCallback"
#define XmNmustMatch "mustMatch"
#define XmNmwmDecorations "mwmDecorations"
```

Figure 6-304: <Xm/XmStrDefs.h>*, Part 17 of 34

```
#define XmNmwmFunctions "mwmFunctions"
#define XmNmwmInputMode "mwmInputMode"
#define XmNmwmMenu "mwmMenu"
#define XmNmwmMessages "mwmMessages"
#define XmNnavigationType "navigationType"
#define XmNneedsMotion "needsMotion"
#define XmNnoMatchCallback "noMatchCallback"
#define XmNnoMatchString "noMatchString"
#define XmNnoResize "noResize"
#define XmNnoneCursorForeground "noneCursorForeground"
#define XmNnotifyProc "notifyProc"
#define XmNnumColumns "numColumns"
#define XmNnumDropRectangles "numDropRectangles"
#define XmNnumDropTransfers "numDropTransfers"
#define XmNnumExportTargets "numExportTargets"
#define XmNnumImportTargets "numImportTargets"
#define XmNnumRectangles "numRectangles"
#define XmNoffsetX "offsetX"
#define XmNoffsetY "offsetY"
#define XmNokCallback "okCallback"
#define XmNokLabelString "okLabelString"
#define XmNoperationChangedCallback "operationChangedCallback"
#define XmNoperationCursorIcon "operationCursorIcon"
#define XmNoptionLabel "optionLabel"
#define XmNoptionMnemonic "optionMnemonic"
#define XmNoutputCreate "outputCreate"
#define XmNpacking "packing"
#define XmNpageDecrementCallback "pageDecrementCallback"
#define XmNpageIncrement "pageIncrement"
#define XmNpageIncrementCallback "pageIncrementCallback"
#define XmNpaneMaximum "paneMaximum"
#define XmNpaneMinimum "paneMinimum"
#define XmNpattern "pattern"
#define XmNpendingDelete "pendingDelete"
#define XmNpopupEnabled "popupEnabled"
```

Figure 6-305: <Xm/XmStrDefs.h>*, Part 18 of 34

```
#define XmNpositionIndex "positionIndex"
#define XmNpostFromButton "postFromButton"
#define XmNpostFromCount "postFromCount"
#define XmNpostFromList "postFromList"
#define XmNpreeditType "preeditType"
#define XmNprocessingDirection "processingDirection"
#define XmNpromptString "promptString"
#define XmNprotocolCallback "protocolCallback"
#define XmNpushButtonEnabled "pushButtonEnabled"
#define XmNqualifySearchDataProc "qualifySearchDataProc"
#define XmNradioAlwaysOne "radioAlwaysOne"
#define XmNradioBehavior "radioBehavior"
#define XmNrealizeCallback "realizeCallback"
#define XmNrecomputeSize "recomputeSize"
#define XmNrectangles "rectangles"
#define XmNrefigureMode "refigureMode"
#define XmNrepeatDelay "repeatDelay"
#define XmNresizable "resizable"
#define XmNresizeCallback "resizeCallback"
#define XmNresizeHeight "resizeHeight"
#define XmNresizePolicy "resizePolicy"
#define XmNresizeWidth "resizeWidth"
#define XmNrightAttachment "rightAttachment"
#define XmNrightOffset "rightOffset"
#define XmNrightPosition "rightPosition"
#define XmNrightWidget "rightWidget"
#define XmNrowColumnType "rowColumnType"
#define XmNrows "rows"
#define XmNrubberPositioning "rubberPositioning"
#define XmNsashHeight "sashHeight"
#define XmNsashIndent "sashIndent"
#define XmNsashShadowThickness "sashShadowThickness"
#define XmNsashWidth "sashWidth"
#define XmNscaleHeight "scaleHeight"
#define XmNscaleMultiple "scaleMultiple"
```

Figure 6-306: <Xm/XmStrDefs.h>*, Part 19 of 34

```
#define XmNscaleWidth "scaleWidth"
#define XmNscrollBarDisplayPolicy "scrollBarDisplayPolicy"
#define XmNscrollBarPlacement "scrollBarPlacement"
#define XmNscrollHorizontal "scrollHorizontal"
#define XmNscrollLeftSide "scrollLeftSide"
#define XmNscrollTopSide "scrollTopSide"
#define XmNscrollVertical "scrollVertical"
#define XmNscrolledWindowMarginHeight "scrolledWindowMarginHeight"
#define XmNscrolledWindowMarginWidth "scrolledWindowMarginWidth"
#define XmNscrollingPolicy "scrollingPolicy"
#define XmNselectColor "selectColor"
#define XmNselectInsensitivePixmap "selectInsensitivePixmap"
#define XmNselectPixmap "selectPixmap"
#define XmNselectThreshold "selectThreshold"
#define XmNselectedItemCount "selectedItemCount"
#define XmNselectedItems "selectedItems"
#define XmNselectionArrayCount "selectionArrayCount"
#define XmNselectionLabelString "selectionLabelString"
#define XmNselectionPolicy "selectionPolicy"
#define XmNseparatorOn "separatorOn"
#define XmNseparatorType "separatorType"
#define XmNset "set"
#define XmNshadow "shadow"
#define XmNshadowThickness "shadowThickness"
#define XmNshadowType "shadowType"
#define XmNshellUnitType "shellUnitType"
#define XmNshowArrows "showArrows"
#define XmNshowAsDefault "showAsDefault"
#define XmNshowSeparator "showSeparator"
#define XmNshowValue "showValue"
#define XmNsimpleCallback "simpleCallback"
#define XmNsingleSelectionCallback "singleSelectionCallback"
#define XmNsizePolicy "sizePolicy"
#define XmNskipAdjust "skipAdjust"
#define XmNsliderSize "sliderSize"
```


Figure 6-307: <Xm/XmStrDefs.h>*, Part 20 of 34

```
#define XmNsource "source"
#define XmNsourceCursorIcon "sourceCursorIcon"
#define XmNsourceIsExternal "sourceIsExternal"
#define XmNsourcePixmapIcon "sourcePixmapIcon"
#define XmNsourceWidget "sourceWidget"
#define XmNsourceWindow "sourceWindow"
#define XmNspacing "spacing"
#define XmNspotLocation "spotLocation"
#define XmNstartTime "startTime"
#define XmNstateCursorIcon "stateCursorIcon"
#define XmNstringDirection "stringDirection"
#define XmNsubMenuId "subMenuId"
#define XmNsymbolPixmap "symbolPixmap"
#define XmNtearOffMenuActivateCallback "tearOffMenuActivateCallback"
#define XmNtearOffMenuDeactivateCallback "tearOffMenuDeactivateCallback"
#define XmNtearOffModel "tearOffModel"
#define XmNtextAccelerators "textAccelerators"
#define XmNtextColumns "textColumns"
#define XmNtextFontList "textFontList"
#define XmNtextString "textString"
#define XmNtextTranslations "textTranslations"
#define XmNtextValue "textValue"
#define XmNtitleString "titleString"
#define XmNtoBottomCallback "toBottomCallback"
#define XmNtoPositionCallback "toPositionCallback"
#define XmNtoTopCallback "toTopCallback"
#define XmNtopAttachment "topAttachment"
#define XmNtopCharacter "topCharacter"
#define XmNtopItemPosition "topItemPosition"
#define XmNtopLevelEnterCallback "topLevelEnterCallback"
#define XmNtopLevelLeaveCallback "topLevelLeaveCallback"
#define XmNtopOffset "topOffset"
#define XmNtopPosition "topPosition"
#define XmNtopShadowColor "topShadowColor"
#define XmNtopShadowPixmap "topShadowPixmap"
```

Figure 6-308: <Xm/XmStrDefs.h>*, Part 21 of 34

```
#define XmNtopWidget "topWidget"
#define XmNtransferProc "transferProc"
#define XmNtransferStatus "transferStatus"
#define XmNtraversalCallback "traversalCallback"
#define XmNtraversalOn "traversalOn"
#define XmNtraversalType "traversalType"
#define XmNtraverseObscuredCallback "traverseObscuredCallback"
#define XmNtreeUpdateProc "treeUpdateProc"
#define XmNtroughColor "troughColor"
#define XmNunitType "unitType"
#define XmNunmapCallback "unmapCallback"
#define XmNunpostBehavior "unpostBehavior"
#define XmNunselectPixmap "unselectPixmap"
#define XmNupdateSliderSize "updateSliderSize"
#define XmNuseAsyncGeometry "useAsyncGeometry"
#define XmNuserData "userData"
#define XmNvalidCursorForeground "validCursorForeground"
#define XmNvalueChangedCallback "valueChangedCallback"
#define XmNvalueWcs "valueWcs"
#define XmNverifyBell "verifyBell"
#define XmNverticalFontUnit "verticalFontUnit"
#define XmNverticalScrollBar "verticalScrollBar"
#define XmNverticalSpacing "verticalSpacing"
#define XmNvisibleItemCount "visibleItemCount"
#define XmNvisibleWhenOff "visibleWhenOff"
#define XmNvisualPolicy "visualPolicy"
#define XmNwhichButton "whichButton"
#define XmNwordWrap "wordWrap"
#define XmNworkWindow "workWindow"
#define XmRAlignment "Alignment"
#define XmRAnimationMask "AnimationMask"
#define XmRAnimationPixmap "AnimationPixmap"
#define XmRAnimationStyle "AnimationStyle"
#define XmRArrowDirection "ArrowDirection"
#define XmRAtomList "AtomList"
```

Figure 6-309: <Xm/XmStrDefs.h>*, Part 22 of 34

```
#define XmRAattachment "Attachment"
#define XmRAudibleWarning "AudibleWarning"
#define XmRAvailability "Availability"
#define XmRBackgroundPixmap "BackgroundPixmap"
#define XmRBlendModel "BlendModel"
#define XmRBooleanDimension "BooleanDimension"
#define XmRBottomShadowPixmap "BottomShadowPixmap"
#define XmRButtonType "ButtonType"
#define XmRCallbackProc "CallbackProc"
#define XmRChar "Char"
#define XmRCharSetTable "CharSetTable"
#define XmRChildHorizontalAlignment "ChildHorizontalAlignment"
#define XmRChildPlacement "ChildPlacement"
#define XmRChildType "ChildType"
#define XmRChildVerticalAlignment "ChildVerticalAlignment"
#define XmRCommandWindowLocation "CommandWindowLocation"
#define XmRCompoundText "CompoundText"
#define XmRDefaultButtonType "DefaultButtonType"
#define XmRDeleteResponse "DeleteResponse"
#define XmRDialogStyle "DialogStyle"
#define XmRDialogType "DialogType"
#define XmRDoubleClickInterval "DoubleClickInterval"
#define XmRDragInitiatorProtocolStyle "DragInitiatorProtocolStyle"
#define XmRDragReceiverProtocolStyle "DragReceiverProtocolStyle"
#define XmRDropSiteActivity "DropSiteActivity"
#define XmRDropSiteOperations "DropSiteOperations"
#define XmRDropSiteType "DropSiteType"
#define XmRDropTransfers "DropTransfers"
#define XmRExtensionType "ExtensionType"
#define XmRFileTypeMask "FileTypeMask"
#define XmRFontList "FontList"
#define XmRGadgetPixmap "GadgetPixmap"
#define XmRHighlightPixmap "HighlightPixmap"
#define XmRHorizontalDimension "HorizontalDimension"
#define XmRHorizontalInt "HorizontalInt"
```

Figure 6-310: <Xm/XmStrDefs.h>*, Part 23 of 34

```
#define XmRHorizontalPosition "HorizontalPosition"
#define XmRIconAttachment "IconAttachment"
#define XmRImportTargets "ImportTargets"
#define XmRIndicatorType "IndicatorType"
#define XmRItemCount "ItemCount"
#define XmRItems "Items"
#define XmRKeySym "KeySym"
#define XmRKeySymTable "KeySymTable"
#define XmRKeyboardFocusPolicy "KeyboardFocusPolicy"
#define XmRLabelType "LabelType"
#define XmRListMarginHeight "ListMarginHeight"
#define XmRListMarginWidth "ListMarginWidth"
#define XmRListSizePolicy "ListSizePolicy"
#define XmRListSpacing "ListSpacing"
#define XmRManBottomShadowPixmap "ManBottomShadowPixmap"
#define XmRManForegroundPixmap "ManForegroundPixmap"
#define XmRManHighlightPixmap "ManHighlightPixmap"
#define XmRManTopShadowPixmap "ManTopShadowPixmap"
#define XmRMenuWidget "MenuWidget"
#define XmRMnemonic "Mnemonic"
#define XmRMultiClick "MultiClick"
#define XmRNavigationType "NavigationType"
#define XmRPacking "Packing"
#define XmRPrimForegroundPixmap "PrimForegroundPixmap"
#define XmRProc "Proc"
#define XmRProcessingDirection "ProcessingDirection"
#define XmRRectangleList "RectangleList"
#define XmRResizePolicy "ResizePolicy"
#define XmRRowColumnType "RowColumnType"
#define XmRScrollBarDisplayPolicy "ScrollBarDisplayPolicy"
#define XmRScrollBarPlacement "ScrollBarPlacement"
#define XmRScrollingPolicy "ScrollingPolicy"
#define XmRSelectedItemCount "SelectedItemCount"
#define XmRSelectedItems "SelectedItems"
#define XmRSelectionPolicy "SelectionPolicy"
```

Figure 6-311: <Xm/XmStrDefs.h>*, Part 24 of 34

```
#define XmRSelectionType "SelectionType"
#define XmRSeparatorType "SeparatorType"
#define XmRShadowType "ShadowType"
#define XmRShellHorizDim "ShellHorizDim"
#define XmRShellHorizPos "ShellHorizPos"
#define XmRShellUnitType "ShellUnitType"
#define XmRShellVertDim "ShellVertDim"
#define XmRShellVertPos "ShellVertPos"
#define XmRSizePolicy "SizePolicy"
#define XmRStringDirection "StringDirection"
#define XmRTearOffModel "TearOffModel"
#define XmRTopShadowPixmap "TopShadowPixmap"
#define XmRTransferStatus "TransferStatus"
#define XmRTraversalType "TraversalType"
#define XmRUnitType "UnitType"
#define XmRUNpostBehavior "UnpostBehavior"
#define XmRValueWcs "ValueWcs"
#define XmRVerticalAlignment "VerticalAlignment"
#define XmRVerticalDimension "VerticalDimension"
#define XmRVerticalInt "VerticalInt"
#define XmRVerticalPosition "VerticalPosition"
#define XmRVirtualBinding "VirtualBinding"
#define XmRVisibleItemCount "VisibleItemCount"
#define XmRVisualPolicy "VisualPolicy"
#define XmRWhichButton "WhichButton"
#define XmRXmBackgroundPixmap "XmBackgroundPixmap"
#define XmRXmString "XmString"
#define XmRXmStringCharSet "XmStringCharSet"
#define XmRXmStringTable "XmStringTable"
#define XmVosfActivate "osfActivate"
#define XmVosfAddMode "osfAddMode"
#define XmVosfBackSpace "osfBackSpace"
#define XmVosfBeginLine "osfBeginLine"
#define XmVosfCancel "osfCancel"
#define XmVosfClear "osfClear"
```

Figure 6-312: <Xm/XmStrDefs.h>*, Part 25 of 34

```
#define XmVosfCopy "osfCopy"
#define XmVosfCut "osfCut"
#define XmVosfDelete "osfDelete"
#define XmVosfDown "osfDown"
#define XmVosfEndLine "osfEndLine"
#define XmVosfHelp "osfHelp"
#define XmVosfInsert "osfInsert"
#define XmVosfLeft "osfLeft"
#define XmVosfMenu "osfMenu"
#define XmVosfMenuBar "osfMenuBar"
#define XmVosfPageDown "osfPageDown"
#define XmVosfPageLeft "osfPageLeft"
#define XmVosfPageRight "osfPageRight"
#define XmVosfPageUp "osfPageUp"
#define XmVosfPaste "osfPaste"
#define XmVosfPrimaryPaste "osfPrimaryPaste"
#define XmVosfQuickPaste "osfQuickPaste"
#define XmVosfRight "osfRight"
#define XmVosfSelect "osfSelect"
#define XmVosfUndo "osfUndo"
#define XmVosfUp "osfUp"
#define XmSFONTLIST_DEFAULT_TAG_STRING "FONTLIST_DEFAULT_TAG_STRING"
#define XmSXmFONTLIST_DEFAULT_TAG_STRING "XmFONTLIST_DEFAULT_TAG_STRING"
#define _XmConst /**/

#define XmSTRING_DEFAULT_CHARSET      XmS
#define XmSTRING_ISO8859_1           "ISO8859-1"
#define XmFONTLIST_DEFAULT_TAG        XmSFONTLIST_DEFAULT_TAG_STRING
#define XmFONTLIST_DEFAULT_TAG_STRING XmSXmFONTLIST_DEFAULT_TAG_STRING

#define XmVaCASCADEBUTTON             "cascadeButton"
#define XmVaCHECKBUTTON               "checkButton"
#define XmVaDOUBLE_SEPARATOR          "doubleSeparator"
#define XmVaPUSHBUTTON                "pushButton"
#define XmVaRADIOBUTTON               "radioButton"
```

Figure 6-313: <Xm/XmStrDefs.h>*, Part 26 of 34

```
#define XmVaSEPARATOR          "separator"
#define XmVaSINGLE_SEPARATOR    "singleSeparator"
#define XmVaTOGGLEBUTTON       "checkButton"
#define XmVaTITLE               XtNtitle

#define XtCKeyboardFocusPolicy  XmCKeyboardFocusPolicy
#define XtCShellUnitType        XmCShellUnitType
#define XtNkeyboardFocusPolicy  XmNkeyboardFocusPolicy
#define XtNshellUnitType        XmNshellUnitType
#define XtRKeyboardFocusPolicy  XmRKeyboardFocusPolicy

#define XmRPrimBottomShadowPixmap XmRBottomShadowPixmap
#define XmRPrimHighlightPixmap    XmRHighlightPixmap
#define XmRPrimTopShadowPixmap    XmRTopShadowPixmap

#define XmCAccelerators         XtCAccelerators
#define XmCAllowShellResize     XtCAllowShellResize
#define XmCArgc                  XtCArgc
#define XmCArgv                  XtCArgv
#define XmCBackground           XtCBackground
#define XmCBaseHeight            XtCBaseHeight
#define XmCBaseHeight            XtCBaseHeight
#define XmCBaseWidth             XtCBaseWidth
#define XmCBaseWidth             XtCBaseWidth
#define XmCBitmap                XtCBitmap
#define XmCBoolean               XtCBoolean
#define XmCBorderColor           XtCBorderColor
#define XmCBorderWidth           XtCBorderWidth
#define XmCCallback              XtCCallback
#define XmCColor                 XtCColor
#define XmCColormap              XtCColormap
#define XmCCreatePopupChildProc  XtCCreatePopupChildProc
#define XmCCursor                XtCCursor
#define XmCDepth                 XtCDepth
#define XmCDimension             XtRDimension
```

Figure 6-314: <Xm/XmStrDefs.h>*, Part 27 of 34

```
#define XmCEditMode           XtREditMode
#define XmCEditType          XtCEditType
#define XmCEventBindings     XtCEventBindings
#define XmCFile              XtCFile
#define XmCFont              XtCFont
#define XmCFontSet           XtCFontSet
#define XmCForeground        XtCForeground
#define XmCFraction          XtCFraction
#define XmCFunction          XtCFunction
#define XmCGeometry          XtCGeometry
#define XmCHSpace            XtCHSpace
#define XmCHeight            XtCHeight
#define XmCHeightInc         XtCHeightInc
#define XmCIconMask          XtCIconMask
#define XmCIconName          XtCIconName
#define XmCIconNameEncoding  XtCIconNameEncoding
#define XmCIconPixmap        XtCIconPixmap
#define XmCIconWindow        XtCIconWindow
#define XmCIconX             XtCIconX
#define XmCIconY             XtCIconY
#define XmCIconic            XtCIconic
#define XmCIndex             XtCIndex
#define XmCInitialResourcesPersistent XtCInitialResourcesPersistent
#define XmCInitialState      XtCInitialState
#define XmCInput             XtCInput
#define XmCInsertPosition    XtCInsertPosition
#define XmCInterval          XtCInterval
#define XmCJustify           XtCJustify
#define XmCLabel             XtCLabel
#define XmCLength            XtCLength
#define XmCMappedWhenManaged XtCMappedWhenManaged
#define XmCMargin            XtCMargin
#define XmCMaxAspectX        XtCMaxAspectX
#define XmCMaxAspectY        XtCMaxAspectY
#define XmCMaxHeight         XtCMaxHeight
```


Figure 6-315: <Xm/XmStrDefs.h>*, Part 28 of 34

#define XmCMaxWidth	XtCMaxWidth
#define XmCMenuEntry	XtCMenuEntry
#define XmCMinAspectX	XtCMinAspectX
#define XmCMinAspectY	XtCMinAspectY
#define XmCMinHeight	XtCMinHeight
#define XmCMinWidth	XtCMinWidth
#define XmCNotify	XtCNotify
#define XmCOrientation	XtCOrientation
#define XmCOverrideRedirect	XtCOverrideRedirect
#define XmCParameter	XtCParameter
#define XmCPixmap	XtCPixmap
#define XmCPosition	XtCPosition
#define XmCReadOnly	XtCReadOnly
#define XmCResize	XtCResize
#define XmCReverseVideo	XtCReverseVideo
#define XmCSaveUnder	XtCSaveUnder
#define XmCScreen	XtCScreen
#define XmCScrollDCursor	XtCScrollDCursor
#define XmCScrollHCursor	XtCScrollHCursor
#define XmCScrollLCursor	XtCScrollLCursor
#define XmCScrollProc	XtCScrollProc
#define XmCScrollRCursor	XtCScrollRCursor
#define XmCScrollUCursor	XtCScrollUCursor
#define XmCScrollVCursor	XtCScrollVCursor
#define XmCSelection	XtCSelection
#define XmCSelectionArray	XtCSelectionArray
#define XmCSensitive	XtCSensitive
#define XmCSpace	XtCSpace
#define XmCString	XtCString
#define XmCTextOptions	XtCTextOptions
#define XmCTextPosition	XtCTextPosition
#define XmCTextSink	XtCTextSink
#define XmCTextSource	XtCTextSource
#define XmCThickness	XtCThickness
#define XmCThumb	XtCThumb

Figure 6-316: <Xm/XmStrDefs.h>*, Part 29 of 34

#define XmCTitle	XtCTitle
#define XmCTitleEncoding	XtCTitleEncoding
#define XmCTransient	XtCTransient
#define XmCTransientFor	XtCTransientFor
#define XmCTranslations	XtCTranslations
#define XmCVSpace	XtCVSpace
#define XmCValue	XtCValue
#define XmCVisual	XtCVisual
#define XmCWaitForWm	XtCWaitForWm
#define XmCWidget	XtRWidget
#define XmCWidth	XtCWidth
#define XmCWidthInc	XtCWidthInc
#define XmCWinGravity	XtCWinGravity
#define XmCWindow	XtCWindow
#define XmCWindowGroup	XtCWindowGroup
#define XmCwmTimeout	XtCwmTimeout
#define XmCX	XtCX
#define XmCY	XtCY
#define XmNaccelerators	XtNaccelerators
#define XmNallowShellResize	XtNallowShellResize
#define XmNancestorSensitive	XtNancestorSensitive
#define XmNargc	XtNargc
#define XmNargv	XtNargv
#define XmNbackground	XtNbackground
#define XmNbackgroundPixmap	XtNbackgroundPixmap
#define XmNbaseHeight	XtNbaseHeight
#define XmNbaseHeight	XtNbaseHeight
#define XmNbaseWidth	XtNbaseWidth
#define XmNbaseWidth	XtNbaseWidth
#define XmNbitmap	XtNbitmap
#define XmNborder	XtNborder
#define XmNborderColor	XtNborderColor
#define XmNborderPixmap	XtNborderPixmap
#define XmNborderWidth	XtNborderWidth
#define XmNcallback	XtNcallback

Figure 6-317: <Xm/XmStrDefs.h>*, Part 30 of 34

#define XmNchildren	XtNchildren
#define XmNcolormap	XtNcolormap
#define XmNcreatePopupChildProc	XtNcreatePopupChildProc
#define XmNdepth	XtNdepth
#define XmNdestroyCallback	XtNdestroyCallback
#define XmNeditType	XtNeditType
#define XmNfile	XtNfile
#define XmNfont	XtNfont
#define XmNfontSet	XtNfontSet
#define XmNforceBars	XtNforceBars
#define XmNforeground	XtNforeground
#define XmNfunction	XtNfunction
#define XmNgeometry	XtNgeometry
#define XmNheight	XtNheight
#define XmNheightInc	XtNheightInc
#define XmNhighlight	XtNhighlight
#define XmNiconMask	XtNiconMask
#define XmNiconName	XtNiconName
#define XmNiconNameEncoding	XtNiconNameEncoding
#define XmNiconPixmap	XtNiconPixmap
#define XmNiconWindow	XtNiconWindow
#define XmNiconX	XtNiconX
#define XmNiconY	XtNiconY
#define XmNiconic	XtNiconic
#define XmNindex	XtNindex
#define XmNinitialResourcesPersistent	XtNinitialResourcesPersistent
#define XmNinitialState	XtNinitialState
#define XmNinnerHeight	XtNinnerHeight
#define XmNinnerWidth	XtNinnerWidth
#define XmNinnerWindow	XtNinnerWindow
#define XmNinput	XtNinput
#define XmNinsertPosition	XtNinsertPosition
#define XmNinternalHeight	XtNinternalHeight
#define XmNinternalWidth	XtNinternalWidth
#define XmNjumpProc	XtNjumpProc

Figure 6-318: <Xm/XmStrDefs.h>*, Part 31 of 34

```
#define XmNjustify          XtNjustify
#define XmNlength          XtNlength
#define XmNlowerRight      XtNlowerRight
#define XmNmappedWhenManaged XtNmappedWhenManaged
#define XmNmaxAspectX      XtNmaxAspectX
#define XmNmaxAspectY      XtNmaxAspectY
#define XmNmaxHeight       XtNmaxHeight
#define XmNmaxWidth        XtNmaxWidth
#define XmNmenuEntry       XtNmenuEntry
#define XmNminAspectX      XtNminAspectX
#define XmNminAspectY      XtNminAspectY
#define XmNminHeight       XtNminHeight
#define XmNminWidth        XtNminWidth
#define XmNname            XtNname
#define XmNnotify          XtNnotify
#define XmNnumChildren     XtNnumChildren
#define XmNorientation     XtNorientation
#define XmNoverrideRedirect XtNoverrideRedirect
#define XmNparameter       XtNparameter
#define XmNpixmap          XtNpixmap
#define XmNpopdownCallback XtNpopdownCallback
#define XmNpopupCallback   XtNpopupCallback
#define XmNresize          XtNresize
#define XmNreverseVideo     XtNreverseVideo
#define XmNsaveUnder       XtNsaveUnder
#define XmNscreen          XtNscreen
#define XmNscrollDCursor   XtNscrollDCursor
#define XmNscrollHCursor   XtNscrollHCursor
#define XmNscrollLCursor   XtNscrollLCursor
#define XmNscrollProc      XtNscrollProc
#define XmNscrollRCursor   XtNscrollRCursor
#define XmNscrollUCursor   XtNscrollUCursor
#define XmNscrollVCursor   XtNscrollVCursor
#define XmNselection       XtNselection
#define XmNselectionArray   XtNselectionArray
```

Figure 6-319: <Xm/XmStrDefs.h>*, Part 32 of 34

#define XmNsensitive	XtNsensitive
#define XmNshown	XtNshown
#define XmNspace	XtNspace
#define XmNstring	XtNstring
#define XmNtextOptions	XtNtextOptions
#define XmNtextSink	XtNtextSink
#define XmNtextSource	XtNtextSource
#define XmNthickness	XtNthickness
#define XmNthumb	XtNthumb
#define XmNthumbProc	XtNthumbProc
#define XmNtitle	XtNtitle
#define XmNtitleEncoding	XtNtitleEncoding
#define XmNtop	XtNtop
#define XmNtransient	XtNtransient
#define XmNtransientFor	XtNtransientFor
#define XmNtransientFor	XtNtransientFor
#define XmNtranslations	XtNtranslations
#define XmNupdate	XtNupdate
#define XmNuseBottom	XtNuseBottom
#define XmNuseRight	XtNuseRight
#define XmNvalue	XtNvalue
#define XmNvisual	XtNvisual
#define XmNwaitForWm	XtNwaitForWm
#define XmNwidth	XtNwidth
#define XmNwidthInc	XtNwidthInc
#define XmNwinGravity	XtNwinGravity
#define XmNwindow	XtNwindow
#define XmNwindowGroup	XtNwindowGroup
#define XmNwmTimeout	XtNwmTimeout
#define XmNx	XtNx
#define XmNy	XtNy
#define XmRAcceleratorTable	XtRAcceleratorTable
#define XmRAtom	XtRAtom
#define XmRBitmap	XtRBitmap
#define XmRBool	XtRBool

Figure 6-320: <Xm/XmStrDefs.h>*, Part 33 of 34

#define XmRBoolean	XtRBoolean
#define XmRCallProc	XtRCallProc
#define XmRCallback	XtRCallback
#define XmRCardinal	XtRCardinal
#define XmRColor	XtRColor
#define XmRColormap	XtRColormap
#define XmRCursor	XtRCursor
#define XmRDimension	XtRDimension
#define XmRDisplay	XtRDisplay
#define XmREditMode	XtREditMode
#define XmREnum	XtREnum
#define XmRFile	XtRFile
#define XmRFloat	XtRFloat
#define XmRFont	XtRFont
#define XmRFontSet	XtRFontSet
#define XmRFontStruct	XtRFontStruct
#define XmRFunction	XtRFunction
#define XmRGeometry	XtRGeometry
#define XmRImmediate	XtRImmediate
#define XmRInitialState	XtRInitialState
#define XmRInt	XtRInt
#define XmRJustify	XtRJustify
#define XmRLongBoolean	XtRLongBoolean
#define XmROrientation	XtROrientation
#define XmRObject	XtRObject
#define XmRPixel	XtRPixel
#define XmRPixmap	XtRPixmap
#define XmRPointer	XtRPointer
#define XmRPosition	XtRPosition
#define XmRScreen	XtRScreen
#define XmRShort	XtRShort
#define XmRString	XtRString
#define XmRStringArray	XtRStringArray
#define XmRStringTable	XtRStringTable
#define XmRTextPosition	XtCTextPosition

Figure 6-321: <Xm/XmStrDefs.h>*, Part 34 of 34

#define XmRTranslationTable	XtRTranslationTable
#define XmRUnsignedChar	XtRUnsignedChar
#define XmRVisual	XtRVisual
#define XmRWidget	XtRWidget
#define XmRWidgetClass	XtRWidgetClass
#define XmRWidgetList	XtRWidgetList
#define XmRWindow	XtRWindow

TCP/IP Data Definitions

NOTE

This section is new to the Third Edition of this document, but will not be marked with the "G" diff-mark.

This section contains standard data definitions that describe system data for the optional TCP/IP Interfaces. These data definitions are referred to by their names in angle brackets: `<name.h>` and `<sys/name.h>`. Included in these data definitions are macro definitions and structure definitions. While an ABI-conforming system may provide TCP/IP interfaces, it need not contain the actual data definitions referenced here. Programmers should observe that the sources of the structures defined in these data definitions are defined in SVID.

Figure 6-322: <netinet/in.h>

```
#define      IPPROTO_IP      0
#define      IPPROTO_TCP    6

struct in_addr {
    union {
        struct { u_char s_b1,s_b2,s_b3,s_b4; } S_un_b;
        struct { u_short s_w1,s_w2; } S_un_w;
        u_long S_addr;
    } S_un;
#define      s_addr          S_un.S_addr
};

#define      INADDR_ANY      (u_long)0x00000000
#define      INADDR_LOOPBACK (u_long)0x7F000001
#define      INADDR_BROADCAST (u_long)0xffffffff

#define      IN_SET_LOOPBACK_ADDR(a) \
    {(a)->sin_addr.s_addr = htonl(INADDR_LOOPBACK); \
    (a)->sin_family = AF_INET;}

struct sockaddr_in {
    short      sin_family;
    u_short    sin_port;
    struct in_addr sin_addr;
    char       sin_zero[8];
};

#define      IP_OPTIONS      1
```

M

Figure 6-323: <netinet/ip.h>

#define	IPOPT_EOL	0
#define	IPOPT_NOP	1
#define	IPOPT_LSRR	131
#define	IPOPT_SSRR	137

Figure 6-324: <netinet/tcp.h>

#define	TCP_NODELAY	0x01
---------	-------------	------

7 DEVELOPMENT ENVIRONMENT

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Development Commands

NOTE

THE FACILITIES AND INTERFACES DESCRIBED IN THIS SECTION ARE OPTIONAL COMPONENTS OF THE *System V Application Binary Interface*.

NOTE

This section is new to the Third Edition of this document, but will not be marked with the "G" diff-mark.

The Development Environment for Intel386 implementations of UnixWare® 2.0 will contain all of the development commands required by the System V ABI, namely; M

as	cc	ld
m4	lex	yacc

Each command accepts all of the options required by the System V ABI, as defined in the SD_CMD section of the *System V Interface Definition, Edition 4*.

PATH Access to Development Tools

The development environment for the Intel386 System V implementations is accessible using the system default value for PATH. The default if no options are given to the CC command is to use the libraries and object file formats that are required for ABI compliance.

Software Packaging Tools

The development environment for i386 implementations of the System V ABI shall include each of the following commands as defined in the AS_CMD section of - *System V Interface Definition, Edition 4*.

pkgproto pkgtrans pkgmk

System Headers

Systems that do not have an ABI Development Environment may not have system header files. If an ABI Development Environment is supported, system header files will be included with the Development Environment. The primary source for contents of header files is always the *System V Interface Definition, Edition 4*. In those cases where SVID Fourth Edition doesn't specify the contents of system headers, Chapter 6 "Data Definitions" of this document shall define the associations of data elements to system headers for compilation. For greatest source portability, applications should only depend on header file contents defined in SVID.

Static Archives

Level 1 interfaces defined in *System V Interface Definition, Edition 4*, for each of the following libraries, may be statically linked safely into applications. The resulting executable will not be made non-compliant to the ABI solely because of the static linkage of such members in the executable.

libcurses libm

The archive libcurses.a is located in /usr/lib on conforming i386 development environments. The archive libm.a is located in /usr/lib on conforming i386 development environments. M

8 EXECUTION ENVIRONMENT

Application Environment

The /dev Subtree

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Application Environment

NOTE

This section is new to the Third Edition of this document, but will not be marked with the "G" diff-mark.

This section specifies the execution environment information available to application programs running on an i386 ABI-conforming computer.

The /dev Subtree

All networking device files described in the Generic ABI shall be supported on all i386 ABI-conforming computers. In addition, the following device files are required to be present on all i386 ABI-conforming computers.

/dev/null This device file is a special “null” device that may be used to test programs or provide a data sink. This file is writable by all processes.

/dev/tty This device file is a special one that directs all output to the controlling TTY of the current process group. This file is readable and writable by all processes.

/dev/sxtXX

/dev/ttyXX These device files, where XX represents a two-digit integer, represent device entries for terminal sessions. All these device files must be examined by the `ttynam()` call. Applications must not have the device names of individual terminals hard-coded within them. The sxt entries are optional in the system but, if present must be included in the library routine’s search.

The following device files are required to be present on all i386 ABI-conforming computers that support the corresponding hardware devices.

`/dev/lpX` This device file is the lineprinter device. The letter "X" represents a one-digit integer that identifies the particular lineprinter device.

`/dev/dsk/`
`/dev/rdisk/` These directories contain the raw and block disk device files. They are of the form:

`f[01][t]`
`f[01][35][dh][t]`
`c#t#d#s#`

where 'c' is followed by a controller number,
't' is followed by a target number,
'd' is followed by a disk unit number,
's' is followed by a disk slice number.

`/dev/rmt/` These directories contain the raw and block tape device files. The devices guaranteed to be in this directory are:
`ctape1`
`ntape1`

`/dev/cdrom/`
`/dev/rcdrom/` These directories contain the raw and block CD-ROM disk device files. They are of the form:

`c#t#l#`
`c#t#l#`
`cdrom#`

The letter 'c' is followed by a controller number. The letter 't' is followed by a target number on the controller. The letter 'l' is followed by a logical unit number on the target. The device "cdrom" is followed by a sequential number as nodes are created.

No leading zeroes are used in the numbers (target four is t4 not t04). The numbering for 'c', 't' and 'l' begins at zero and the numbering for 'n' begins at one.

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